# Mathematics

# Class 10 (English Medium)

# Solved Exercises — Sindh Textbook Board

14 lessons — 923 questions

# 17. عمل سیٹ

Question: Exercise 17.1, Question 1: Write the following sets in tabular form. (i) A = the set of integers between -3 and 3

Answer: The integers between -3 and 3 are -2, -1, 0, 1, 2.

So A = {-2, -1, 0, 1, 2}.

Question: Exercise 17.1, Question 1: (ii) B = the set of composite numbers less than 11

Answer: The composite numbers less than 11 are 4, 6, 8, 9, 10.

So B = {4, 6, 8, 9, 10}.

Question: Exercise 17.1, Question 1: (iii) C = {x | x ∈ P ∧ 5 < x ≤ 13}

Answer: The prime numbers greater than 5 and up to 13 are 7, 11, 13.

So C = {7, 11, 13}.

Question: Exercise 17.1, Question 1: (iv) D = {y | y ∈ O ∧ 7 < y < 17}

Answer: The odd numbers between 7 and 17 are 9, 11, 13, 15.

So D = {9, 11, 13, 15}.

Question: Exercise 17.1, Question 1: (v) E = {z | z ∈ R ∧ $z^2$ = 121}

Answer: From $z^2$ = 121 we get z = 11 or z = -11, because $11^2$ = 121 and $(-11)^2$ = 121.

So E = {-11, 11}.

Question: Exercise 17.1, Question 1: (vi) F = {p | p ∈ Q ∧ $p^2$ = -1}

Answer: There is no rational number whose square is -1, because the

square of any real number is never negative.

So F = ∅ (the empty set).

Question: Exercise 17.1, Question 2: Write the following sets in set-builder form. (i) A = the set of rational numbers between 5 and 6

Answer: A = {x | x ∈ Q ∧ 5 < x < 6}.

Question: Exercise 17.1, Question 2: (ii) B = {1, 2, 3, 4, 6, 12}

Answer: These are all the exact divisors of 12.

So B = {x | x ∈ N ∧ x is a divisor of 12}.

Question: Exercise 17.1, Question 2: (iii) C = {0, ±1, ±2, …, ±40}

Answer: These are all the integers from -40 to 40.

So C = {x | x ∈ Z ∧ -40 ≤ x ≤ 40}.

Question: Exercise 17.1, Question 2: (iv) D = {-4, -2, 0, 2, 4}

Answer: These are all the even numbers from -4 to 4.

So D = {x | x ∈ E ∧ -4 ≤ x ≤ 4}.

Question: Exercise 17.1, Question 2: (v) E = {1, 4, 9, 16, 25}

Answer: These are the squares of the natural numbers from 1 to 5: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$, $5^2 = 25$.

So E = {x | x = n², n ∈ N ∧ 1 ≤ n ≤ 5}.

Question: Exercise 17.1, Question 2: (vi) F = {-1, -3, -5, -7, …}

Answer: These are all the negative odd numbers. So F = {x | x is a negative odd number}.

Question: Exercise 17.1, Question 3: Give five examples of the empty set.

Answer: (1) {x | x ∈ P ∧ 14 < x < 16} — there is no prime number

between 14 and 16.

(2) $\{x \mid x \in N \wedge x^2 = 2\}$ — no natural number is the square root of 2.

(3) $\{x \mid x \in R \wedge x^2 = -4\}$ — no real number has a negative square.

(4) The set of months with 32 days — there is no such month.

(5) $\{x \mid x \in Z \wedge 3 < x < 4\}$ — there is no integer between 3 and 4.

Each has no members, so each is the empty set.

Question: Exercise 17.1, Question 4: Separate the finite and infinite sets from the following. (i) The set of Asian countries

Answer: It is a finite set, because the number of countries in Asia can be counted and is limited.

Question: Exercise 17.1, Question 4: (ii) The set of all the universities of the world

Answer: It is a finite set, because the number of universities in the world is limited, though large.

Question: Exercise 17.1, Question 4: (iii) The set of all real numbers between 6 and 9

Answer: It is an infinite set, because there are countless real numbers between 6 and 9, which cannot be counted.

Question: Exercise 17.1, Question 4: (iv) The set of all prime numbers

Answer: It is an infinite set, because prime numbers never end.

Question: Exercise 17.1, Question 4: (v) The set of all odd numbers less than 5

Answer: It is a finite set (as natural odd numbers: {1, 3}).

Question: Exercise 17.1, Question 5: For each of the following sets, write one equivalent set, one non-equivalent set, one proper subset and one improper subset. (i) P = {a, e, i, o, u}

Answer: P has 5 members.

Equivalent set (5 members): {1, 2, 3, 4, 5}.

Non-equivalent set (not 5 members): {1, 2, 3}.

Proper subset: {a, e} (part of P but not equal to P).

Improper subset: {a, e, i, o, u} = P itself.

Question: Exercise 17.1, Question 5: (ii) Q = {x | x ∈ Z ∧ -2 ≤ x ≤ 2}

Answer: Q = {-2, -1, 0, 1, 2}, which has 5 members.

Equivalent set: {10, 20, 30, 40, 50}.

Non-equivalent set: {10, 20}.

Proper subset: {-2, -1, 0}.

Improper subset: {-2, -1, 0, 1, 2} = Q itself.

Question: Exercise 17.1, Question 6: Give two examples of a singleton set in set-builder form.

Answer: (1) {x | x ∈ N ∧ 2 < x < 4} = {3} — it has only one member.

(2) {x | x ∈ P ∧ 10 < x < 13} = {11} — it has only one member.

Question: Exercise 17.1, Question 7: Find the power sets of the following

sets. (i) A = {5, 10, 15}

Answer: A has 3 members, so the power set has $2^3 = 8$ subsets: P(A) = {∅, {5}, {10}, {15}, {5, 10}, {5, 15}, {10, 15}, {5, 10, 15}}.

Question: Exercise 17.1, Question 7: (ii) B = {x | x ∈ Z ∧ -1 < x < 4}

Answer: B = {0, 1, 2, 3}, which has 4 members, so the power set has $2^4 = 16$ subsets: P(B) = {∅, {0}, {1}, {2}, {3}, {0, 1}, {0, 2}, {0, 3}, {1, 2}, {1, 3}, {2, 3}, {0, 1, 2}, {0, 1, 3}, {0, 2, 3}, {1, 2, 3}, {0, 1, 2, 3}}.

Question: Exercise 17.1, Question 8: Write a set. (i) Which has only proper subsets

Answer: No such set exists, because every set is an improper subset of itself (A ⊆ A is always true). [This part of the printed question has no solution — noted.]

Question: Exercise 17.1, Question 8: (ii) Which has only one proper subset

Answer: {a} — its only proper subset is ∅. (Itself, {a}, is the improper subset.)

Question: Exercise 17.1, Question 8: (iii) Which has no proper subset

Answer: ∅ — the empty set has no proper subset, because its only subset is ∅ itself, which is improper.

Question: Exercise 17.2, Question 1: State whether the following are disjoint, overlapping, exhaustive or partitions. (i) {1, 2, 3, 5, 7} and {4, 6, 8, 9, 10}

Answer: They are disjoint sets, because they have no common member: {1, 2, 3, 5, 7} ∩ {4, 6, 8, 9, 10} = ∅.

Question: Exercise 17.2, Question 1: (ii) {1, 2, 3, 6} and {1, 2, 4, 8}

Answer: They are overlapping sets, because the common members are {1, 2} and neither set is a subset of the other.

Question: Exercise 17.2, Question 1: (iii) E and O when U = Z

Answer: They form a partition (and are exhaustive): E ∩ O = ∅ and E ∪ O

= Z = U. Even and odd numbers together make all integers, and no number is in both.

Question: Exercise 17.2, Question 1: (iv) N and W when U = {x | x ∈ Z ∧ x ≥ 0}

Answer: They are exhaustive sets: N ∪ W = W = U, because W = {0, 1, 2, ...} = U.

But they are not disjoint (N ⊂ W) and not a partition, because they share members.

Question: Exercise 17.2, Question 1: (v) Q and Q' when U = R

Answer: They form a partition (and are exhaustive): Q ∩ Q' = ∅ and Q ∪ Q' = R = U. Every real number is either rational or irrational, never both.

Question: Exercise 17.2, Question 2: Given A = {1, 2, 3, 4, 5, 6} and B = {2, 4, 6, 8, 10}, find: (i) A ∪ B

Answer: A ∪ B = {1, 2, 3, 4, 5, 6, 8, 10} — all members of both sets, without repetition.

Question: Exercise 17.2, Question 2: (ii) B ∩ A

Answer: B ∩ A = {2, 4, 6} — the members that are in both sets.

Question: Exercise 17.2, Question 2: (iii) A - B

Answer: A - B = {1, 3, 5} — the members of A that are not in B.

Question: Exercise 17.2, Question 2: (iv) B - A

Answer: B - A = {8, 10} — the members of B that are not in A.

Question: Exercise 17.2, Question 2: (v) AΔB

Answer: AΔB = (A ∪ B) - (A ∩ B) = {1, 2, 3, 4, 5, 6, 8, 10} - {2, 4, 6} = {1, 3, 5, 8, 10}.

Question: Exercise 17.2, Question 3: Given U = {1, 2, 3, ..., 10}, A = {1, 2, 3, 4, 5} and B = {1, 3, 5, 7, 9}, find: (i) A'

Answer: A' = U - A = {6, 7, 8, 9, 10}.

Question: Exercise 17.2, Question 3: (ii) B'

Answer: B' = U - B = {2, 4, 6, 8, 10}.

Question: Exercise 17.2, Question 3: (iii) A' ∪ B'

Answer: A' ∪ B' = {6, 7, 8, 9, 10} ∪ {2, 4, 6, 8, 10} = {2, 4, 6, 7, 8, 9, 10}.

Question: Exercise 17.2, Question 3: (iv) A' ∩ B'

Answer: A' ∩ B' = {6, 7, 8, 9, 10} ∩ {2, 4, 6, 8, 10} = {6, 8, 10}.

Question: Exercise 17.2, Question 3: (v) (A ∪ B)'

Answer: A ∪ B = {1, 2, 3, 4, 5, 7, 9}, so (A ∪ B)' = U - (A ∪ B) = {6, 8, 10}.

Question: Exercise 17.2, Question 3: (vi) (A ∩ B)'

Answer: A ∩ B = {1, 3, 5}, so (A ∩ B)' = U - (A ∩ B) = {2, 4, 6, 7, 8, 9, 10}.

Question: Exercise 17.2, Question 3: (vii) A'ΔB'

Answer: A'ΔB' = (A' ∪ B') - (A' ∩ B') = {2, 4, 6, 7, 8, 9, 10} - {6, 8, 10} = {2, 4, 7, 9}.

Question: Exercise 17.2, Question 3: (viii) (AΔB)'

Answer: AΔB = (A ∪ B) - (A ∩ B) = {1, 2, 3, 4, 5, 7, 9} - {1, 3, 5} = {2, 4, 7, 9}, so (AΔB)' = U - {2, 4, 7, 9} = {1, 3, 5, 6, 8, 10}.

Question: Exercise 17.2, Question 3: (ix) A - B'

Answer: A - B' = {1, 2, 3, 4, 5} - {2, 4, 6, 8, 10} = {1, 3, 5}, since B' = {2, 4, 6, 8, 10}.

Question: Exercise 17.2, Question 3: (x) A' - B

Answer: A' = {6, 7, 8, 9, 10}, so A' - B = {6, 7, 8, 9, 10} - {1, 3, 5, 7, 9} = {6, 8, 10}.

Question: Exercise 17.2, Question 4: Given U = {x | x ∈ Z ∧ -4 < x < 6}, P = {p | p ∈ E ∧ -4 < p < 8} and Q = {q | q ∈ P ∧ q < 6}, verify: (i) P - Q = P ∩ Q'

Answer: U = {-3, -2, -1, 0, 1, 2, 3, 4, 5}, P = {-2, 0, 2, 4, 6}, Q = {-2, 0, 2, 4}, Q' = U - Q = {-3, -1, 1, 3, 5}.

Left side: P - Q = {6}.

Right side: P ∩ Q' = ∅.

The two sides are not equal, because 6 is in P but not in U.

If P is restricted to U (P = {-2, 0, 2, 4}), both sides are ∅ and equal.

[This part of the printed question has the problem of the element 6 — noted.]

Question: Exercise 17.2, Question 4: (ii) Q - P = Q ∩ P'

Answer: Q - P = ∅, because every member of Q is in P.

P' = U - P = {-3, -1, 1, 3, 5}, so Q ∩ P' = {-2, 0, 2, 4} ∩ {-3, -1, 1, 3, 5} = ∅.

Both sides are equal.

Question: Exercise 17.2, Question 4: (iii) (P ∪ Q)' = P' ∩ Q'

Answer: P ∪ Q = {-2, 0, 2, 4, 6}, so (P ∪ Q)' = U - (P ∪ Q) = {-3, -1, 1, 3, 5}.

P' ∩ Q' = {-3, -1, 1, 3, 5} ∩ {-3, -1, 1, 3, 5} = {-3, -1, 1, 3, 5}.

Both sides are equal.

Question: Exercise 17.2, Question 4: (iv) (P ∩ Q)' = P' ∪ Q'

Answer: P ∩ Q = {-2, 0, 2, 4}, so (P ∩ Q)' = U - (P ∩ Q) = {-3, -1, 1, 3, 5}.

P' ∪ Q' = {-3, -1, 1, 3, 5} ∪ {-3, -1, 1, 3, 5} = {-3, -1, 1, 3, 5}.

Both sides are equal.

Question: Exercise 17.2, Question 5: Given A = {2n | n ∈ N}, B = {3n | n ∈ N} and C = {4n | n ∈ N}, find: (i) A ∩ B

Answer: A = {2, 4, 6, 8, ...} (multiples of 2) and B = {3, 6, 9, 12, ...} (multiples of 3).

The common members are those that are multiples of both 2 and 3, i.e. multiples of 6.

So A ∩ B = {6n | n ∈ N}.

Question: Exercise 17.2, Question 5: (ii) A ∪ C

Answer: C = {4, 8, 12, ...} (multiples of 4) is a subset of A, because every multiple of 4 is also a multiple of 2.

So A ∪ C = A = {2n | n ∈ N}.

Question: Exercise 17.2, Question 5: (iii) B ∩ C

Answer: B = multiples of 3 and C = multiples of 4.

The common members are those that are multiples of both 3 and 4, i.e. multiples of 12.

So B ∩ C = {12n | n ∈ N}.

Question: Exercise 17.3, Question 1: Verify the commutative property of union and intersection for the following sets. (i) A = {a, b, c, d, e} and B = {a, e, i, o, u}

Answer: A ∪ B = {a, b, c, d, e, i, o, u} and B ∪ A = {a, e, i, o, u, b, c, d} = {a, b, c, d, e, i, o, u}. So A ∪ B = B ∪ A. A ∩ B = {a, e} and B ∩ A = {a, e}. So A ∩ B = B ∩ A. The commutative property is verified.

Question: Exercise 17.3, Question 1: (ii) $P = \{x \mid x \in Z \wedge -3 < x < 3\}$ and $Q = \{y \mid y \in E^+ \wedge y \leq 4\}$

Answer: $P = \{-2, -1, 0, 1, 2\}$ and $Q = \{2, 4\}$.

$P \cup Q = \{-2, -1, 0, 1, 2, 4\} = Q \cup P$.

$P \cap Q = \{2\} = Q \cap P$.

The commutative property is verified.

Question: Exercise 17.3, Question 2: Verify the associative property of union and intersection for the following sets. (i) $A = \{1, 2, 4, 5, 10, 20\}$, $B = \{5, 10, 15, 20\}$, $C = \{1, 2, 5, 10\}$

Answer: Union: $A \cup (B \cup C) = \{1, 2, 4, 5, 10, 20\} \cup \{1, 2, 5, 10, 15, 20\} = \{1, 2, 4, 5, 10, 15, 20\}$.

$(A \cup B) \cup C = \{1, 2, 4, 5, 10, 15, 20\} \cup \{1, 2, 5, 10\} = \{1, 2, 4, 5, 10, 15, 20\}$.

Both are equal.

Intersection: $A \cap (B \cap C) = \{1, 2, 4, 5, 10, 20\} \cap \{5, 10\} = \{5, 10\}$.

$(A \cap B) \cap C = \{5, 10, 20\} \cap \{1, 2, 5, 10\} = \{5, 10\}$.

Both are equal.

The associative property is verified.

Question: Exercise 17.3, Question 2: (ii) $A = N$, $B = P$, $C = Z$

Answer: Union: B ∪ C = P ∪ Z = Z (since P ⊂ Z), so A ∪ (B ∪ C) = N ∪ Z = Z. (A ∪ B) ∪ C = (N ∪ P) ∪ Z = N ∪ Z = Z (since P ⊂ N). Both are equal. Intersection: B ∩ C = P ∩ Z = P, so A ∩ (B ∩ C) = N ∩ P = P. (A ∩ B) ∩ C = (N ∩ P) ∩ Z = P ∩ Z = P. Both are equal. The associative property is verified.

Question: Exercise 17.3, Question 3: For the following sets verify: (a) the distributive property of union over intersection (b) the distributive property of intersection over union. (i) A = {1, 2, 3, ..., 10}, B = {2, 3, 5, 7}, C = {1, 3, 5, 7, 9}

Answer: (a) B ∩ C = {3, 5}, so A ∪ (B ∩ C) = {1, 2, 3, ..., 10} ∪ {3, 5} = {1, 2, 3, ..., 10}.

(A ∪ B) ∩ (A ∪ C) = {1, ..., 10} ∩ {1, ..., 10} = {1, ..., 10} (since B, C ⊂ A).

Both are equal.

(b) B ∪ C = {1, 2, 3, 5, 7, 9}, so A ∩ (B ∪ C) = {1, 2, 3, 5, 7, 9}.

(A ∩ B) ∪ (A ∩ C) = {2, 3, 5, 7} ∪ {1, 3, 5, 7, 9} = {1, 2, 3, 5, 7, 9}.

Both are equal.

Question: Exercise 17.3, Question 3: (ii) A = N, B = P, C = W

Answer: (a) B ∩ C = P ∩ W = P (every prime is a natural and whole number), so A ∪ (B ∩ C) = N ∪ P = N. (A ∪ B) ∩ (A ∪ C) = (N ∪ P) ∩ (N ∪ W) = N ∩ W = N. Both are equal.

(b) B ∪ C = P ∪ W = W (P ⊂ W), so A ∩ (B ∪ C) = N ∩ W = N. (A ∩ B) ∪ (A ∩ C) = (N ∩ P) ∪ (N ∩ W) = P ∪ N = N. Both are equal.

Question: Exercise 17.3, Question 4: Given U = {1, 2, 3, ..., 12}, A = {1, 2, 3, 4, 6, 12} and B = {2, 4, 6, 8}, verify De Morgan's laws.

Answer: A' = {5, 7, 9, 10, 11} and B' = {1, 3, 5, 7, 9, 10, 11, 12}.

(i) A ∪ B = {1, 2, 3, 4, 6, 8, 12}, so (A ∪ B)' = {5, 7, 9, 10, 11}.

A' ∩ B' = {5, 7, 9, 10, 11}.

They are equal.

(ii) A ∩ B = {2, 4, 6}, so (A ∩ B)' = {1, 3, 5, 7, 8, 9, 10, 11, 12}.

A' ∪ B' = {1, 3, 5, 7, 9, 10, 11, 12}.

They are equal.

De Morgan's laws are verified.

Question: Exercise 17.3, Question 5: If A and B are subsets of U, prove: (i) A ∪ (A ∩ B) = A ∩ (A ∪ B)

Answer: Left side: A ∩ B ⊆ A, so A ∪ (A ∩ B) = A. Right side: A ⊆ A ∪ B, so A ∩ (A ∪ B) = A. Both sides equal A, so they are equal.

Question: Exercise 17.3, Question 5: (ii) A ∪ B = A ∪ (A' ∩ B)

Answer: Right side, by the distributive property: A ∪ (A' ∩ B) = (A ∪ A') ∩ (A ∪ B) = U ∩ (A ∪ B) = A ∪ B = left side. Proved.

Question: Exercise 17.3, Question 5: (iii) B = (A ∩ B) ∪ (A' ∩ B)

Answer: Right side, by the distributive property: (A ∩ B) ∪ (A' ∩ B) = (A ∪ A') ∩ B = U ∩ B = B = left side. Proved.

Question: Exercise 17.3, Question 5: (iv) If A ⊆ B then B = A ∪ (A' ∩ B)

Answer: From part (iii), B = (A ∩ B) ∪ (A' ∩ B). When A ⊆ B, A ∩ B = A, so B = A ∪ (A' ∩ B). Proved.

Question: Exercise 17.4, Question 1: Given A and B are subsets of the universal set U, find AΔB, A∩B, A∪B and draw the general Venn diagrams for A - B when: (i) A and B are disjoint sets

Answer: For disjoint sets, A ∩ B = ∅. A ∪ B = all members of A and B together. AΔB = (A ∪ B) - (A ∩ B) = A ∪ B. A - B = A (since no member of A is in B). Venn diagram: two separate circles; for A - B the whole A circle is shaded.

Question: Exercise 17.4, Question 1: (ii) A and B are overlapping sets

Answer: A ∩ B = the members common to both. A ∪ B = all members of both sets. AΔB = the members in exactly one set = (A ∪ B) - (A ∩ B). A - B = the members of A not in B. Venn diagram: two overlapping circles; for A - B only the A-only part (the crescent) is shaded.

Question: Exercise 17.4, Question 1: (iii) A is a subset of B

Answer: A ∩ B = A (since every member of A is in B). A ∪ B = B. AΔB = B - A (the members of B not in A). A - B = ∅ (no member of A is outside B). Venn diagram: the A circle inside the B circle; there is nothing to shade for A - B.

Question: Exercise 17.4, Question 2: Verify the commutative property of union and intersection by means of Venn diagrams. (i) A = {a, b, c, d, e} and B = {a, e, i, o, u}

Answer: A ∪ B = {a, b, c, d, e, i, o, u} = B ∪ A. A ∩ B = {a, e} = B ∩ A. Venn diagram: two overlapping circles; for the union both circles are shaded — changing the order gives the same figure. For the intersection only the common part is shaded — changing the order gives the same part.

Question: Exercise 17.4, Question 2: (ii) P = {1, 2, 3, ..., 10} and Q = {2,

4, 6, 8, 10}

Answer: Q ⊂ P.

P ∪ Q = {1, 2, 3, …, 10} = Q ∪ P.

P ∩ Q = {2, 4, 6, 8, 10} = Q ∩ P.

Venn diagram: the Q circle inside the P circle; for the union the whole large circle is shaded and for the intersection the inner circle — the same result in both orders.

Question: Exercise 17.4, Question 3: Given U = {1, 2, 3, …, 10}, A = {1, 3, 5, 7, 9} and B = {5, 6, 7, 8}, verify De Morgan's laws by means of Venn diagrams.

Answer: A' = {2, 4, 6, 8, 10}, B' = {1, 2, 3, 4, 9, 10}.

(i) A ∪ B = {1, 3, 5, 6, 7, 8, 9}, so (A ∪ B)' = {2, 4, 10}.

A' ∩ B' = {2, 4, 10}.

They are equal.

(ii) A ∩ B = {5, 7}, so (A ∩ B)' = {1, 2, 3, 4, 6, 8, 9, 10}.

A' ∪ B' = {1, 2, 3, 4, 6, 8, 9, 10}.

They are equal.

Venn diagrams: in (i) the region outside both circles is shaded; in (ii) the

union of the two complements shades the same outer region — the two figures match.

Question: Exercise 17.4, Question 4: Given A = {1, 2, 3, ..., 8}, B = {4, 6, 7, 8, 9, 10} and C = {5, 7, 9, 11}, verify the distributive laws by means of Venn diagrams.

Answer: (a) B ∩ C = {7, 9}, so A ∪ (B ∩ C) = {1, 2, 3, 4, 5, 6, 7, 8, 9}.

(A ∪ B) ∩ (A ∪ C) = {1, ..., 10} ∩ {1, ..., 9, 11} = {1, 2, 3, 4, 5, 6, 7, 8, 9}.

They are equal.

(b) B ∪ C = {4, 5, 6, 7, 8, 9, 10, 11}, so A ∩ (B ∪ C) = {4, 5, 6, 7, 8}.

(A ∩ B) ∪ (A ∩ C) = {4, 6, 7, 8} ∪ {5, 7} = {4, 5, 6, 7, 8}.

They are equal.

Venn diagrams: three overlapping circles; the shaded regions on both sides match.

Question: Exercise 17.5, Question 1: Find the values of x and y given: (i) (x - 5, 10) = (11, y - 7)

Answer: From equality of ordered pairs: x - 5 = 11 and 10 = y - 7.

From x - 5 = 11, x = 16.

From 10 = y - 7, y = 17.

Question: Exercise 17.5, Question 1: (ii) (5x + 8, 5y - 4) = (3x + 10, 2y + 2)

Answer: From 5x + 8 = 3x + 10, 2x = 2, so x = 1.

From 5y - 4 = 2y + 2, 3y = 6, so y = 2.

Question: Exercise 17.5, Question 1: (iii) (2x - 3y, 5x + y) = (3, 16)

Answer: 2x - 3y = 3 and 5x + y = 16.

From the second equation, y = 16 - 5x.

Substituting in the first: 2x - 3(16 - 5x) = 3, i.e. 2x - 48 + 15x = 3, i.e. 17x = 51, so x = 3.

Then y = 16 - 5(3) = 1.

Check: 2(3) - 3(1) = 3 and 5(3) + 1 = 16, both correct.

Question: Exercise 17.5, Question 2: Given set P has 10 members and set Q has 15 members, find the number of members of P×Q and Q×P.

Answer: O(P×Q) = O(P) × O(Q) = 10 × 15 = 150.

O(Q×P) = O(Q) × O(P) = 15 × 10 = 150.

Question: Exercise 17.5, Question 3: Given A = {1, 2}, B = {2, 3} and C = {1, 3, 5}, find: (i) A×B

Answer: A×B = {(1, 2), (1, 3), (2, 2), (2, 3)}.

Question: Exercise 17.5, Question 3: (ii) B×C

Answer: B×C = {(2, 1), (2, 3), (2, 5), (3, 1), (3, 3), (3, 5)}.

Question: Exercise 17.5, Question 3: (iii) A×(B∪C)

Answer: B ∪ C = {1, 2, 3, 5}, so A×(B∪C) = {(1, 1), (1, 2), (1, 3), (1, 5), (2, 1), (2, 2), (2, 3), (2, 5)}.

Question: Exercise 17.5, Question 3: (iv) B×(A∪C)

Answer: A ∪ C = {1, 2, 3, 5}, so B×(A∪C) = {(2, 1), (2, 2), (2, 3), (2, 5), (3, 1), (3, 2), (3, 3), (3, 5)}.

Question: Exercise 17.5, Question 3: (v) (A∩B)×(B∩C)

Answer: A ∩ B = {2} and B ∩ C = {3}, so (A∩B)×(B∩C) = {(2, 3)}.

Question: Exercise 17.5, Question 4: Given A = {5, 6} and B = {1, 2, 3}, write: (i) two binary relations in A×B

Answer: R1 = {(5, 1), (6, 2)} and R2 = {(5, 3), (6, 1)} — both are subsets of A×B, so both are binary relations from A to B.

Question: Exercise 17.5, Question 4: (ii) four binary relations in B×A

Answer: S1 = {(1, 5)}, S2 = {(2, 6)}, S3 = {(1, 5), (2, 6)}, S4 = {(1, 5), (2, 5), (3, 5), (1, 6), (2, 6), (3, 6)} = B×A — all are subsets of B×A.

Question: Exercise 17.5, Question 4: (iii) five binary relations in B

Answer: T1 = {(1, 1)}, T2 = {(1, 2), (2, 1)}, T3 = {(1, 1), (2, 2), (3, 3)}, T4 = {(1, 2), (2, 3), (3, 1)}, T5 = ∅ — all are subsets of B×B.

Question: Exercise 17.5, Question 4: (iv) all binary relations in A

Answer: A×A = {(5, 5), (5, 6), (6, 5), (6, 6)} has 4 pairs, so there are $2^4$ = 16 binary relations: ∅, {(5, 5)}, {(5, 6)}, {(6, 5)}, {(6, 6)}, {(5, 5), (5, 6)}, {(5, 5), (6, 5)}, {(5, 5), (6, 6)}, {(5, 6), (6, 5)}, {(5, 6), (6, 6)}, {(6, 5), (6, 6)}, {(5, 5), (5, 6), (6, 5)}, {(5, 5), (5, 6), (6, 6)}, {(5, 5), (6, 5), (6, 6)}, {(5, 6), (6, 5), (6, 6)}, {(5, 5), (5, 6), (6, 5), (6, 6)}.

Question: Exercise 17.5, Question 5: For two sets A and B, if O(A) = 3 and O(B) = 4, find the number of binary relations from A to B in A×B.

Answer: A×B has 3 × 4 = 12 ordered pairs.

Each pair may or may not be in a relation, so the number of relations = $2^{12}$ = 4096.

Question: Exercise 17.5, Question 6: Given A = {0, 1, 2, 3} and B = {2, 4, 6, 8}, write the following relations of A×B in tabular form: (i) R1 = {(a, b) |

b < 5}

Answer: For b < 5, b = 2 or 4, and a can be anything.

So R1 = {(0, 2), (0, 4), (1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)}.

Question: Exercise 17.5, Question 6: (ii) R2 = {(a, b) | a + b = 9}

Answer: The possible pairs with a + b = 9 are (1, 8) and (3, 6).

So R2 = {(1, 8), (3, 6)}.

Question: Exercise 17.5, Question 6: (iii) R3 = {(a, b) | a - b = 1}

Answer: For a - b = 1, a = b + 1: b = 2 gives a = 3;

b = 4 gives a = 5, which is not in A.

So R3 = {(3, 2)}.

Question: Exercise 17.5, Question 7: Given the relation R = {(x, y) | y = 2x + 5} and Z, find: (i) the range when the domain is {-2, -1, 0, 1, 2}

Answer: y = 2x + 5: x = -2 gives y = 1;

x = -1 gives y = 3;

x = 0 gives y = 5;

x = 1 gives y = 7;

x = 2 gives y = 9.

So the range = {1, 3, 5, 7, 9}.

Question: Exercise 17.5, Question 7: (ii) the domain when the range is

{11, 13, 15, 17}

Answer: From y = 2x + 5, x = (y - 5)/2: y = 11 gives x = 3;

y = 13 gives x = 4;

y = 15 gives x = 5;

y = 17 gives x = 6.

So the domain = {3, 4, 5, 6}.

Question: Exercise 17.5, Question 8: Given x, y are members of W, find the domain and range of the following relations: (i) {(x, y) | 3x + y = 11}

Answer: y = 11 - 3x and y ≥ 0 (since y ∈ W), so x = 0, 1, 2, 3 and y = 11, 8, 5, 2.

Domain = {0, 1, 2, 3}, Range = {2, 5, 8, 11}.

Question: Exercise 17.5, Question 8: (ii) {(x, y) | x - y = 6}

Answer: y = x - 6 and y ≥ 0, so x ≥ 6: x = 6, 7, 8, ... and y = 0, 1, 2, ....

Domain = {x | x ∈ W ∧ x ≥ 6}, Range = W = {0, 1, 2, ...}.

Question: Exercise 17.5, Question 9: Given f: A → B is a function with f = {(1, 5), (2, 6), (3, 7), (4, 8)} where A = {1, 2, 3, 4} and B = N, find the domain, co-domain and range. Find the values of f(2) and f(4).

Answer: Domain = {1, 2, 3, 4}, Co-domain = N, Range = {5, 6, 7, 8}.

f(2) = 6 and f(4) = 8.

Question: Exercise 17.5, Question 10: Given A = {0, 1, 2, 3, 4} and B = {2, 4, 6, 8, 10}, check whether the relations from A to B are functions or

relations, and find the type of each function: (i) R1 = {(0, 2), (1, 4), (2, 6), (3, 8)}

Answer: R1 is not a function, only a relation, because the domain = {0, 1, 2, 3} ≠ A — 4 has no image.

Question: Exercise 17.5, Question 10: (ii) R2 = {(0, 10), (1, 8), (2, 6), (3, 4), (4, 2)}

Answer: R2 is a function: the domain = A and each member has exactly one image.

The images are all distinct, so it is a one-one function.

Range = {2, 4, 6, 8, 10} = B, so it is an onto function.

Hence it is a bijective (one-one and onto) function.

Question: Exercise 17.5, Question 10: (iii) R3 = {(0, 2), (1, 4), (2, 6), (3, 8), (4, 8)}

Answer: R3 is a function: the domain = A and each member has exactly one image.

It is not one-one, because 3 and 4 both map to 8.

Range = {2, 4, 6, 8} ≠ B, so it is an into function.

Question: Exercise 17.5, Question 10: (iv) R4 = {(0, 2), (1, 2), (2, 2), (3, 2), (4, 8)}

Answer: R4 is a function: the domain = A and each member has exactly one image.

It is not one-one, because 0, 1, 2, 3 all map to 2.

Range = {2, 8} ≠ B, so it is an into function.

Question: Exercise 17.5, Question 10: (v) R5 = {(0, 2), (1, 4), (2, 6), (3, 8), (4, 10)}

Answer: R5 is a function: the domain = A and each member has exactly one image.

The images are all distinct, so it is a one-one function.

Range = {2, 4, 6, 8, 10} = B, so it is an onto function.

Hence it is a bijective function.

Question: Exercise 17.5, Question 11: State which of the following is a one-one correspondence or not. (i) Map from A = {0, 1, 2} to B = {5, 6, 7}: 0→5, 1→6, 2→7

Answer: It is a one-one correspondence: each member has exactly one image, the images are distinct, and the range = {5, 6, 7} = B (onto).

Question: Exercise 17.5, Question 11: (ii) Map from A = {0, 1, 2} to B = {5, 6, 7}: 0→5, 1→6, 2→6

Answer: It is not a one-one correspondence: although it is a function, it is not one-one because 1 and 2 both map to 6.

Range = {5, 6} ≠ B.

Question: Exercise 17.5, Question 11: (iii) Map from A = {0, 1, 2} to B = {4, 5, 6, 7}: 0→4, 1→5 and 1→6, 2→7

Answer: It is not a one-one correspondence: it is not even a function, because 1 has two images (5 and 6).

Question: Exercise 17.5, Question 12: Given P = {a, b, c}, Q = {x, y, z} and R = {p, q, r, s}, write: (i) an into function f from P to Q

Answer: f = {(a, x), (b, x), (c, y)} — range = {x, y}, which is a proper subset of Q, so it is an into function.

Question: Exercise 17.5, Question 12: (ii) an onto function g from R to P

Answer: g = {(p, a), (q, b), (r, c), (s, c)} — range = {a, b, c} = P, so it is an onto function.

Question: Exercise 17.5, Question 12: (iii) an injective function h from P to R

Answer: h = {(a, p), (b, q), (c, r)} — the images are distinct, so it is a one-one (injective) function.

Question: Exercise 17.5, Question 12: (iv) a bijective function k from Q to P

Answer: k = {(x, a), (y, b), (z, c)} — the images are distinct (one-one) and the range = {a, b, c} = P (onto), so it is a bijective function.

Exercise 17 Review, Question 1: Circle the correct answer: (i) Which is a subset of N?

(a) Z

(b) Q

(c) E

✓ (d) P

Exercise 17 Review, Question 1: (ii) The tabular form of $A = \{x \mid x \in Z \wedge x^2 = 25\}$ is ___.

(a) {1, 5}

✓ (b) {-5, 5}

(c) {1, 5, 25}

(d) {-5}

Exercise 17 Review, Question 1: (iii) For two sets A and B, if O(A) = O(B) then ___.

(a) $A \subseteq B$

(b) B = A

(c) A ~ B

(d) A ⊂ B

Exercise 17 Review, Question 1: (iv) {x | x ∈ N ∧ x < 1} is ___.

(a) singleton set

(b) infinite set

✓ (c) empty set

(d) superset

Exercise 17 Review, Question 1: (v) If A = {1, 2, 3, 4} and B = {2, 4, 6} then AΔB = ___.

(a) {1, 2}

(b) {6}

(c) {1, 3}

✓ (d) {1, 3, 6}

Exercise 17 Review, Question 1: (vi) Which operation does not have the commutative property?

(a) symmetric difference

(b) union

✓ (c) difference

(d) intersection

Exercise 17 Review, Question 1: (vii) {x | x ∈ P ∨ x ∈ Q} = ___.

(a) P ∩ Q

✓ (b) P ∪ Q

(c) P - Q

(d) PΔQ

Exercise 17 Review, Question 1: (viii) {x | x ∈ U ∧ x ∉ A ∩ B} = ___.

(a) A ∩ B

(d) A' ∩ B'

Exercise 17 Review, Question 1: (ix) If A and B are two non-empty sets

✓

with A ∪ B = U and A ∩ B = ∅, then A and B are ___.

✓ (a) partitions

(b) overlapping sets

(c) equal sets

(d) none

Exercise 17 Review, Question 1: (x) ((A')') = ___.

(b) A'

✓ (c) A

(d) B

Exercise 17 Review, Question 1: (xi) If A ⊇ B then A ∪ B = ___.

✓ (a) A

(b) B

(c) ∅

(d) U

Exercise 17 Review, Question 1: (xii) In Venn diagrams the universal set is shown by using ___.

✓ (a) rectangle

(b) circle

(c) semi-circle

(d) set

Exercise 17 Review, Question 1: (xiii) If (x, 6) = (2, y - 6) then x + y = ___.

(a) 8

(b) 10

(c) 12

✓ (d) 14

Exercise 17 Review, Question 1: (xiv) If O(A×B) = 100 and O(B) = 5 then O(A) = ___.

(a) 10

(b) 15

✓ (c) 20

(d) 25

Exercise 17 Review, Question 1: (xv) For a surjective function, Range ___ Co-domain.

(a) ⊇

(b) ⊆

✓ (c) =

(d) these

Exercise 17 Review, Question 1: (xvi) For an into function, Range ___ Co-domain.

(a) ⊇

✓ (b) ⊂

(c) =

(d) none

Question: Exercise 17 Review, Question 1: (xvii) Which of the following is a function? (a) {(2, 5), (2, 7), (3, 8)} (b) {(6, 7), (7, 6), (6, 8)} (c) {(0, 5), (6, 0), (5, 6)} (d) none

Answer: (c) {(0, 5), (6, 0), (5, 6)} — each first component has exactly one image. In (a), 2 has two images, and in (b), 6 has two images.

Exercise 17 Review, Question 1: (xviii) Which of the following is always bijective?

(a) onto function

(b) one-one function

(c) into function

✓ (d) one-one correspondence

Exercise 17 Review, Question 1: (xix) If f: N → R then which is not possible?

Exercise 17 Review, Question 1: (xx) Which of the following has the

commutative property?

(a) difference

(b) union

(c) Cartesian product

(d) none

Question: Exercise 17 Review, Question 2: Given U = {1, 2, 3, 4, 5, 6}, A = {1, 2, 5} and B = {2, 4, 6}, find: (a) A ∪ B

Answer: A ∪ B = {1, 2, 4, 5, 6}.

Question: Exercise 17 Review, Question 2: (b) A ∩ B

Answer: A ∩ B = {2}.

Question: Exercise 17 Review, Question 2: (c) AΔB

Answer: AΔB = (A ∪ B) - (A ∩ B) = {1, 2, 4, 5, 6} - {2} = {1, 4, 5, 6}.

Question: Exercise 17 Review, Question 2: (d) A - B

Answer: A - B = {1, 5}.

Question: Exercise 17 Review, Question 2: (e) B'

Answer: B' = U - B = {1, 3, 5}.

Question: Exercise 17 Review, Question 3: Verify De Morgan's laws for the sets in Question 2.

Answer: A' = {3, 4, 6}, B' = {1, 3, 5}.

(i) A ∪ B = {1, 2, 4, 5, 6}, so (A ∪ B)' = {3}.

A' ∩ B' = {3}.

They are equal.

(ii) A ∩ B = {2}, so (A ∩ B)' = {1, 3, 4, 5, 6}.

A' ∪ B' = {1, 3, 4, 5, 6}.

They are equal.

Question: Exercise 17 Review, Question 4: For two sets A = {a, b, d} and B = {a, d, e}, verify the commutative property of union and intersection.

Answer: A ∪ B = {a, b, d, e} = B ∪ A. A ∩ B = {a, d} = B ∩ A. The commutative property is verified.

Question: Exercise 17 Review, Question 5: For two sets P = {1, 3, 5} and Q = {1, 3, 7}, show the following by means of Venn diagrams: (a) P ∪ Q

Answer: P ∪ Q = {1, 3, 5, 7}.

Venn diagram: two overlapping circles, the whole of both circles is shaded.

Question: Exercise 17 Review, Question 5: (b) P ∩ Q

Answer: P ∩ Q = {1, 3}.

Venn diagram: two overlapping circles, only the common part is shaded.

Question: Exercise 17 Review, Question 5: (c) PΔQ

Answer: PΔQ = {5, 7}.

Venn diagram: two overlapping circles, both parts except the common part are shaded.

Question: Exercise 17 Review, Question 5: (d) P - Q

Answer: P - Q = {5}.

Venn diagram: two overlapping circles, only the P-only part is shaded.

Question: Exercise 17 Review, Question 6: Given A = {a, c} and B = {b, d}, write: (a) two binary relations in A

Answer: R1 = {(a, a)}, R2 = {(a, c), (c, a)} — both are subsets of A×A.

Question: Exercise 17 Review, Question 6: (b) two binary relations in B

Answer: S1 = {(b, b)}, S2 = {(b, d), (d, b)} — both are subsets of B×B.

Question: Exercise 17 Review, Question 6: (c) two binary relations in A×B

Answer: T1 = {(a, b)}, T2 = {(a, b), (c, d)} — both are subsets of A×B.

Question: Exercise 17 Review, Question 6: (d) two binary relations in A∪B

Answer: A ∪ B = {a, b, c, d}.

U1 = {(a, b), (c, d)}, U2 = {(a, a), (b, b), (c, c), (d, d)} — both are subsets of (A∪B)×(A∪B).

Question: Exercise 17 Review, Question 7: Given A = {1, 2, 5} and B = {6, 8}, find a function from A to B which is: (a) an onto function

Answer: f = {(1, 6), (2, 8), (5, 6)} — range = {6, 8} = B, so it is an onto function.

Question: Exercise 17 Review, Question 7: (b) an into function

Answer: g = {(1, 6), (2, 6), (5, 6)} — range = {6}, which is a proper subset of B, so it is an into function.

# 18. قیرار یا متغیر

Question: Exercise 18.1, Question 1: Find the ratio of the following. (i) 70 kg and 28 kg

Answer: 70:28 = 5:2 (dividing both by 14).

Question: Exercise 18.1, Question 1: (ii) 60 cm and 1 metre

Answer: 1 metre = 100 cm, so 60:100 = 3:5.

Question: Exercise 18.1, Question 1: (iii) 40 seconds and 3 minutes

Answer: 3 minutes = 180 seconds, so 40:180 = 2:9.

Question: Exercise 18.1, Question 1: (iv) 200 ml and 2 litres

Answer: 2 litres = 2000 ml, so 200:2000 = 1:10.

Question: Exercise 18.1, Question 1: (v) 360° and 135°

Answer: 360:135 = 8:3 (dividing both by 45).

Question: Exercise 18.1, Question 1: (vi) 3.5 kg and 5 kg and 200 g

Answer: In grams: 3500:5000:200 = 35:50:2 (dividing all by 100).

Question: Exercise 18.1, Question 2: A factory has 120 workers, of whom 45 are women. Find the ratio of: (i) men to women

Answer: Men = 120 - 45 = 75.

The ratio of men to women = 75:45 = 5:3.

Question: Exercise 18.1, Question 2: (ii) women to men

Answer: The ratio of women to men = 45:75 = 3:5.

Question: Exercise 18.1, Question 2: (iii) women to all workers

Answer: The ratio of women to all workers = 45:120 = 3:8.

Question: Exercise 18.1, Question 2: (iv) men to all workers

Answer: The ratio of men to all workers = 75:120 = 5:8.

Question: Exercise 18.1, Question 3: Given 5(4x - 2y) = 3x - 4y, find x:y.

Answer: From 20x - 10y = 3x - 4y, 20x - 3x = 10y - 4y, i.e. 17x = 6y, so x/y = 6/17, i.e. x:y = 6:17.

Question: Exercise 18.1, Question 4: Given the ratios 3a+4 : 2a+5 are equal to 4:3, find the value of "a".

Answer: From (3a+4)/(2a+5) = 4/3, 3(3a+4) = 4(2a+5), i.e. 9a + 12 = 8a + 20, so a = 8.

Question: Exercise 18.1, Question 5: What number must be added to each term of the ratio 5:27 so that it becomes equal to 1:3?

Answer: Suppose the added number is x.

From (5+x)/(27+x) = 1/3, 3(5+x) = 27+x, i.e. 15 + 3x = 27 + x, i.e. 2x = 12, so x = 6.

Question: Exercise 18.1, Question 6: Given a:b = 5:8, find the value of 3a+4b : 5a+7b.

Answer: Let a = 5k and b = 8k.

3a + 4b = 15k + 32k = 47k and 5a + 7b = 25k + 56k = 81k.

So the ratio = 47:81.

Question: Exercise 18.1, Question 7: Find the value of x from the following. (i) 2x+5 : 5 :: 3x-2 : 7

Answer: From (2x+5)/5 = (3x-2)/7, 7(2x+5) = 5(3x-2), i.e. 14x + 35 = 15x - 10, so x = 45.

Question: Exercise 18.1, Question 7: (ii) (4x-3)/5 : 3/4 :: (4x)/3 : 7/2

Answer: From ((4x-3)/5)/(3/4) = ((4x)/3)/(7/2), 4(4x-3)/15 = 8x/21.

Multiplying by 105: 28(4x-3) = 40x, i.e. 112x - 84 = 40x, i.e. 72x = 84, so x = 7/6.

Question: Exercise 18.1, Question 7: (iii) (x-3)/2 : 5/(x-1) :: (x-1)/3 : 4/(x+4)

Answer: From ((x-3)/2)/(5/(x-1)) = ((x-1)/3)/(4/(x+4)), (x-3)(x-1)/10 = (x-1)(x+4)/12.

Multiplying by 60: 6(x-3)(x-1) = 5(x-1)(x+4).

Since x ≠ 1 (x-1 is in a denominator), 6(x-3) = 5(x+4), i.e. 6x - 18 = 5x + 20, so x = 38.

Question: Exercise 18.1, Question 7: (iv) $(a^2 - ab + b^2) : x :: (a^3+b^3)/(a-b) : (a+b)^2$

Answer: $a^3+b^3 = (a+b)(a^2-ab+b^2)$. So the right ratio = $(a+b)(a^2-ab+b^2)/((a-b)(a+b)^2) = (a^2-ab+b^2)/((a-b)(a+b))$. From the proportion, $(a^2-ab+b^2)/x = (a^2-ab+b^2)/((a-b)(a+b))$, so $x = (a-b)(a+b) = a^2-b^2$.

Question: Exercise 18.1, Question 7: (v) 11-x : 8-x :: 25-x : 16-x

Answer: From (11-x)/(8-x) = (25-x)/(16-x), (11-x)(16-x) = (25-x)(8-x), i.e. $176 - 27x + x^2 = 200 - 33x + x^2$, i.e. 6x = 24, so x = 4.

Check: 7:4 :: 21:12, correct.

Question: Exercise 18.2, Question 1: Given y varies directly as x and y = 10 when x = 3, find: (i) the equation in the form y = kx

Answer: y = kx;

from 10 = 3k, k = 10/3.

So y = (10/3)x.

Question: Exercise 18.2, Question 1: (ii) y when x = 6

Answer: y = (10/3)(6) = 20.

Question: Exercise 18.2, Question 1: (iii) x when y = 15

Answer: From 15 = (10/3)x, x = 45/10 = 9/2.

Question: Exercise 18.2, Question 2: Given V ∝ T and V = 15 when T = 24, find: (i) the equation relating T and V

Answer: V = kT;

from 15 = 24k, k = 15/24 = 5/8.

So V = (5/8)T.

Question: Exercise 18.2, Question 2: (ii) V when T = 30

Answer: V = (5/8)(30) = 150/8 = 75/4.

Question: Exercise 18.2, Question 2: (iii) T when V = 10

Answer: From 10 = (5/8)T, T = 80/5 = 16.

Question: Exercise 18.2, Question 3: Given u ∝ $\sqrt[3]{v}$ and u = 4 when v = 64, find u when v = 216; and find v when u = 5.

Answer: u = k$\sqrt[3]{v}$;

from 4 = k$\sqrt[3]{64}$ = 4k, k = 1, so u = $\sqrt[3]{v}$.

For v = 216, u = $\sqrt[3]{216}$ = 6.

For u = 5, 5 = $\sqrt[3]{v}$, so v = 125.

Question: Exercise 18.2, Question 4: Given F varies directly as $m^3$ and F = 81 when m = 3, find F when m = 5.

Answer: F = $km^3$;

from 81 = 27k, k = 3, so F = $3m^3$.

For m = 5, F = 3(125) = 375.

Question: Exercise 18.2, Question 5: Given y varies inversely as x and y = 10 when x = 3, find y when x = 10.

Answer: $y = k/x$;

from $10 = k/3$, $k = 30$, so $y = 30/x$.

For x = 10, y = 3.

Question: Exercise 18.2, Question 6: Given the quantity V varies inversely as the cube of P and V = 12 when P = 9, find P when V = 4.

Answer: $V = k/P^3$;

from $12 = k/729$, $k = 8748$.

For V = 4, $4 = 8748/P^3$, i.e. $P^3 = 2187 = 3^7$, so $P = 9\sqrt[3]{3}$ (about 12.98).

Question: Exercise 18.2, Question 7: Given $F \propto 1/r^2$ and F = 8 when r = 2, find: (i) F when r = 5

Answer: $F = k/r^2$;

from $8 = k/4$, $k = 32$, so $F = 32/r^2$.

For r = 5, F = 32/25.

Question: Exercise 18.2, Question 7: (ii) r when F = 24

Answer: From $24 = 32/r^2$, $r^2 = 32/24 = 4/3$, so $r = 2/\sqrt{3} = 2\sqrt{3}/3$.

Question: Exercise 18.2, Question 8: Given x varies directly as the square of y and x = 8 when y = 3/2, find x when y = 3.

Answer: $x = ky^2$;

from 8 = k(9/4), k = 32/9, so x = (32/9)y².

For y = 3, x = (32/9)(9) = 32.

Question: Exercise 18.2, Question 9: The force F between two bodies varies inversely as the square of the distance d between their centres. F = 2 when d = 3. Find d when F = 72.

Answer: F = k/d²;

from 2 = k/9, k = 18, so F = 18/d².

For F = 72, 72 = 18/d², i.e. d² = 1/4, so d = 1/2.

Question: Exercise 18.2, Question 10: Given y varies inversely as (x-5) and y = 6 when x = 8, find y when x = 10.

Answer: y = k/(x-5);

from 6 = k/3, k = 18, so y = 18/(x-5).

For x = 10, y = 18/5.

Question: Exercise 18.3, Question 1: Find the third proportional when the first, second and fourth proportionals are given below: (i) 6, 18, 54

Answer: From 6:18 :: x:54, 18x = 6 × 54 = 324, so x = 18.

Question: Exercise 18.3, Question 1: (ii) a-b, a+b, a²-b²

Answer: From (a-b):(a+b) :: x:(a²-b²), (a+b)x = (a-b)(a²-b²) = (a-b)²(a+b), so x = (a-b)².

Question: Exercise 18.3, Question 1: (iii) x+y, x³+y³, (x+y)²

Answer: From (x+y):(x³+y³) :: x':(x+y)², (x³+y³)x' = (x+y)³. Since x³+y³ = (x+y)(x²-xy+y²), x' = (x+y)³/((x+y)(x²-xy+y²)) = (x+y)²/(x²-xy+y²).

Question: Exercise 18.3, Question 1: (iv) a+b, (a²-ab+b²)/(a-b),

$(a^3+b^3)/(a^2-b^2)$

Answer: From $(a+b) : (a^2-ab+b^2)/(a-b) :: x : (a^3+b^3)/(a^2-b^2)$, $x = (a+b)(a^3+b^3)(a-b)/((a^2-b^2)(a^2-ab+b^2))$. Substituting $a^3+b^3 = (a+b)(a^2-ab+b^2)$ and $a^2-b^2 = (a-b)(a+b)$ gives $x = a+b$.

Question: Exercise 18.3, Question 2: Find the fourth proportional to the following. (i) 8, 4, 2

Answer: From $8:4 :: 2:x$, $8x = 8$, so $x = 1$.

Question: Exercise 18.3, Question 2: (ii) $a^3+b^3$, $a^2-b^2$, $a^2-ab+b^2$

Answer: From $(a^3+b^3):(a^2-b^2) :: (a^2-ab+b^2):x$, $x = (a^2-b^2)(a^2-ab+b^2)/(a^3+b^3)$. Substituting $a^3+b^3 = (a+b)(a^2-ab+b^2)$ and $a^2-b^2 = (a-b)(a+b)$ gives $x = a-b$.

Question: Exercise 18.3, Question 2: (iii) $a^2-8a+12$, $a-2$, $2a^3-12a^2$

Answer: $a^2-8a+12 = (a-2)(a-6)$ and $2a^3-12a^2 = 2a^2(a-6)$.

So $x = (a-2)(2a^2)(a-6)/((a-2)(a-6)) = 2a^2$.

Question: Exercise 18.3, Question 2: (iv) $(a^2-b^2)(a^2-ab+b^2)$, $a^3+b^3$, $a^3-b^3$

Answer: $x = (a^3+b^3)(a^3-b^3)/((a^2-b^2)(a^2-ab+b^2))$. Substituting $a^3+b^3 = (a+b)(a^2-ab+b^2)$, $a^3-b^3 = (a-b)(a^2+ab+b^2)$ and $a^2-b^2 = (a-b)(a+b)$ gives $x = a^2+ab+b^2$.

Question: Exercise 18.3, Question 3: Find the mean proportional to the following. (i) 8, 18

Answer: $m^2 = 8 \times 18 = 144$, so $m = 12$.

Question: Exercise 18.3, Question 3: (ii) $5ab^2$, $28a^3b^2$

Answer: $m^2 = 5ab^2 \times 28a^3b^2 = 140a^4b^4$, so $m = 2\sqrt{35}\ a^2b^2$.

Question: Exercise 18.3, Question 3: (iii) $a^4-b^4$, $(a^2-b^2)/(a^2+b^2)$

Answer: $m^2 = (a^4-b^4)(a^2-b^2)/(a^2+b^2) = (a^2-b^2)^2$, so $m = a^2-b^2$.

Question: Exercise 18.3, Question 3: (iv) $a^3-b^3$, $(a-b)/(a^2-ab+b^2)$

Answer: $m^2 = (a^3-b^3)(a-b)/(a^2-ab+b^2) = (a-b)^2(a^2+ab+b^2)/(a^2-ab+b^2)$, so m

= (a-b)√((a²+ab+b²)/(a²-ab+b²)).

Question: Exercise 18.3, Question 4: The following numbers are in continued proportion, find the value of x. (i) 45, x, 5

Answer: $x^2 = 45 \times 5 = 225$, so x = 15.

Question: Exercise 18.3, Question 4: (ii) 16, x, 9

Answer: $x^2 = 16 \times 9 = 144$, so x = 12.

Question: Exercise 18.3, Question 4: (iii) 12, 3x-6, 27

Answer: From $(3x-6)^2 = 12 \times 27 = 324$, 3x-6 = 18 or 3x-6 = -18, so x = 8 or x = -4.

Question: Exercise 18.3, Question 4: (iv) 7, x-3, 112

Answer: From $(x-3)^2 = 7 \times 112 = 784$, x-3 = 28 or x-3 = -28, so x = 31 or x = -25.

Question: Exercise 18.4, Question 1: Given x:y = z:w, prove: (i) (4x+3y)/(4x-3y) = (4z+3w)/(4z-3w)

Answer: Let x/y = z/w = k, i.e. x = ky and z = kw.

Left side = (4ky+3y)/(4ky-3y) = (4k+3)/(4k-3).

Right side = (4kw+3w)/(4kw-3w) = (4k+3)/(4k-3).

Both are equal.

Proved.

Question: Exercise 18.4, Question 1: (ii) (5x-3y)/(5x+3y) = (5z-3w)/(5z+3w)

Answer: Left side = (5ky-3y)/(5ky+3y) = (5k-3)/(5k+3).

Right side = (5kw-3w)/(5kw+3w) = (5k-3)/(5k+3).

Both are equal.

Proved.

Question: Exercise 18.4, Question 1: (iii) $(x^3+y^3)/(x^3-y^3) = (z^3+w^3)/(z^3-w^3)$

Answer: Left side = $(k^3y^3+y^3)/(k^3y^3-y^3) = (k^3+1)/(k^3-1)$.

Right side = $(k^3w^3+w^3)/(k^3w^3-w^3) = (k^3+1)/(k^3-1)$.

Both are equal.

Proved.

Question: Exercise 18.4, Question 1: (iv) $(3x+2y)/(3x-2y) = (3z+2w)/(3z-2w)$

Answer: Left side = $(3ky+2y)/(3ky-2y) = (3k+2)/(3k-2)$.

Right side = $(3kw+2w)/(3kw-2w) = (3k+2)/(3k-2)$.

Both are equal.

Proved.

Question: Exercise 18.4, Question 1: (v) $(2x+3y+2z+3w)/(2x+3y-2z-3w) = (2x-3y+2z-3w)/(2x-3y-2z+3w)$

Answer: $2x+3y = y(2k+3)$, $2z+3w = w(2k+3)$, $2x-3y = y(2k-3)$, $2z-3w = w(2k-3)$.

Left side = $((y+w)(2k+3))/((y-w)(2k+3)) = (y+w)/(y-w)$.

Right side = $((y+w)(2k-3))/((y-w)(2k-3)) = (y+w)/(y-w)$.

Both are equal.

Proved.

Question: Exercise 18.4, Question 2: Given a:b = c:d, prove: (i) $a^2-b^2$ : $a^2+b^2$ = ac-bd : ac+bd

Answer: Putting a = bk and c = dk, the left ratio = $(b^2k^2-b^2)/(b^2k^2+b^2)$ = $(k^2-1)/(k^2+1)$.

The right ratio = $(bk\cdot dk-bd)/(bk\cdot dk+bd)$ = $(k^2-1)/(k^2+1)$.

Both are equal.

Proved.

Question: Exercise 18.4, Question 2: (ii) (a+b+c+d)/(a+b-c-d) = (a-b+c-d)/(a-b-c+d)

Answer: Putting a = bk and c = dk, the left side = ((b+d)(k+1))/((b-d)(k+1)) = (b+d)/(b-d).

The right side = ((b+d)(k-1))/((b-d)(k-1)) = (b+d)/(b-d).

Both are equal.

Proved.

Question: Exercise 18.4, Question 3: Solve the following equations using componendo and dividendo: (i) $((x+3)^2 - (x-5)^2)/((x+3)^2 + (x-5)^2) = 4/5$

Answer: By componendo and dividendo, $(2(x+3)^2)/(-2(x-5)^2) = 9/(-1)$, i.e. $(x+3)^2/(x-5)^2 = 9$, so (x+3)/(x-5) = 3 or -3.

From (x+3)/(x-5) = 3, x = 9.

From (x+3)/(x-5) = -3, x = 3.

Solution set = {9, 3}.

Question: Exercise 18.4, Question 3: (ii) (√(x+1)+√(x-1))/(√(x+1)-√(x-1)) = 1/2

Answer: As printed, this equation has no solution: the left side is always greater than 1 for x > 1 (the numerator exceeds the denominator), while the right side is 1/2.

Componendo and dividendo gives √(x+1)/√(x-1) = -3, which is impossible (a square root cannot be negative).

[If the right side were 2/1, the answer would be x = 5/4 — the difference in the printed question is noted.]

Question: Exercise 18.4, Question 3: (iii) (√(x+5)-√(x-5))/(√(x+5)+√(x-5)) = 1/10

Answer: Inverting gives (√(x+5)+√(x-5))/(√(x+5)-√(x-5)) = 10/1.

By componendo and dividendo, (2√(x+5))/(2√(x-5)) = 11/9, i.e. (x+5)/(x-5) = 121/81.

From 81x + 405 = 121x - 605, 40x = 1010, so x = 101/4.

Check: √(101/4+5) = 11/2 and √(101/4-5) = 9/2, (11/2-9/2)/(11/2+9/2) = 1/10, correct.

Question: Exercise 18.4, Question 3: (iv) ((x+3)² + (x-1)³)/((x+3)² - (x-1)³)

= 11/7

Answer: By componendo and dividendo, $(2(x+3)^2)/(2(x-1)^3) = 18/4$, i.e. $(x+3)^2/(x-1)^3 = 9/2$, i.e. $2(x+3)^2 = 9(x-1)^3$.

Expanding gives $9x^3 - 29x^2 + 15x - 27 = 0$, which factors as $(x-3)(9x^2-2x+9)$.

The quadratic factor has negative discriminant, so the only real solution is $x = 3$.

Check: $(36+8)/(36-8) = 44/28 = 11/7$, correct.

Question: Exercise 18.5, Question 1: Given y varies directly as $x^2$ and z, and $y = 6$ when $x = 4$ and $z = 9$, write y as a function of x and z. Find y when $z = 12$ and $x = -8$.

Answer: $y = kx^2z$;

from $6 = k(16)(9) = 144k$, $k = 1/24$, so $y = (1/24)x^2z$.

For $z = 12$ and $x = -8$, $y = (1/24)(64)(12) = 32$.

Question: Exercise 18.5, Question 2: Given y varies directly as $x^2$ and $u^2$ and inversely as t and v, and $y = 40$ when $x = 8$, $u = 5$, $v = 3$ and $t = 2$, write y as a function of x, u, v, t. Find y when $x = -2$, $u = 4$, $v = 3$ and $t = -1$.

Answer: $y = kx^2u^2/(tv)$;

from $40 = k(64)(25)/(2\times3) = k(800/3)$, $k = 3/20$, so $y = (3/20)x^2u^2/(tv)$.

For $x = -2$, $u = 4$, $v = 3$, $t = -1$, $y = (3/20)(4)(16)/((-1)(3)) = -16/5$.

Question: Exercise 18.5, Question 3: Given w varies directly as $u^2$ and inversely as the cube root of v, and w = 216 when u = 6 and v = 27, find w when u = 10 and v = 125.

Answer: $w = ku^2/\sqrt[3]{v}$;

from $216 = k(36)/3 = 12k$, $k = 18$, so $w = 18u^2/\sqrt[3]{v}$.

For u = 10 and v = 125, $w = 18(100)/5 = 360$.

Question: Exercise 18.5, Question 4: The time T of a simple pendulum varies directly as the square root of its length L and inversely as the square root of the acceleration due to gravity g. T = 2 seconds when L = 100 m and g = 9.8 $m/s^2$. Find the time of the pendulum when L = 200 m and g = 7.6 $m/s^2$.

Answer: $T = k\sqrt{(L/g)}$;

from $2 = k\sqrt{(100/9.8)}$, $k = 2/\sqrt{(100/9.8)} = \sqrt{9.8}/5$.

For L = 200 and g = 7.6, $T = (\sqrt{9.8}/5)\sqrt{(200/7.6)} = (1/5)\sqrt{(1960/7.6)} = 14/\sqrt{19} \approx 3.21$ seconds.

Question: Exercise 18.5, Question 5: The volume V of a certain gas varies directly as the temperature T and inversely as the pressure P. V = 100 when T = 30 and P = 64. Find V when T = 60 and P = 81.

Answer: $V = kT/P$;

from $100 = k(30)/64$, $k = 640/3$.

For T = 60 and P = 81, $V = (640/3)(60)/81 = 12800/81 \approx 158.02$.

Question: Exercise 18.6, Question 1: Given p:q = r:s, prove: (i)

(8p-3q)/(8p+3q) = (8r-3s)/(8r+3s)

Answer: Let p/q = r/s = k, i.e. p = qk and r = sk.

Left side = (8qk-3q)/(8qk+3q) = (8k-3)/(8k+3).

Right side = (8sk-3s)/(8sk+3s) = (8k-3)/(8k+3).

Both are equal.

Proved.

Question: Exercise 18.6, Question 1: (ii) $\sqrt[3]{((p^3+r^3)/(q^3+s^3))} = p/q$

Answer: $(p^3+r^3)/(q^3+s^3) = (q^3k^3+s^3k^3)/(q^3+s^3) = k^3$, so the cube root = k = p/q. Proved.

Question: Exercise 18.6, Question 1: (iii) $(p^2+q^2) : p^3/(p+q) = (r^2+s^2) : r^3/(r+s)$

Answer: The left ratio = $(p^2+q^2)(p+q)/p^3 = (q^2k^2+q^2)(qk+q)/(q^3k^3) = (k^2+1)(k+1)/k^3$.

The right ratio = $(s^2k^2+s^2)(sk+s)/(s^3k^3) = (k^2+1)(k+1)/k^3$.

Both are equal.

Proved.

Question: Exercise 18.6, Question 1: (iv) $p^5+r^5 : q^5+s^5 = p^3r^2 : q^3s^2$

Answer: The left ratio = $(p^5+r^5)/(q^5+s^5) = (q^5k^5+s^5k^5)/(q^5+s^5) = k^5$. The right ratio = $p^3r^2/(q^3s^2) = (q^3k^3)(s^2k^2)/(q^3s^2) = k^5$. Both are equal. Proved.

Question: Exercise 18.6, Question 1: (v) $((p-q)/p) : (q/(p+q)) = ((r-s)/r) : (s/(r+s))$

Answer: Left side = ((p-q)/p)·((p+q)/q) = (p²-q²)/(pq) = (q²k²-q²)/(q²k) = (k²-1)/k.

Right side = (r²-s²)/(rs) = (s²k²-s²)/(s²k) = (k²-1)/k.

Both are equal.

Proved.

Question: Exercise 18.6, Question 2: Given a:b = c:d = e:f, prove: (i) (a⁴b² + a²e² - e⁴f)/(b⁶ + b²f² - f⁵) = a⁴/b⁴

Answer: Putting a = bk, c = dk, e = fk, the numerator = b⁶k⁴ + b²f²k⁴ - f⁵k⁴ = k⁴(b⁶ + b²f² - f⁵), so the ratio = k⁴ = a⁴/b⁴. Proved.

Question: Exercise 18.6, Question 2: (ii) (a²b + c²d + e²f)/(ab² + cd² + ef²) = (a+c+e)/(b+d+f)

Answer: The numerator = k²(b³+d³+f³) and the denominator = k(b³+d³+f³), so the ratio = k. The right side = (bk+dk+fk)/(b+d+f) = k. Both are equal. Proved.

Question: Exercise 18.6, Question 2: (iii) ac/bd + ce/df + ea/fb = a²/b² + c²/d² + e²/f²

Answer: ac/bd = k², ce/df = k², ea/fb = k², so the left side = 3k².

a²/b² = k², c²/d² = k², e²/f² = k², so the right side = 3k².

Both are equal.

Proved.

Question: Exercise 18.7, Question 1: The electric current I in a wire varies directly as the voltage V and inversely as the resistance R. I = 6 amperes

when V = 220 volts and R = 5 ohms. Find I when V = 180 volts and R = 8 ohms.

Answer: $I = kV/R$;

from $6 = k(220)/5$, $k = 3/22$.

For V = 180 and R = 8, $I = (3/22)(180)/8 = 135/44 \approx 3.07$ amperes.

Question: Exercise 18.7, Question 2: The intensity of light spreading from its source at a distance varies inversely as the square of the distance. The intensity of a bulb is 0.08 foot-candles when the distance is 3 metres. Find the intensity when the distance is 8 metres.

Answer: $I = k/d^2$;

from $0.08 = k/9$, $k = 0.72$.

For d = 8, $I = 0.72/64 = 0.01125$ foot-candles.

Question: Exercise 18.7, Question 3: The strength S of a rectangular beam varies directly as the breadth b and the square of the depth d. A beam 9 cm broad and 12 cm deep can support 1200 pounds. How much weight can a beam 12 cm broad and 9 cm deep support?

Answer: $S = kbd^2$;

from $1200 = k(9)(144) = 1296k$, $k = 25/27$.

For b = 12 and d = 9, $S = (25/27)(12)(81) = 900$ pounds.

Question: Exercise 18.7, Question 4: The amount C varies jointly as the number n of workers and the number d of days. The wages for 100 workers for 20 days is 9000 rupees. Find the wages for 300 workers for 15

days.

Answer: C = knd;

from 9000 = k(100)(20) = 2000k, k = 4.5.

For n = 300 and d = 15, C = 4.5 × 300 × 15 = 20250 rupees.

Question: Exercise 18.7, Question 5: The sales tax on buying a new car varies directly as the price of the car. A new car bought for 2,000,000 rupees has a sales tax of 40,000 rupees. Find the sales tax on a car priced at 2,800,000 rupees.

Answer: T = kP;

from 40000 = k(2000000), k = 0.02.

For P = 2800000, T = 0.02 × 2800000 = 56000 rupees.

Exercise 18 Review, Question 1: Circle the correct answer: (i) In the proportion p:q::r:s, p is called the ___.

(a) third proportional

(b) mean proportional

(c) fourth proportional

✓ (d) first proportional

Exercise 18 Review, Question 1: (ii) In a ratio m:n, m is called the ___.

(a) consequent

✓ (b) antecedent

(c) relation

(d) none

Exercise 18 Review, Question 1: (iii) In the ratio u:v, v is called the ___.

(a) antecedent

(b) consequent

(c) relation

(d) none

Exercise 18 Review, Question 1: (iv) If a, b, c are in continued proportion, b is called the ___ proportional.

(a) first

✓ (b) mean

(c) third

(d) none

Exercise 18 Review, Question 1: (v) The mean proportional of $a^2$ and $b^2$ is ___.

✓ (b) ab

(c) a/b

(d) -ab

Exercise 18 Review, Question 1: (vi) If x+5 : x+7 = 5:7 then x is equal to ___.

(a) 2

(b) -1

✓ (c) 0

(d) 1

Exercise 18 Review, Question 1: (vii) If 1, 9, x, 45 are in proportion then x = ___.

(a) 27

(b) 1/5

✓ (c) 5

(d) 405

Exercise 18 Review, Question 1: (viii) If p:q = r:5 then p:r = q:5 is called the ___ law.

✓

(a) componendo

(b) invertendo

(c) dividendo

✓ (d) alternendo

Exercise 18 Review, Question 1: (ix) If x:y = z:w then by the componendo law:

Exercise 18 Review, Question 1: (x) If a:b = c:d then by the alternendo law:

(a) a/b = c/d

(c) b/a = d/c

✓ (d) a/c = b/d

Exercise 18 Review, Question 1: (xi) The fourth proportional to 3, 5, 12 is ___.

✓ (a) 20

(b) 15

(c) 60

(d) 36

Exercise 18 Review, Question 1: (xii) If 2x, 3y, 6z are in continued proportion then:

(a) $y^2 = 12xy$

(b) $9y^2 = xz$

✓ (c) $9y^2 = 12xz$

(d) $3y^2 = 4xz$

Exercise 18 Review, Question 1: (xiii) If x/y = w/z then by the dividendo law:

(d) none

Exercise 18 Review, Question 1: (xiv) Force and acceleration are in ___ proportion.

(a) direct

(b) joint

(c) inverse

(d) none

Exercise 18 Review, Question 1: (xv) If a:4 = 15:5 then a = ___.

(a) 20

(b) 15



(c) 12

(d) 10

Question: Exercise 18 Review, Question 2: Find the ratio of the following:

(i) 100 metres and 500 cm

Answer: 100 metres = 10000 cm, so 10000:500 = 20:1.

Question: Exercise 18 Review, Question 2: (ii) 50 kg and 300 g

Answer: 50 kg = 50000 g, so 50000:300 = 500:3.

Question: Exercise 18 Review, Question 3: Find the value of x from the following. (i) 9 : x-10 = x+13 : 12

Answer: From 9 × 12 = (x-10)(x+13), 108 = $x^2$ + 3x - 130, i.e. $x^2$ + 3x - 238 = 0.

x = (-3 ± 31)/2, so x = 14 or x = -17.

Question: Exercise 18 Review, Question 3: (ii) 5 : x-3 = x+11 : 3

Answer: From 5 × 3 = (x-3)(x+11), 15 = $x^2$ + 8x - 33, i.e. $x^2$ + 8x - 48 = 0.

x = (-8 ± 16)/2, so x = 4 or x = -12.

Question: Exercise 18 Review, Question 4: Given y varies directly as x and y = 25 when x = 75, find y when x = 144.

Answer: y = kx;



from 25 = 75k, k = 1/3.

For x = 144, y = 144/3 = 48.

Question: Exercise 18 Review, Question 5: Given y varies inversely as x and y = 100 when x = 1/2, find y when x = 4.

Answer: y = k/x;

from 100 = k/(1/2), k = 50.

For x = 4, y = 50/4 = 25/2.

Question: Exercise 18 Review, Question 6: Given x:y = z:w, prove (7x+5y)/(7x-5y) = (7z+5w)/(7z-5w).

Answer: Putting x/y = z/w = k, the left side = (7k+5)/(7k-5) and the right side = (7k+5)/(7k-5).

Both are equal.

Proved.

Question: Exercise 18 Review, Question 7: Solve the following equation using componendo and dividendo: ((x-3)(x-5))/((x-7)(x-2)) = ((x-6)(x-2))/((x-1)(x-8))

Answer: By componendo and dividendo, $(2x^2-17x+29)/(x+1) = (2x^2-17x+20)/(x+4)$.

Cross-multiplying gives $3(2x^2-17x) + 9x + 96 = 0$, i.e. $2x^2 - 14x + 32 = 0$, i.e. $x^2 - 7x + 16 = 0$.

The discriminant = 49 - 64 = -15 < 0, so there is no real solution.

[As printed, the equation has no real solution — noted.]

Question: Exercise 18 Review, Question 8: Given x varies directly as y and inversely as z, and x = 30 when y = 15 and z = 2, find x when y = 30 and z = 12.

Answer: $x = ky/z$;

from $30 = k(15)/2$, $k = 4$, so $x = 4y/z$.

For $y = 30$ and $z = 12$, $x = 4(30)/12 = 10$.

Question: Exercise 18 Review, Question 9: The electric current in a circuit varies inversely as the resistance. The current in the circuit is 40 amperes when the resistance is 10 ohms. Find the current in the circuit when the resistance is 12 ohms.

Answer: $I = k/R$;

from $40 = k/10$, $k = 400$.

For $R = 12$, $I = 400/12 = 100/3 \approx 33.33$ amperes.

# 19. قالب قطعي ملهه

Question: Exercise 19.1, Question 1: Name the type of each of the following matrices. (i) [[2, -1], [3, 0]]

Answer: Square matrix (2x2, number of rows equals number of columns).

Question: Exercise 19.1, Question 1: (ii) [[3], [-1], [4]]

Answer: Column matrix (only one column).

Question: Exercise 19.1, Question 1: (iii) [[-2, 0]]

Answer: Row matrix (only one row).

Question: Exercise 19.1, Question 1: (iv) [[λ, 0], [0, λ]]

Answer: Scalar matrix (all diagonal entries equal and the rest zero).

Question: Exercise 19.1, Question 1: (v) [[a, 0, 0], [0, b, 0], [0, 0, c]]

Answer: Diagonal matrix (all entries off the diagonal are zero).

Question: Exercise 19.1, Question 2: State the order of each of the following matrices. (i) [[0, 1, -1], [2, 0, 3], [-1, -3, 2]]

Answer: 3x3 (three rows and three columns).

Question: Exercise 19.1, Question 2: (ii) [[3, 4], [-1, 2], [0, 3]]

Answer: 3x2 (three rows and two columns).

Question: Exercise 19.1, Question 2: (iii) [[1, 1, 1], [2, 2, 2]]

Answer: 2x3 (two rows and three columns).

Question: Exercise 19.1, Question 2: (iv) [[-2], [1], [-1]]

Answer: 3x1 (three rows and one column).

Question: Exercise 19.1, Question 2: (v) [[0]]

Answer: 1x1 (one row and one column).

Question: Exercise 19.1, Question 3: Check whether the following are symmetric or skew-symmetric matrices. (i) [[1, 1, -1], [1, 2, 0], [-1, 0, 5]]

Answer: It is a symmetric matrix, because its transpose equals itself: a12

= a21 = 1, a13 = a31 = -1, a23 = a32 = 0.

Question: Exercise 19.1, Question 3: (ii) [[0, 1, -2], [-1, 0, 3], [2, -3, 0]]

Answer: It is a skew-symmetric matrix, because its transpose equals its negative: diagonal entries are zero and a12 = -a21 = 1, a13 = -a31 = -2, a23 = -a32 = 3.

Question: Exercise 19.1, Question 3: (iii) [[1, 7, 3], [7, 4, -5], [3, -5, 6]]

Answer: It is a symmetric matrix, because a12 = a21 = 7, a13 = a31 = 3, a23 = a32 = -5.

Question: Exercise 19.1, Question 3: (iv) [[0, -6, 4], [-6, 0, 7], [4, 7, 0]]

Answer: It is a symmetric matrix, because a12 = a21 = -6, a13 = a31 = 4, a23 = a32 = 7.

(Not skew-symmetric, because the off-diagonal entries do not equal their negatives.)

Question: Exercise 19.1, Question 4: Given A = [[1, -1], [0, 1]], B = [[2, -1], [1, 0]], C = [[2, 3], [-1, 2]], D = [[1, 2, 3], [1, 2, 3]], E = [[1, 2, -1], [-1, 0, 2], [2, 1, 0]], find the following if possible. (i) A + C

Answer: A + C = [[1+2, -1+3], [0-1, 1+2]] = [[3, 2], [-1, 3]].

Question: Exercise 19.1, Question 4: (ii) A + E

Answer: Not possible — A is of order 2x2 and E is of order 3x3, so addition is not possible.

Question: Exercise 19.1, Question 4: (iii) B - D

Answer: Not possible — B is of order 2x2 and D is of order 2x3, so subtraction is not possible.

Question: Exercise 19.1, Question 4: (iv) 2B + 3A

Answer: 2B + 3A = [[4, -2], [2, 0]] + [[3, -3], [0, 3]] = [[7, -5], [2, 3]].

Question: Exercise 19.1, Question 4: (v) B - C

Answer: B - C = [[2-2, -1-3], [1+1, 0-2]] = [[0, -4], [2, -2]].

Question: Exercise 19.1, Question 4: (vi) $A^2$

Answer: $A^2$ = [[1, -1], [0, 1]] × [[1, -1], [0, 1]] = [[1, -2], [0, 1]].

Question: Exercise 19.1, Question 4: (vii) $B^2$

Answer: $B^2$ = [[2, -1], [1, 0]] × [[2, -1], [1, 0]] = [[3, -2], [2, -1]].

Question: Exercise 19.1, Question 4: (viii) D + E

Answer: Not possible — D is of order 2x3 and E is of order 3x3, so addition is not possible.

Question: Exercise 19.1, Question 5: Given A = [[-1, 0, 1], [2, 1, 0], [3, 2, -1]], B = [[4, 2, 3], [-2, 4, 1], [3, 2, 1]], find: (i) A + B

Answer: A + B = [[3, 2, 4], [0, 5, 1], [6, 4, 0]].

Question: Exercise 19.1, Question 5: (ii) A - B

Answer: A - B = [[-5, -2, -2], [4, -3, -1], [0, 0, -2]].

Question: Exercise 19.1, Question 5: (iii) 3A + 2B

Answer: 3A + 2B = [[-3, 0, 3], [6, 3, 0], [9, 6, -3]] + [[8, 4, 6], [-4, 8, 2], [6, 4, 2]] = [[5, 4, 9], [2, 11, 2], [15, 10, -1]].

Question: Exercise 19.1, Question 5: (iv) AB

Answer: AB = [[-1, 0, -2], [6, 8, 7], [5, 12, 10]] (for example, first row: -1x4+0x(-2)+1x3 = -1, -1x2+0x4+1x2 = 0, -1x3+0x1+1x1 = -2).

Question: Exercise 19.1, Question 5: (v) BA

Answer: BA = [[9, 8, 1], [13, 6, -3], [4, 4, 2]] (for example, first row: 4x(-1)+2x2+3x3 = 9, 4x0+2x1+3x2 = 8, 4x1+2x0+3x(-1) = 1).

Question: Exercise 19.1, Question 5: (vi) $A^2$

Answer: $A^2$ = [[4, 2, -2], [0, 1, 2], [-2, 0, 4]] (for example, first row: -1x(-1)+0x2+1x3 = 4, -1x0+0x1+1x2 = 2, -1x1+0x0+1x(-1) = -2).

Question: Exercise 19.1, Question 6: Verify (i) A + B = B + A, given A = [[2, -1], [-7, 4]], B = [[-3, 0], [7, -4]].

Answer: A + B = [[-1, -1], [0, 0]] = B + A.

Verified.

Question: Exercise 19.1, Question 6: Verify (ii) C - D ≠ D - C, given C = [[3, 1, -4], [4, 3, 1], [1, 4, -3]], D = [[2, 7, -5], [-2, 1, 0], [6, 3, 4]].

Answer: C - D = [[1, -6, 1], [6, 2, 1], [-5, 1, -7]] and D - C = [[-1, 6, -1], [-6, -2, -1], [5, -1, 7]] = -(C - D) ≠ C - D.

Verified.

Question: Exercise 19.1, Question 7: Prove that AB ≠ BA given A = [[8, 2, 1], [2, 1, 2], [2, 3, 0]], B = [[1, 4, 4], [2, 5, 7], [0, -4, 3]].

Answer: AB = [[12, 38, 49], [4, 5, 21], [8, 23, 29]] and BA = [[24, 18, 9], [40, 30, 12], [-2, 5, -8]].

AB ≠ BA.

Proved.

Question: Exercise 19.1, Question 8: Given A = [[-3, 0], [7, -4]], B = [[2, -1], [-7, 4]], C = [[1, 0], [-2, -4]], find 2A - 3B + 4C.

Answer: 2A - 3B + 4C = [[-6, 0], [14, -8]] + [[-6, 3], [21, -12]] + [[4, 0], [-8, -16]] = [[-8, 3], [27, -36]].

Question: Exercise 19.1, Question 9: Given [[a, b], [c, d]] + [[-3, 1], [1, 0]] = [[1, 0], [0, 1]], find the values of a, b, c, d.

Answer: From a - 3 = 1, a = 4;

from b + 1 = 0, b = -1;

from c + 1 = 0, c = -1;

from d + 0 = 1, d = 1.

Question: Exercise 19.1, Question 10: Given [[x, -1, y], [2, 0, 3], [z, 3, 2]] = [[3, a, x], [b, 0, c], [2, 3, d]], find the values of a, b, c, d, x, y, z.

Answer: Comparing corresponding entries: x = 3, a = -1, y = x = 3, b = 2, c = 3, z = 2, d = 2.

Question: Exercise 19.1, Question 11: State whether multiplication is possible for the following matrices. A = [[1, -2], [3, 1]], B = [[2, 1, 0], [1, 0, -1]], C = [[0, -1, 2], [-1, 0, -4], [3, 1, 2]]

Answer: A is of order 2x2, B is 2x3 and C is 3x3.

AB is possible (2x2 · 2x3 = 2x3);

BC is possible (2x3 · 3x3 = 2x3);

so ABC is possible.

BA is not possible (3 ≠ 2); CA is not possible (3 ≠ 2); AC is not possible (2 ≠ 3); CB is not possible (3 ≠ 2).

Question: Exercise 19.1, Question 12: Find the product: [[1, 0, 3], [2, 1, 2], [1, 3, 1]] × [[2, 2, 0], [1, 3, 2], [3, 2, 0]]

Answer: [[11, 8, 0], [11, 11, 2], [8, 13, 6]] (for example, first row: 1x2+0x1+3x3 = 11, 1x2+0x3+3x2 = 8, 1x0+0x2+3x0 = 0).

Question: Exercise 19.1, Question 13: Given A = [[3, 1], [1, 2]], B = [[2, -1, 0], [3, 1, 2]], C = [[1, 1, 2], [2, 2, 3]], verify: (i) A(B+C) = AB + AC

Answer: B + C = [[3, 0, 2], [5, 3, 5]], A(B+C) = [[14, 3, 11], [13, 6, 12]].

AB = [[9, -2, 2], [8, 1, 4]], AC = [[5, 5, 9], [5, 5, 8]], AB + AC = [[14, 3, 11], [13, 6, 12]] = A(B+C).

Verified.

Question: Exercise 19.1, Question 13: (ii) (B+C)A = BA + CA

Answer: As printed, this equality cannot be formed: B + C is of order 2x3 and A is of order 2x2, so none of the products (B+C)A, BA and CA is possible (the number of columns of the first matrix does not equal the number of rows of the second). [The difference in the printed question is noted.]

Question: Exercise 19.1, Question 14: Given A = [[6, 5, 2], [4, 6, 3], [0, 1, 7]], B = [[6, 8, 0], [5, 1, 2], [3, 6, 4]], C = [[1, 2, 3], [1, 2, 3], [1, 2, 3]], verify: (a) A + B = B + A

Answer: A + B = [[12, 13, 2], [9, 7, 5], [3, 7, 11]] = B + A.

Verified.

Question: Exercise 19.1, Question 14: (b) (A+B) + C = A + (B+C)

Answer: (A+B) + C = [[13, 15, 5], [10, 9, 8], [4, 9, 14]];

B + C = [[7, 10, 3], [6, 3, 5], [4, 8, 7]], A + (B+C) = [[13, 15, 5], [10, 9, 8], [4, 9, 14]].

Both are equal.

Verified.

Question: Exercise 19.1, Question 14: (c) (A+B)C = AC + BC

Answer: (A+B)C = [[27, 54, 81], [21, 42, 63], [21, 42, 63]];

AC = [[13, 26, 39], [13, 26, 39], [8, 16, 24]], BC = [[14, 28, 42], [8, 16, 24], [13, 26, 39]], AC + BC = [[27, 54, 81], [21, 42, 63], [21, 42, 63]].

Both are equal.

Verified.

Question: Exercise 19.1, Question 14: (d) A(B+C) = AB + AC

Answer: A(B+C) = [[80, 91, 57], [76, 82, 63], [34, 59, 54]];

AB = [[67, 65, 18], [63, 56, 24], [26, 43, 30]], AC = [[13, 26, 39], [13, 26, 39], [8, 16, 24]], AB + AC = [[80, 91, 57], [76, 82, 63], [34, 59, 54]].

Both are equal.

Verified.

Question: Exercise 19.1, Question 15: Given A = [[2, 3, 1], [1, -1, 1], [0, 2, 2]], B = [[1, 2, 1], [-1, 2, -1], [2, 0, 2]], C = [[1, 0, 1], [0, 1, 1], [1, 0, 2]], prove A(BC) = (AB)C.

Answer: BC = [[2, 2, 5], [-2, 2, -1], [4, 0, 6]], A(BC) = [[2, 10, 13], [8, 0, 12], [4, 4, 10]].

AB = [[1, 10, 1], [4, 0, 4], [2, 4, 2]], (AB)C = [[2, 10, 13], [8, 0, 12], [4, 4, 10]] = A(BC).

Proved.

Question: Exercise 19.2, Question 1: Find the value of the determinant of each of the following. (i) |-5 -3; 3 -4|

Answer: (-5)(-4) - (-3)(3) = 20 + 9 = 29.

Question: Exercise 19.2, Question 1: (ii) |-4 3; 1 -3|

Answer: (-4)(-3) - (3)(1) = 12 - 3 = 9.

Question: Exercise 19.2, Question 1: (iii) |-1 -5; 2 3|

Answer: (-1)(3) - (-5)(2) = -3 + 10 = 7.

Question: Exercise 19.2, Question 1: (iv) |2 -5; 2 -1|

Answer: (2)(-1) - (-5)(2) = -2 + 10 = 8.

Question: Exercise 19.2, Question 1: (v) |3 -3 -4; 4 1 -5; 0 -1 -4|

Answer: Expanding along the first row: 3(1x(-4) - (-5)x(-1)) + 3(4x(-4) - (-5)x0) - 4(4x(-1) - 1x0) = 3(-9) + 3(-16) - 4(-4) = -27 - 48 + 16 = -59.

Question: Exercise 19.2, Question 1: (vi) |-3 4 -5; 2 -3 -5; 1 3 5|

Answer: Expanding along the first row: -3((-3)(5) - (-5)(3)) - 4(2x5 - (-5)x1) - 5(2x3 - (-3)x1) = -3(0) - 4(15) - 5(9) = -105.

Question: Exercise 19.2, Question 1: (vii) |0 5 -4; -3 4 -5; 1 0 -5|

Answer: Expanding along the first row: 0 - 5((-3)(-5) - (-5)(1)) - 4((-3)(0) - 4x1) = -5(20) - 4(-4) = -100 + 16 = -84.

Question: Exercise 19.2, Question 1: (viii) |4 3 -5; -5 -1 -5; 2 4 4|

Answer: Expanding along the first row: 4((-1)(4) - (-5)(4)) - 3((-5)(4) - (-5)(2)) - 5((-5)(4) - (-1)(2)) = 4(16) - 3(-10) - 5(-18) = 64 + 30 + 90 = 184.

Question: Exercise 19.2, Question 1: (ix) |1 0 4; -2 3 -5; 3 5 3|

Answer: Expanding along the first row: 1(3x3 - (-5)(5)) + 4((-2)(5) - 3x3) = 34 + 4(-19) = 34 - 76 = -42.

Question: Exercise 19.2, Question 1: (x) |0 a b; 0 c d; 0 x y|

Answer: All entries of the first column are zero, so the determinant = 0.

Question: Exercise 19.2, Question 1: (xi) |-2 0 0; -6 x 1; -4 0 -1|

Answer: Expanding along the first row: -2(xx(-1) - 1x0) = -2(-x) = 2x.

Question: Exercise 19.2, Question 2: Given A = [[2, 1, 1], [1, 1, 1], [1, -1, 2]], find the values of M12, M21, M22, A12, A21, A22.

Answer: M12 = |1 1;

1 2| = 1, A12 = $(-1)^3$ x 1 = -1.

M21 = |1 1;

-1 2| = 3, A21 = $(-1)^3$ x 3 = -3.

M22 = |2 1;

1 2| = 3, A22 = $(-1)^4$ x 3 = 3.

Question: Exercise 19.2, Question 3: For which value of x will [[5-x, x+1], [2, 4]] be a singular matrix?

Answer: The determinant = 4(5-x) - 2(x+1) = 18 - 6x.

For singular, the determinant = 0, i.e. 18 - 6x = 0, so x = 3.

Question: Exercise 19.2, Question 4: Which of the following square matrices is singular and which is non-singular: A = [[0, 2, -1], [3, -2, 1], [3, 2, -1]], B = [[0, 2, -1], [3, -2, 1], [3, 2, 1]]

Answer: |A| = -2(3x(-1) - 1x3) - (3x2 - (-2)x3) = -2(-6) - 12 = 0, so A is singular.

|B| = -2(3x1 - 1x3) - (3x2 - (-2)x3) = 0 - 12 = -12 ≠ 0, so B is non-singular.

Question: Exercise 19.2, Question 5: Find the adjoints of the following. A = [[1, 6], [4, 7]]

Answer: For a 2x2 matrix, adj(A) = [[7, -6], [-4, 1]].

Question: Exercise 19.2, Question 5: B = [[-3, 2, -5], [-1, 0, -2], [3, -4, 1]]

Answer: The cofactor matrix = [[-8, -5, 4], [18, 12, -6], [-4, -1, 2]], so adj(B) = [[-8, 18, -4], [-5, 12, -1], [4, -6, 2]].

Question: Exercise 19.2, Question 5: C = [[0, -2, 0], [6, 2, -6], [-3, 0, 6]]

Answer: The cofactor matrix = [[12, -18, 6], [12, 0, 6], [12, 0, 12]], so adj(C) = [[12, 12, 12], [-18, 0, 0], [6, 6, 12]].

Question: Exercise 19.2, Question 6: Given A = [[2, 3], [-4, -6]], verify A(adj A) = |A| I.

Answer: |A| = 2x(-6) - 3x(-4) = 0.

adj(A) = [[-6, -3], [4, 2]].

A(adj A) = [[-12+12, -6+6], [24-24, 12-12]] = [[0, 0], [0, 0]] = 0·I = |A| I.

Verified.

Question: Exercise 19.2, Question 7: Find the multiplicative inverse of the following matrices by the adjoint method. (i) A = [[3, 2], [-2, -1]]

Answer: |A| = 3x(-1) - 2x(-2) = 1.

adj(A) = [[-1, -2], [2, 3]].

$A^{-1}$ = (1/1)[[-1, -2], [2, 3]] = [[-1, -2], [2, 3]].

Question: Exercise 19.2, Question 7: (ii) B = [[3, 6], [5, 10]]

Answer: |B| = 3x10 - 6x5 = 0, so B is singular and its multiplicative inverse does not exist.

[The inverse of the printed question does not exist — noted.]

Question: Exercise 19.2, Question 7: (iii) C = [[2, 1, 1], [3, 2, 1], [2, 1, 2]]

Answer: |C| = 2(4-1) - (6-2) + (3-4) = 1.

The cofactor matrix = [[3, -4, -1], [-1, 2, 0], [-1, 1, 1]], adj(C) = [[3, -1, -1], [-4, 2, 1], [-1, 0, 1]].

$C^{-1}$ = [[3, -1, -1], [-4, 2, 1], [-1, 0, 1]].

Question: Exercise 19.2, Question 7: (iv) D = [[1, 0, 1], [-4, 1, -1], [6, -2, 1]]

Answer: |D| = (1-2) + (8-6) = 1.

The cofactor matrix = [[-1, -2, 2], [-2, -5, 2], [-1, -3, 1]], adj(D) = [[-1, -2, -1], [-2, -5, -3], [2, 2, 1]].

$D^{-1}$ = [[-1, -2, -1], [-2, -5, -3], [2, 2, 1]].

Question: Exercise 19.2, Question 7: (v) E = [[1, -1, 3], [2, 1, 2], [-2, -2, 1]]

Answer: |E| = (1+4) + (2+4) + 3(-4+2) = 5.

The cofactor matrix = [[5, -6, -2], [-5, 7, 4], [-5, 4, 3]], adj(E) = [[5, -5, -5], [-6, 7, 4], [-2, 4, 3]].

$E^{-1}$ = (1/5)[[5, -5, -5], [-6, 7, 4], [-2, 4, 3]].

Question: Exercise 19.2, Question 8: Solve the following equations by Cramer's rule and the multiplicative inverse method. 1) 2x + 3y = 14, -4x + y = 28

Answer: Cramer: |A| = 2 + 12 = 14.

x = (14 - 84)/14 = -5, y = (56 + 56)/14 = 8.

Inverse method: $A^{-1}$ = (1/14)[[1, -3], [4, 2]], X = (1/14)[[-70], [112]] = [[-5], [8]].

Solution set = {(-5, 8)}.

Question: Exercise 19.2, Question 8: 2) 2x - 4y = -12, 2y + 3x = 0

Answer: The equations: 3x + 2y = 0, 2x - 4y = -12.

Cramer: |A| = -12 - 4 = -16.

x = 24/(-16) = -3/2, y = (-36)/(-16) = 9/4.

Inverse method: $A^{-1}$ = (1/-16)[[-4, -2], [-2, 3]], X = [[-3/2], [9/4]].

Solution set = {(-3/2, 9/4)}.

Exercise 19 Review, Question 1: Circle the correct answer. (i) If m denotes the number of rows and n the number of columns, then when m = n the matrix is called a ___ matrix.

(a) rectangular

(b) equal

✓ (c) square

(d) zero

Exercise 19 Review, Question 1: (ii) If A = [[i, 0], [0, -i]] then $A^2$ = ___.

(a) $I_2$

(b) $I_3$

✓ (c) $-I_2$

(d) 0

Exercise 19 Review, Question 1: (iii) If A is a square matrix such that $A^t$ = -A then A is called a ___ matrix.

(a) diagonal

(b) scalar

(c) symmetric

✓ (d) skew-symmetric

Exercise 19 Review, Question 1: (iv) If A, B and C are three matrices then $(ABC)^t$ = ___.

(a) $A^t \cdot B^t \cdot C^t$

(b) $C^t \cdot B^t \cdot A^t$

(c) $C^t \cdot A^t \cdot B^t$

Exercise 19 Review, Question 1: (v) Given A = [[i, 0], [0, -i]], B = [[0, $i^2$], [$-i^2$, 0]], C = [[0, i], [i, 0]] then ___.

(a) $A^2 = -I$

(b) $B^2 = -I$

(c) $C^2 = -I$

✓ (d) all

Exercise 19 Review, Question 1: (vi) For two matrices A and B, given A = [[sinθ, -cosθ], [cosθ, sinθ]] and B = [[sinθ, cosθ], [-cosθ, sinθ]], AB = ___.

✓ (a) [[1, 0], [0, 1]]

(b) [[-1, 0], [0, -1]]

(c) [[1, 1], [1, 1]]

(d) [[0, 0], [0, 0]]

Exercise 19 Review, Question 1: (vii) 2[[x], [3], [4]] + 3[[-1], [y], [0]] = 4[[0], [4], [z]], find the values of x, y and z.

(a) 2, 3/2, 10/3

(b) 3/2, 2, 10/3

✓ (c) 3/2, 10/3, 2

(d) 1, 2, 3

Exercise 19 Review, Question 1: (viii) $(A^{-1})^{-1}$ = ___.

(a) $A^{-2}$

✓ (b) A

(c) $A^{-1}$

(d) $A^2$

Exercise 19 Review, Question 1: (ix) |5 1; 2 x| = x + 4, find the value of x.

✓ (a) 3/2

(b) 2/3

✓

(c) 0

(d) none

Exercise 19 Review, Question 1: (x) If [[λ, -3, 4], [-3, 0, 1], [-1, 3, 2]] is a non-singular matrix then λ ≠ ___.

(a) -15



(b) -17

(c) -16

(d) none

Question: Exercise 19 Review, Question 2: Define row and column matrices.

Answer: A row matrix is a matrix with only one row, such as [1 2 3]. A column matrix is a matrix with only one column, such as [[1], [2], [3]].

Question: Exercise 19 Review, Question 3: Find the order of the following matrices. (i) [[1, 5, 8]]

Answer: 1x3 (one row and three columns).

Question: Exercise 19 Review, Question 3: (ii) [[1, 2, 3], [4, 5, 8]]

Answer: 2x3 (two rows and three columns).

Question: Exercise 19 Review, Question 3: (iii) [[7, 2], [9, 3], [8, 1]]

Answer: 3x2 (three rows and two columns).

Question: Exercise 19 Review, Question 4: Define the following. (i) Square matrix

Answer: A matrix in which the number of rows equals the number of columns, such as a 2x2 or 3x3 matrix.

Question: Exercise 19 Review, Question 4: (ii) Rectangular matrix

Answer: A matrix in which the number of rows does not equal the number of columns, such as a 2x3 matrix.

Question: Exercise 19 Review, Question 4: (iii) Diagonal matrix

Answer: A square matrix in which all entries off the diagonal are zero.

Question: Exercise 19 Review, Question 4: (iv) Scalar matrix

Answer: A diagonal matrix in which all diagonal entries are equal, such as [[k, 0], [0, k]].

Question: Exercise 19 Review, Question 4: (v) Symmetric matrix

Answer: A square matrix whose transpose equals itself, i.e. $A^t = A$.

Question: Exercise 19 Review, Question 4: (vi) Skew-symmetric matrix

Answer: A square matrix whose transpose equals its negative, i.e. $A^t = -A$ (diagonal entries are zero).

Question: Exercise 19 Review, Question 5: Given A = [[1, 8, 9], [2, 1, 0], [-2, 1, 7]], B = [[2, 5, 8], [1, 3, 4], [7, 2, 4]], find: (i) A + B

Answer: A + B = [[3, 13, 17], [3, 4, 4], [5, 3, 11]].

Question: Exercise 19 Review, Question 5: (ii) A - B

Answer: A - B = [[-1, 3, 1], [1, -2, -4], [-9, -1, 3]].

Question: Exercise 19 Review, Question 5: (iii) AB

Answer: AB = [[73, 47, 76], [5, 13, 20], [46, 7, 16]] (for example, first row: 1x2+8x1+9x7 = 73, 1x5+8x3+9x2 = 47, 1x8+8x4+9x4 = 76).

Question: Exercise 19 Review, Question 5: (iv) BA

Answer: BA = [[-4, 29, 74], [-1, 15, 37], [3, 62, 91]] (for example, first row: 2x1+5x2+8x(-2) = -4, 2x8+5x1+8x1 = 29, 2x9+5x0+8x7 = 74).

Question: Exercise 19 Review, Question 5: (v) 5A

Answer: 5A = [[5, 40, 45], [10, 5, 0], [-10, 5, 35]].

Question: Exercise 19 Review, Question 5: (vi) 7B

Answer: 7B = [[14, 35, 56], [7, 21, 28], [49, 14, 28]].

Question: Exercise 19 Review, Question 5: (vii) 8A - 9B

Answer: 8A - 9B = [[-10, 19, 0], [7, -19, -36], [-79, -10, 20]].

Question: Exercise 19 Review, Question 6: Find the value of |1 8 9; 2 0 -1; -7 8 -10|.

Answer: Expanding along the first row: 1(0x(-10) - (-1)(8)) - 8(2x(-10) -

(-1)(-7)) + 9(2x8 - 0x(-7)) = 8 - 8(-27) + 9(16) = 8 + 216 + 144 = 368.

Question: Exercise 19 Review, Question 7: Define singular and non-singular matrices with examples.

Answer: A singular matrix is a square matrix whose determinant is zero, such as [[1, 2], [2, 4]] (determinant = 4 - 4 = 0).

A non-singular matrix is a square matrix whose determinant is not zero, such as [[1, 2], [3, 4]] (determinant = 4 - 6 = -2).

Question: Exercise 19 Review, Question 8: Given A = [[1, 4, 2], [7, 0, 9], [0, 2, -3]], find $A^{-1}$.

Answer: |A| = -18 + 84 + 28 = 94.

The cofactor matrix = [[-18, 21, 14], [16, -3, -2], [36, 5, -28]], adj(A) = [[-18, 16, 36], [21, -3, 5], [14, -2, -28]].

$A^{-1}$ = (1/94)[[-18, 16, 36], [21, -3, 5], [14, -2, -28]].

Question: Exercise 19 Review, Question 9: Solve the given pair of equations by the following methods: 2x + 5y = 27, 7x + y = 12. (i) Multiplicative inverse method

Answer: |A| = 2 - 35 = -33.

$A^{-1}$ = (1/-33)[[1, -5], [-7, 2]].

X = (1/-33)[[27-60], [-189+24]] = (1/-33)[[-33], [-165]] = [[1], [5]].

Solution set = {(1, 5)}.

Question: Exercise 19 Review, Question 9: (ii) Cramer's rule

Answer: x = (27x1 - 5x12)/(-33) = (-33)/(-33) = 1, y = (2x12 - 27x7)/(-33)

= (-165)/(-33) = 5.

Solution set = {(1, 5)}.

# 20. پ درجي مساواتن جو نظريو

Question: Exercise 20.1, Question 1: Using the discriminant, find the nature of the roots of the following equations. (i) $x^2 + 5x - 6 = 0$

Answer: $\Delta = 25 - 4(1)(-6) = 49 > 0$ and a perfect square, so the roots are real, rational and unequal.

Question: Exercise 20.1, Question 1: (ii) $x^2 = 6x$

Answer: $x^2 - 6x = 0$, $\Delta = 36 - 0 = 36 > 0$ and a perfect square, so the roots are real, rational and unequal (0 and 6).

Question: Exercise 20.1, Question 1: (iii) $2x^2 + 8 = 6x$

Answer: $2x^2 - 6x + 8 = 0$, $\Delta = 36 - 64 = -28 < 0$, so the roots are imaginary (non-real).

Question: Exercise 20.1, Question 1: (iv) $1 + 6x + 9x^2 = 0$

Answer: $9x^2 + 6x + 1 = 0$, $\Delta = 36 - 36 = 0$, so the roots are real, rational and equal (-1/3).

Question: Exercise 20.1, Question 1: (v) $24x^2 + 12x + 36 = 0$

Answer: $\Delta = 144 - 3456 = -3312 < 0$, so the roots are imaginary (non-real).

Question: Exercise 20.1, Question 1: (vi) $x = x^2 + 1$

Answer: $x^2 - x + 1 = 0$, $\Delta = 1 - 4 = -3 < 0$, so the roots are imaginary (non-real).

Question: Exercise 20.1, Question 1: (vii) $3x^2 + 6 = 5x$

Answer: $3x^2 - 5x + 6 = 0$, $\Delta = 25 - 72 = -47 < 0$, so the roots are imaginary (non-real).

Question: Exercise 20.1, Question 1: (viii) $3x^2 + 9 = 0$

Answer: $\Delta = 0 - 108 = -108 < 0$, so the roots are imaginary (non-real).

Question: Exercise 20.1, Question 2: Find the real value of k for which the

following quadratic equations have (i) equal real roots (ii) distinct real roots. (i) $x^2 - 3x + k = 0$

Answer: $\Delta = 9 - 4k$.

(i) For equal roots $\Delta = 0$, i.e. $k = 9/4$.

(ii) For distinct real roots $\Delta > 0$, i.e. $k < 9/4$.

Question: Exercise 20.1, Question 2: (ii) $x^2 + k = 4$

Answer: $x^2 + (k - 4) = 0$, $\Delta = 16 - 4k$.

(i) For equal roots $k = 4$.

(ii) For distinct real roots $k < 4$.

Question: Exercise 20.1, Question 2: (iii) $x^2 + kx + 2 = 0$

Answer: $\Delta = k^2 - 8$.

(i) For equal roots $k^2 = 8$, i.e. $k = \pm 2\sqrt{2}$.

(ii) For distinct real roots $k^2 > 8$, i.e. $k > 2\sqrt{2}$ or $k < -2\sqrt{2}$.

Question: Exercise 20.1, Question 2: (iv) $(k-1)x^2 = 4x + 2$

Answer: $(k-1)x^2 - 4x - 2 = 0$, $\Delta = 16 + 8(k-1) = 8k + 8$.

(i) For equal roots $k = -1$.

(ii) For distinct real roots $k > -1$ (and $k \neq 1$ for it to be quadratic).

Question: Exercise 20.1, Question 2: (v) $x^2 + kx + 4 = 0$

Answer: $\Delta = k^2 - 16$.

(i) For equal roots k = ±4.

(ii) For distinct real roots k > 4 or k < -4.

Question: Exercise 20.1, Question 2: (vi) $9x^2 + kx = -16$

Answer: $9x^2 + kx + 16 = 0$, $\Delta = k^2 - 576$.

(i) For equal roots k = ±24.

(ii) For distinct real roots k > 24 or k < -24.

Question: Exercise 20.1, Question 2: (vii) $(k-2)x^2 = 4x + (k+2)$

Answer: $(k-2)x^2 - 4x - (k+2) = 0$, $\Delta = 16 + 4(k^2-4) = 4k^2$.

(i) For equal roots k = 0.

(ii) For distinct real roots k ≠ 0 (and k ≠ 2 for it to be quadratic).

Question: Exercise 20.1, Question 2: (viii) $x^2 + 1 = kx$

Answer: $x^2 - kx + 1 = 0$, $\Delta = k^2 - 4$.

(i) For equal roots k = ±2.

(ii) For distinct real roots k > 2 or k < -2.

Question: Exercise 20.1, Question 3: Find m if the roots of the given quadratic are equal. (i) $(m+1)x^2 + 2(m+3)x + (2m+3) = 0$, with the condition $m \neq -1$

Answer: $\Delta = 4(m+3)^2 - 4(m+1)(2m+3) = 4(-m^2+m+6)$.

From $\Delta = 0$, $m^2 - m - 6 = 0$, i.e. $(m-3)(m+2) = 0$, so m = 3 or m = -2.

Question: Exercise 20.1, Question 3: (ii) $9x^2 + mx + 16 = 0$

Answer: From $\Delta = m^2 - 576 = 0$, $m = \pm 24$.

Question: Exercise 20.1, Question 4: Prove that the following quadratics have real roots. (i) $x^2 - 2(k + 1/k)x + 3 = 0$, for all k in R - {0}

Answer: $\Delta = 4(k+1/k)^2 - 12 = 4[(k+1/k)^2 - 3]$.

For $k \neq 0$, $|k+1/k| \geq 2$, i.e. $(k+1/k)^2 \geq 4$, so $\Delta \geq 4 > 0$, so the roots are real.

Question: Exercise 20.1, Question 4: (ii) $2nx^2 + 2(l+m+n)x + (l+m) = 0$, for all l, m, n in R, $m \neq 0$

Answer: $\Delta = 4(l+m+n)^2 - 8n(l+m) = 4[(l+m)^2 + n^2] \geq 0$, so the roots are real.

Question: Exercise 20.1, Question 5: Prove that the following quadratics have rational roots. (i) $(l-m)x^2 + (m+n-l)x - n = 0$, for all l, m, n in R, $l \neq m$

Answer: $\Delta = (m+n-l)^2 + 4n(l-m) = (l-m)^2 + 2n(l-m) + n^2 = (l-m+n)^2$, which is a perfect square, so the roots are rational.

Question: Exercise 20.1, Question 5: (ii) $(a+c-b)x^2 + 2cx + (b+c-a) = 0$, for all a, b, c in R

Answer: $\Delta = 4c^2 - 4(a+c-b)(b+c-a) = 4c^2 - 4[c^2 - (a-b)^2] = 4(a-b)^2 = [2(a-b)]^2$, which is a perfect square, so the roots are rational.

Question: Exercise 20.1, Question 6: For which values of p and q will the quadratic $x^2 + (2p-4)x - (3q+5) = 0$ have equal roots?

Answer: From $\Delta = (2p-4)^2 + 4(3q+5) = 0$, $4(p-2)^2 + 12q + 20 = 0$, i.e. $q = -((p-2)^2+5)/3$.

For every real p, this value of q gives equal roots.

Question: Exercise 20.1, Question 7: Prove that the given quadratic $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ has real roots and that they are equal when $a = b = c$.

Answer: Expanding: $3x^2 - 2(a+b+c)x + (ab+bc+ca) = 0$.

$\Delta = 4(a+b+c)^2 - 12(ab+bc+ca) = 2[(a-b)^2+(b-c)^2+(c-a)^2] \geq 0$, so the roots are real.

$\Delta = 0$ only when $a = b = c$, when the roots are equal.

Question: Exercise 20.2, Question 1: Find all three cube roots of the following. (i) 64

Answer: $4, 4\omega, 4\omega^2$.

Question: Exercise 20.2, Question 1: (ii) -125

Answer: $-5, -5\omega, -5\omega^2$ (because $x^3 + 125 = 0$ gives $(x+5)(x^2-5x+25) = 0$).

Question: Exercise 20.2, Question 1: (iii) 216

Answer: $6, 6\omega, 6\omega^2$.

Question: Exercise 20.2, Question 2: Find the following. (i) $(1+\omega^2)^4$

Answer: $1+\omega^2 = -\omega$, so $(1+\omega^2)^4 = (-\omega)^4 = \omega^4 = \omega^3 \cdot \omega = \omega$.

Question: Exercise 20.2, Question 2: (ii) $(1-\omega+\omega^2)(1+\omega-\omega^2)$

Answer: $1-\omega+\omega^2 = -2\omega$ and $1+\omega-\omega^2 = -2\omega^2$ (because $1+\omega+\omega^2 = 0$), so the product $= 4\omega^3 = 4$.

Question: Exercise 20.2, Question 2: (iii) $(2+5\omega+2\omega^2)^6$

Answer: $2+5\omega+2\omega^2 = 2(1+\omega^2)+5\omega = -2\omega+5\omega = 3\omega$, so $(3\omega)^6 = 729\omega^6 = 729$.

Question: Exercise 20.2, Question 2: (iv) $(-1+\sqrt{-3})^4 + (-1-\sqrt{-3})^4$

Answer: $-1+\sqrt{-3} = -1+i\sqrt{3} = 2\omega$ and $-1-\sqrt{-3} = 2\omega^2$, so $(2\omega)^4 + (2\omega^2)^4 = 16\omega + 16\omega^2 = 16(\omega+\omega^2) = -16$.

Question: Exercise 20.2, Question 3: Prove: (i) $(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8) = (\omega+\omega^2)^4$

Answer: Left side: $(1+\omega) = -\omega^2$, $(1+\omega^2) = -\omega$, $(1+\omega^4) = (1+\omega) = -\omega^2$, $(1+\omega^8) = (1+\omega^2) = -\omega$, product $= \omega^6 = 1$.

Right side: $(\omega+\omega^2)^4 = (-1)^4 = 1$.

Both are equal.

Question: Exercise 20.2, Question 3: (ii) $(a+b)(a\omega+b\omega^2)(a\omega^2+b\omega) = a^3+b^3$

Answer: $(a\omega+b\omega^2)(a\omega^2+b\omega) = a^2+b^2+ab(\omega+\omega^2) = a^2-ab+b^2$ (because $\omega+\omega^2 = -1$, $\omega^3 = 1$).

$(a+b)(a^2-ab+b^2) = a^3+b^3$.

Question: Exercise 20.2, Question 3: (iii) $(a+\omega b+\omega^2 c)(a+\omega^2 b+\omega c) = a^2+b^2+c^2-ab-bc-ca$

Answer: Expanding: $a^2+b^2+c^2+ab(\omega+\omega^2)+bc(\omega^2+\omega)+ca(\omega+\omega^2) = a^2+b^2+c^2-ab-bc-ca$ (because $\omega+\omega^2 = -1$).

[The middle term in the printed question appears to read ba instead of bc — the correct identity is given here.]

Question: Exercise 20.2, Question 3: (iv) $((-1+i\sqrt{3})/2)^9 + ((-1-i\sqrt{3})/2)^9 - 2 = 0$

Answer: $(-1+i\sqrt{3})/2 = \omega$ and $(-1-i\sqrt{3})/2 = \omega^2$, so $\omega^9 + (\omega^2)^9 - 2 = 1 + 1 - 2 = 0$ (because $\omega^3 = 1$).

Question: Exercise 20.2, Question 3: (v) $(1+\omega-\omega^2)^3 - (1-\omega+\omega^2)^3 = 0$

Answer: $1+\omega-\omega^2 = -2\omega^2$ and $1-\omega+\omega^2 = -2\omega$, so $(-2\omega^2)^3 - (-2\omega)^3 = -8\omega^6 + 8\omega^3 = -8 + 8 = 0$.

Question: Exercise 20.3, Question 1: Solve and find the sum and product of the roots of each of the following quadratics. (i) $x^2 - x - 1 = 0$

Answer: Sum of roots = 1, product of roots = -1.

Question: Exercise 20.3, Question 1: (ii) $2x^2 - 3x - 4 = 0$

Answer: Sum of roots = 3/2, product of roots = -2.

Question: Exercise 20.3, Question 1: (iii) $x^2 - (3/4)a^2 = ax$

Answer: $x^2 - ax - (3/4)a^2 = 0$, sum of roots = a, product of roots = $-3a^2/4$.

Question: Exercise 20.3, Question 1: (iv) $7x^2 - 5kx + 7k = 0$

Answer: Sum of roots = 5k/7, product of roots = k.

Question: Exercise 20.3, Question 2: Find m if: (i) the sum of the roots of $x^2 + (3m-7)x + 5m = 0$ is 3/2 times its product

Answer: Sum of roots = 7-3m, product = 5m.

From 7-3m = (3/2)(5m), 14-6m = 15m, i.e. m = 2/3.

Question: Exercise 20.3, Question 2: (ii) the sum of the roots of $2x^2 - 3x + 4m = 0$ is 6 times its product

Answer: Sum of roots = 3/2, product = 2m.

From 3/2 = 6(2m) = 12m, m = 1/8.

Question: Exercise 20.3, Question 3: Find p if: (i) the sum of the squares of the roots of $x^2 - px + 6 = 0$ is 13

Answer: $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = p^2 - 12 = 13$, so $p^2 = 25$, i.e. $p = \pm 5$.

Question: Exercise 20.3, Question 3: (ii) the sum of the squares of the roots of $x^2 - 2px + (2p-3) = 0$ is 6

Answer: $\alpha^2+\beta^2 = (2p)^2 - 2(2p-3) = 4p^2 - 4p + 6 = 6$, so $4p(p-1) = 0$, i.e. p = 0 or p = 1.

Question: Exercise 20.3, Question 4: Find m if: (i) the roots of $x^2 - 5x + 2m = 0$ differ by 1

Answer: $(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta = 25 - 8m = 1$, so 8m = 24, i.e. m = 3.

Question: Exercise 20.3, Question 4: (ii) the roots of $x^2 - 8x + m + 2 = 0$ differ by 2

Answer: $(\alpha-\beta)^2 = 64 - 4(m+2) = 4$, so 4m = 52, i.e. m = 13.

Question: Exercise 20.3, Question 5: Find k if: (i) the roots of $5x^2 - 7x + k - 2 = 0$ satisfy the relation $2\alpha + 5\beta = 1$

Answer: $\alpha+\beta = 7/5$.

Solving with $2\alpha+5\beta = 1$: $\beta = -3/5$, $\alpha = 2$.

From $\alpha\beta = (k-2)/5 = -6/5$, $k = -4$.

Question: Exercise 20.3, Question 5: (ii) the roots of $3x^2 - 2x + 7k + 2 = 0$ satisfy the relation $7\alpha - 3\beta = 18$

Answer: $\alpha+\beta = 2/3$.

Solving with $7\alpha-3\beta = 18$: $\beta = -4/3$, $\alpha = 2$.

From $\alpha\beta = (7k+2)/3 = -8/3$, $k = -10/7$.

Question: Exercise 20.3, Question 6: Find p if in the given quadratic the sum of the roots equals 4 times the product of the roots. (i) $(2p+3)x^2 + (7p-5)x + (3p-10) = 0$

Answer: From $-(7p-5)/(2p+3) = 4(3p-10)/(2p+3)$, $-7p+5 = 12p-40$, i.e. $p = 45/19$.

Question: Exercise 20.3, Question 6: (ii) $4x^2 - (5p+3)x + 17 - 9p = 0$

Answer: From $(5p+3)/4 = 4(17-9p)/4$, $5p+3 = 68-36p$, i.e. $p = 65/41$.

Question: Exercise 20.4, Question 1: Given $\alpha$ and $\beta$ are the roots of a quadratic, express the following symmetric functions in terms of $\alpha+\beta$ and $\alpha\beta$. (i) $(\alpha-\beta)^2$

Answer: $(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$.

Question: Exercise 20.4, Question 1: (ii) $(\alpha+\beta)^3$

Answer: $(\alpha+\beta)^3$ (already in terms of the sum).

Question: Exercise 20.4, Question 1: (iii) $\alpha^2\beta^{-1} + \beta^2\alpha^{-1}$

Answer: $\alpha^2/\beta + \beta^2/\alpha = (\alpha^3+\beta^3)/(\alpha\beta) = [(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)]/(\alpha\beta)$.

Question: Exercise 20.4, Question 1: (iv) $\alpha^3\beta + \alpha\beta^3$

Answer: $\alpha\beta(\alpha^2+\beta^2) = \alpha\beta[(\alpha+\beta)^2 - 2\alpha\beta]$.

Question: Exercise 20.4, Question 1: (v) $1/\alpha^3 + 1/\beta^3$

Answer: $(\alpha^3+\beta^3)/(\alpha\beta)^3 = [(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)]/(\alpha\beta)^3$.

Question: Exercise 20.4, Question 1: (vi) $(1/\alpha - 1/\beta)^2$

Answer: $(\beta-\alpha)^2/(\alpha\beta)^2 = [(\alpha+\beta)^2 - 4\alpha\beta]/(\alpha\beta)^2$.

Question: Exercise 20.4, Question 2: Given α and β are the roots of $2x^2 - 3x + 7 = 0$, find the values of the following symmetric functions. (i) $(1/\alpha + 1/\beta)^2$

Answer: $\alpha+\beta = 3/2$, $\alpha\beta = 7/2$.

$(1/\alpha+1/\beta)^2 = ((\alpha+\beta)/(\alpha\beta))^2 = (3/7)^2 = 9/49$.

Question: Exercise 20.4, Question 2: (ii) $\alpha/\beta^2 + \beta/\alpha^2$

Answer: $(\alpha^3+\beta^3)/(\alpha\beta)^2 = [(27/8) - 3(7/2)(3/2)]/(49/4) = (-99/8)/(49/4) = -99/98$.

Question: Exercise 20.4, Question 2: (iii) $1/(\alpha^2+1) + 1/(\beta^2+1)$

Answer: $(\alpha^2+\beta^2+2)/(\alpha^2\beta^2+\alpha^2+\beta^2+1) = (-19/4+2)/(49/4-19/4+1) = (-11/4)/(17/2) = -11/34$.

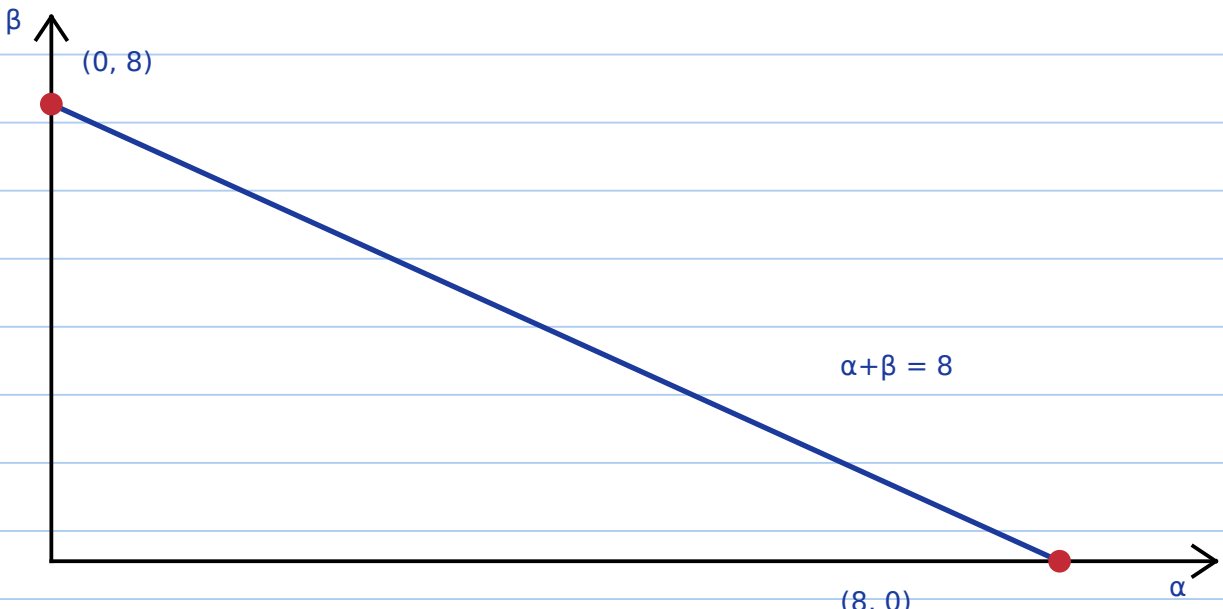
Question: Exercise 20.4, Question 3: Given α and β are the roots of $px^2 + qx + q = 0$, $p \neq 0$, find the value of $\sqrt{(\alpha/\beta)} + \sqrt{(\beta/\alpha)}$.

Answer: $\alpha+\beta = -q/p$, $\alpha\beta = q/p$. $\sqrt{(\alpha/\beta)} + \sqrt{(\beta/\alpha)} = (\alpha+\beta)/\sqrt{(\alpha\beta)} = (-q/p)/\sqrt{(q/p)} = -\sqrt{(q/p)}$.

Question: Exercise 20.4, Question 4: Show the symmetric equation $\alpha+\beta = 8$ graphically, where α and β are the roots of a quadratic.

Answer: In the (α, β) plane, $\alpha+\beta = 8$ is the straight line through (0, 8) and (8, 0).

Every point (α, β) on it gives a quadratic $x^2 - 8x + \alpha\beta = 0$.




Plotted from the values given in the question/answer.

Question: Exercise 20.5, Question 1: Form equations whose roots are: (i) 2, -3

Answer: Sum = -1, product = -6, equation: $x^2 + x - 6 = 0$.

Question: Exercise 20.5, Question 1: (ii) ω, $\omega^2$

Answer: Sum = -1, product = 1, equation: $x^2 + x + 1 = 0$.

Question: Exercise 20.5, Question 1: (iii) 2+i, 2-i

Answer: Sum = 4, product = 5, equation: $x^2 - 4x + 5 = 0$.

Question: Exercise 20.5, Question 1: (iv) $2\sqrt{2}$, $-2\sqrt{2}$

Answer: Sum = 0, product = -8, equation: $x^2 - 8 = 0$.

Question: Exercise 20.5, Question 2: Given α and β are the roots of $6x^2 - 3x + 1 = 0$, form the equations whose roots are as follows. (i) 2α+1, 2β+1

Answer: New sum = 3, new product = 8/3, equation: $3x^2 - 9x + 8 = 0$.

Question: Exercise 20.5, Question 2: (ii) $\alpha^2$, $\beta^2$

Answer: New sum = -1/12, new product = 1/36, equation: $36x^2 + 3x + 1 = 0$.

Question: Exercise 20.5, Question 2: (iii) 1/α, 1/β

Answer: New sum = 3, new product = 6, equation: $x^2 - 3x + 6 = 0$.

Question: Exercise 20.5, Question 2: (iv) α/β, β/α

Answer: New sum = -1/2, new product = 1, equation: $2x^2 + x + 2 = 0$.

Question: Exercise 20.5, Question 2: (v) $\alpha+\beta$, $1/\alpha+1/\beta$

Answer: New sum = 7/2, new product = 3/2, equation: $2x^2 - 7x + 3 = 0$.

Question: Exercise 20.5, Question 3: Find the equation whose roots are the reciprocals of the roots of $px^2 - qx + r = 0$, $p \neq 0$.

Answer: Sum of reciprocals = q/r, product of reciprocals = p/r, equation: $rx^2 - qx + p = 0$.

Question: Exercise 20.5, Question 4: Find the equation whose roots are twice the roots of $x^2 - px + q = 0$.

Answer: Sum of doubles = 2p, product of doubles = 4q, equation: $x^2 - 2px + 4q = 0$.

Question: Exercise 20.5, Question 5: Find the equation whose roots are 2 more than the roots of $px^2 + qx + r = 0$.

Answer: New sum = (4p-q)/p, new product = (4p-2q+r)/p, equation: $px^2 - (4p-q)x + (4p-2q+r) = 0$.

Question: Exercise 20.5, Question 6: Find the condition that one root of $ax^2 + bx + c = 0$, $a \neq 0$, is: (i) the negative of the other

Answer: $\beta = -\alpha$, i.e. sum of roots = 0, so the condition: b = 0.

Question: Exercise 20.5, Question 6: (ii) the square of the other

Answer: From $\beta = \alpha^2$, $\alpha^3 = c/a$ and $\alpha^2+\alpha = -b/a$.

Eliminating α from these gives the condition: $b^3 + ac^2 - 3abc + a^2c = 0$.

Question: Exercise 20.5, Question 6: (iii) the additive inverse of the other

Answer: $\beta = -\alpha$, i.e. sum of roots = 0, so the condition: b = 0.

(This is the same condition as in (i).)

Question: Exercise 20.5, Question 6: (iv) the multiplicative inverse of the other

Answer: $\beta = 1/\alpha$, i.e. product of roots = 1, so the condition: $c = a$.

Question: Exercise 20.6, Question 1: Find the remaining roots of the following cubic equations, given one root. (i) $2x^3 - x^2 - 2x + 1 = 0$, $x = 1$

Answer: By synthetic division the quotient is $2x^2 + x - 1 = 0$, i.e. $(2x-1)(x+1) = 0$.

Remaining roots: 1/2 and -1.

Question: Exercise 20.6, Question 1: (ii) $x^3 - 4x^2 + x + 6 = 0$, $x = 3$

Answer: By synthetic division the quotient is $x^2 - x - 2 = 0$, i.e. $(x-2)(x+1) = 0$.

Remaining roots: 2 and -1.

Question: Exercise 20.6, Question 1: (iii) $x^3 - 28x + 48 = 0$, $x = 2$

Answer: By synthetic division the quotient is $x^2 + 2x - 24 = 0$, i.e. $(x+6)(x-4) = 0$.

Remaining roots: -6 and 4.

Question: Exercise 20.6, Question 2: Find the remaining two roots of the following biquadratic equations, given two roots. (i) $12x^4 - 8x^3 - 7x^2 + 2x + 1 = 0$, $x = 1, -1/2$

Answer: After two synthetic divisions the quotient is $12x^2 - 2x - 2 = 0$, i.e. $6x^2 - x - 1 = 0$, i.e. $(3x+1)(2x-1) = 0$.

Remaining roots: -1/3 and 1/2.

Question: Exercise 20.6, Question 2: (ii) $x^4 + 4x^2 - 5 = 0$, $x = 1, -1$

Answer: Setting $x^2 = y$: $y^2 + 4y - 5 = 0$, i.e. $(y+5)(y-1) = 0$.

From $y = 1$, $x = \pm 1$ (given);

from $y = -5$, $x = \pm i\sqrt{5}$.

Remaining roots: $i\sqrt{5}$ and $-i\sqrt{5}$.

Question: Exercise 20.6, Question 2: (iii) $x^4 + 2x^3 - 13x^2 - 14x + 24 = 0$, $x = 3, -4$

Answer: After two synthetic divisions the quotient is $x^2 + x - 2 = 0$, i.e. $(x+2)(x-1) = 0$.

Remaining roots: -2 and 1.

Question: Exercise 20.6, Question 3: Find m if 3 is a root of $2x^3 - 3mx^2 + 9 = 0$, and find the remaining roots.

Answer: Setting $x = 3$: $54 - 27m + 9 = 0$, i.e. $m = 7/3$.

The quotient: $2x^2 - x - 3 = 0$, i.e. $(2x-3)(x+1) = 0$.

Remaining roots: 3/2 and -1.

Question: Exercise 20.6, Question 4: If 1 is a root of $x^3 - 3ax^2 - x + 6 = 0$, find the value of a and the other two roots.

Answer: Setting $x = 1$: $1 - 3a - 1 + 6 = 0$, i.e. $a = 2$.

The quotient: $x^2 - 5x - 6 = 0$, i.e. $(x-6)(x+1) = 0$.

Remaining roots: 6 and -1.

Question: Exercise 20.6, Question 5: Find the values of a and b if 6 and -2 are the roots of $x^4 - ax^2 + bx + 252 = 0$, and find the remaining two roots.

Answer: Setting $x = 6$: $6a - b = 258$.

Setting $x = -2$: $2a + b = 134$.

Solving: a = 49, b = 36.

The quotient: $x^2 + 4x - 21 = 0$, i.e. $(x+7)(x-3) = 0$.

Remaining roots: -7 and 3.

Question: Exercise 20.7, Question 1: Solve the following simultaneous equations. $2x - y = 3$ and $x^2 + y^2 = 2$

Answer: Substituting $y = 2x - 3$: $5x^2 - 12x + 7 = 0$, i.e. $(5x-7)(x-1) = 0$.

For x = 7/5, y = -1/5;

for x = 1, y = -1.

Solution set = {(7/5, -1/5), (1, -1)}.

Question: Exercise 20.7, Question 2: $2x + y = 4$ and $x^2 - 2x + y^2 = 0$

Answer: Substituting $y = 4 - 2x$: $5x^2 - 18x + 16 = 0$, i.e. $(5x-8)(x-2) = 0$.

For x = 8/5, y = 4/5;

for x = 2, y = 0.

Solution set = {(8/5, 4/5), (2, 0)}.

Question: Exercise 20.7, Question 3: $4/x + 3/y = 2$ and $4x + 3y = 25$

Answer: Substituting $x = (25-3y)/4$: $6y^2 - 43y + 75 = 0$, i.e. $(3y-5)(2y-15) = 0$.

For y = 5/3, x = 5;

for y = 15/2, x = 5/8.

Solution set = {(5, 5/3), (5/8, 15/2)}.

Question: Exercise 20.7, Question 4: $(x-1)^2 + (y+3)^2 = 25$ and $x^2 + (y+1)^2 = 10$

Answer: Expanding and subtracting: x = 2y - 3.

Substituting: $5y^2 - 10y - 1 = 0$, i.e. $y = 1 \pm \sqrt{30}/5$.

Solution set = $\{(-1+2\sqrt{30}/5, 1+\sqrt{30}/5), (-1-2\sqrt{30}/5, 1-\sqrt{30}/5)\}$.

Question: Exercise 20.7, Question 5: $x^2 + y^2 = 25$ and $(4x-3y)(x-y-5) = 0$

Answer: Case (a) 4x-3y = 0 gives (3, 4) and (-3, -4).

Case (b) x-y-5 = 0 gives (5, 0) and (0, -5).

Solution set = {(3, 4), (-3, -4), (5, 0), (0, -5)}.

Question: Exercise 20.7, Question 6: $x^2 + y^2 = 16$ and $2x^2 - 3xy + y^2 = 0$

Answer: From the second equation: (2x-y)(x-y) = 0.

From y = 2x: $(\pm 4\sqrt{5}/5, \pm 8\sqrt{5}/5)$.

From y = x: $(\pm 2\sqrt{2}, \pm 2\sqrt{2})$.

Solution set = $\{(4\sqrt{5}/5, 8\sqrt{5}/5), (-4\sqrt{5}/5, -8\sqrt{5}/5), (2\sqrt{2}, 2\sqrt{2}), (-2\sqrt{2}, -2\sqrt{2})\}$.

Question: Exercise 20.7, Question 7: $x^2 + y^2 = 5$ and $xy = 2$

Answer: From $(x+y)^2 = 9$ and $(x-y)^2 = 1$: solution set = {(2, 1), (1, 2), (-2, -1), (-1, -2)}.

Question: Exercise 20.7, Question 8: $x^2 + xy = 5$ and $x^2 - 2xy = 2$

Answer: Subtracting: xy = 1.

From the first: $x^2 = 4$, i.e. $x = \pm 2$, $y = \pm 1/2$ (with matching signs).

Solution set = {(2, 1/2), (-2, -1/2)}.

Question: Exercise 20.7, Question 9: $x + 4/y = 1$ and $y + 4/x = 25$

Answer: Substituting: $y^2 - 25y + 100 = 0$, i.e. $(y-20)(y-5) = 0$.

For y = 20, x = 4/5;

for y = 5, x = 1/5.

Solution set = {(4/5, 20), (1/5, 5)}.

Question: Exercise 20.7, Question 10: Divide 12 into two parts such that the sum of their squares is 4 more than their product.

Answer: Let the parts be x and 12-x;

then $3x^2 - 36x + 140 = 0$, whose discriminant is $-384 < 0$, so as printed there are no real parts.

[The printed question has no real solution — noted.]

Question: Exercise 20.7, Question 11: The length of a prayer place is 4 metres more than its width; if its area is 36 $m^2$, find its length and width.

Answer: Let the width be w;

then $w(w+4) = 36$, i.e. $w^2 + 4w - 36 = 0$, so $w = 2(\sqrt{10} - 1) \approx 4.32$ m and the length = $2(\sqrt{10} + 1) \approx 8.32$ m.

Question: Exercise 20.7, Question 12: The sum of the squares of two

numbers is 100. One number is 2 more than the other; find the numbers.

Answer: From $x^2 + (x+2)^2 = 100$, $x^2 + 2x - 48 = 0$, i.e. $(x+8)(x-6) = 0$.

The numbers: (6, 8) or (-8, -6).

Question: Exercise 20.7, Question 13: The base of a right-angled triangle is 3 more than its perpendicular. If the hypotenuse of the triangle is 34, find the base and the perpendicular.

Answer: Let the perpendicular be p;

from $p^2 + (p+3)^2 = 34^2$, $2p^2 + 6p - 1147 = 0$, i.e. $p = (\sqrt{2303} - 3)/2 \approx 22.49$ and the base $= (\sqrt{2303} + 3)/2 \approx 25.49$.

Question: Exercise 20.7, Question 14: The altitude of an isosceles triangle is 12 cm, the altitude of the unequal side of the triangle is 12 cm; find the lengths of those sides of the triangle.

Answer: As printed, both given lengths in the question are 12 cm and no other number is given, so the sides of the triangle cannot be determined — the question is not solvable in its printed form. [The data given in the printed question is insufficient — noted.]

Question: Exercise 20.7, Question 15: The difference of two numbers is 5 and the difference of their squares is 275; find the numbers.

Answer: From $x - y = 5$ and $x^2 - y^2 = 275$, $x + y = 55$.

Solving: $x = 30$, $y = 25$.

Exercise 20 Review, Question 1: Circle the correct answer. (i) If the roots of $2x^2 + 5x + 3 = 0$ are p and q then $p + q =$ ___.

(a) 5/3

(b) 3/5

(c) 5/2

(d) -5/2

Exercise 20 Review, Question 1: (ii) If the roots of $ax^2 + bx + c = 0$, $a \neq 0$, are $1/\alpha$ and $1/\beta$ then $\alpha + \beta =$ ___.

(a) -b/a

(b) b/c

✓ (c) -b/c

(d) -c/b

Exercise 20 Review, Question 1: (iii) If the sum of the roots of $(p+1)x^2 + (2p+3)x + (3p+4) = 0$ is -1 then the product of the roots is ___.

(a) 0

(b) 1

✓ (c) 2

(d) 3

Exercise 20 Review, Question 1: (iv) The roots of $ax^2 + bx + c = 0$, $a \neq 0$, are obtained by ___.

(a) sum of roots

(b) product of roots

✓ (c) discriminant

(d) none

Exercise 20 Review, Question 1: (v) If the sum of the roots of a quadratic is b/a and the product of the roots is c/a then the equation is ___.

(a) $ax^2 + bx + c = 0$

(b) $ax^2 + bx - c = 0$

✓ (c) $ax^2 - bx + c = 0$

(d) $ax^2 - bx - c = 0$

Exercise 20 Review, Question 1: (vi) The required equation ___ has roots which are the reciprocals of the roots of $ax^2 + bx + c = 0$.

(a) $ax^2 + bx + c = 0$

✓

(b) $cx^2 + bx + a = 0$

(c) $cx^2 + ax + b = 0$

(d) $cx^2 - bx - a = 0$

Exercise 20 Review, Question 1: (vii) If $\alpha$ and $\beta$ are the roots of $x^2 - 2x - 15 = 0$ then $\alpha^2 + \beta^2 =$ ___.

✓ (a) 34

(b) -34

(c) 26

(d) -26

Exercise 20 Review, Question 1: (viii) If the $\Delta = b^2 - 4ac$ of a quadratic with real coefficients is a perfect square then the type of roots is ___.

(a) real, equal

✓ (b) real, rational, unequal

(c) real, irrational, unequal

(d) imaginary

Exercise 20 Review, Question 1: (ix) If one root of a quadratic is $2 + \sqrt{3}$ then the other root will be ___.

(a) 2

(b) $-2 + \sqrt{3}$

✓ (c) $2 - \sqrt{3}$

(d) $-2 - \sqrt{3}$

Exercise 20 Review, Question 1: (x) The quadratic ___ whose roots are the complex cube roots of unity.

(a) $x^2 - x - 1 = 0$

(b) $x^2 - x + 1 = 0$

✓ (c) $x^2 + x + 1 = 0$

(d) $x^2 + x - 1 = 0$

Question: Exercise 20 Review, Question 2: Find the nature of the roots of

✓

the following quadratics. (i) $x^2 - 7x + 12 = 0$

Answer: $\Delta = 49 - 48 = 1 > 0$ and a perfect square, so the roots are real, rational and unequal.

Question: Exercise 20 Review, Question 2: (ii) $x^2 - 14x + 49 = 0$

Answer: $\Delta = 196 - 196 = 0$, so the roots are real, rational and equal ($x = 7$).

Question: Exercise 20 Review, Question 2: (iii) $x^2 - x + 7 = 0$

Answer: $\Delta = 1 - 28 = -27 < 0$, so the roots are imaginary (non-real).

Question: Exercise 20 Review, Question 2: (iv) $x^2 - 5 = 0$

Answer: $\Delta = 20 > 0$ and not a perfect square, so the roots are real, irrational and unequal ($\pm\sqrt{5}$).

Question: Exercise 20 Review, Question 3: For what value of k does $x^2 + kx + 4 = 0$ have: (i) equal roots

Answer: From $\Delta = k^2 - 16 = 0$, $k = \pm 4$.

Question: Exercise 20 Review, Question 3: (ii) imaginary roots

Answer: From $\Delta < 0$, $-4 < k < 4$.

Question: Exercise 20 Review, Question 3: (iii) real roots

Answer: From $\Delta \geq 0$, $k \leq -4$ or $k \geq 4$.

Question: Exercise 20 Review, Question 3: (iv) rational roots

Answer: $\Delta = k^2 - 16$ must be a perfect square, for example $k = \pm 4$ ($\Delta = 0$) or $k = \pm 5$ ($\Delta = 9$).

Question: Exercise 20 Review, Question 4: Find the cube roots of 729.

Answer: $9, 9\omega, 9\omega^2$.

Question: Exercise 20 Review, Question 5: Find the sum and product of the roots of the following quadratics. (i) $x^2 - 7x + 29 = 0$

Answer: Sum of roots = 7, product of roots = 29.

Question: Exercise 20 Review, Question 5: (ii) $x^2 - px + q = 0$

Answer: Sum of roots = p, product of roots = q.

Question: Exercise 20 Review, Question 5: (iii) $7x - 8 = 5x^2$

Answer: $5x^2 - 7x + 8 = 0$, sum of roots = 7/5, product of roots = 8/5.

Question: Exercise 20 Review, Question 5: (iv) $11x = 9x^2 - 28$

Answer: $9x^2 - 11x - 28 = 0$, sum of roots = 11/9, product of roots = -28/9.

Question: Exercise 20 Review, Question 6: Define the symmetric function of the roots of a quadratic.

Answer: A symmetric function of the roots is a function unchanged when α and β are swapped, i.e. $f(\alpha, \beta) = f(\beta, \alpha)$, such as $\alpha+\beta$, $\alpha\beta$ and $\alpha^2+\beta^2$.

Question: Exercise 20 Review, Question 7: Find the quadratic whose roots are $1 + \sqrt{3}$ and $1 - \sqrt{3}$.

Answer: Sum = 2, product = -2, equation: $x^2 - 2x - 2 = 0$.

Question: Exercise 20 Review, Question 8: Find the equation whose roots are the reciprocals of the roots of $x^2 - 10x + 16 = 0$.

Answer: The reciprocal-roots equation: $16x^2 - 10x + 1 = 0$.

Question: Exercise 20 Review, Question 9: Solve each of the following systems of equations. (i) $x^2 + y^2 = 10$, $x + y = 3$

Answer: From $(x+y)^2 = 9$, $xy = -1/2$.

x and y are the roots of $t^2 - 3t - 1/2 = 0$, i.e. $t = (3\pm\sqrt{11})/2$.

Solution set = $\{((3+\sqrt{11})/2, (3-\sqrt{11})/2), ((3-\sqrt{11})/2, (3+\sqrt{11})/2)\}$.

Question: Exercise 20 Review, Question 9: (ii) $x^2 + y^2 = 37$, $2x + 5y = 32$

Answer: Substituting $x = (32-5y)/2$: $29y^2 - 320y + 876 = 0$, whose discriminant is $784 = 28^2$.

For y = 6, x = 1;

for y = 146/29, x = 99/29.

Solution set = {(1, 6), (99/29, 146/29)}.

# 21. جزوي پاگا

Question: Exercise 21.1, Question 1: Resolve the following into partial fractions. $12/(x^2 - 9)$

Answer: $12/((x-3)(x+3)) = 2/(x-3) - 2/(x+3)$ (because at $x = 3$, $12 = 6A$ and at $x = -3$, $12 = -6B$).

Question: Exercise 21.1, Question 2: $4(x - 4)/(x^2 - 2x - 3)$

Answer: $4(x-4)/((x-3)(x+1)) = -1/(x-3) + 5/(x+1)$ (because at $x = 3$, $-4 = 4A$ and at $x = -1$, $-20 = -4B$).

Question: Exercise 21.1, Question 3: $(x^2 - 3x + 6)/(x(x - 2)(x - 1))$

Answer: $3/x + 2/(x-2) - 4/(x-1)$ (because at $x = 0$, $6 = 2A$;

at $x = 2$, $4 = 2B$;

at $x = 1$, $4 = -C$).

Question: Exercise 21.1, Question 4: $3(2x^2 - 8x - 1)/((x + 4)(x - 1)(2x - 1))$

Answer: $(21/5)/(x+4) - (21/5)/(x-1) + 6/(2x-1)$ (because at $x = -4$, $189 = 45A$;

at $x = 1$, $-21 = 5B$;

at $x = 1/2$, $-27/2 = -9C/4$).

Question: Exercise 21.1, Question 5: $(x^2 + 9x + 8)/(x^2 + x - 6)$

Answer: $2/(x+3) + 6/(x-2)$ (because at $x = 2$, $30 = 5B$ and at $x = -3$, $-10 = -5A$).

Question: Exercise 21.1, Question 6: $(x^2 - x - 14)/(x^2 - 2x - 3)$

Answer: $-2/(x-3) + 3/(x+1)$ (because at $x = 3$, $-8 = 4A$ and at $x = -1$, $-12 =$

-4B).

Question: Exercise 21.1, Question 7: $(3x^3 - 2x^2 - 16x + 20)/((x - 2)(x + 2))$

Answer: Dividing first: $3x - 2 + (-4x+12)/((x-2)(x+2)) = 3x - 2 + 1/(x-2) - 5/(x+2)$.

Question: Exercise 21.2, Question 1: Resolve the following into partial fractions. $(4x - 3)/(x + 1)^2$

Answer: $4/(x+1) - 7/(x+1)^2$ (because from $4x - 3 = A(x+1) + B$, $A = 4$, $B = -7$).

Question: Exercise 21.2, Question 2: $(x^2 + 7x + 3)/(x^2(x + 3))$

Answer: $2/x + 1/x^2 - 1/(x+3)$ (because at $x = 0$, $B = 1$;

at $x = -3$, $C = -1$;

comparing $x^2$ gives $A = 2$).

Question: Exercise 21.2, Question 3: $(5x^2 - 30x + 44)/(x - 2)^3$

Answer: $5/(x-2) - 10/(x-2)^2 + 4/(x-2)^3$ (because at $x = 2$, $C = 4$;

comparing $x^2$ gives $A = 5$;

comparing $x$ gives $B = -10$).

Question: Exercise 21.2, Question 4: $(18 + 21x - x^2)/((x - 5)(x + 2)^2)$

Answer: $2/(x-5) - 3/(x+2) + 4/(x+2)^2$ (because at $x = 5$, $A = 2$;

at $x = -2$, $C = 4$;

comparing $x^2$ gives $B = -3$).

Question: Exercise 21.2, Question 5: $(x^2 - x + 3)/(x - 1)^3$

Answer: $1/(x-1) + 1/(x-1)^2 + 3/(x-1)^3$ (because at $x = 1$, $C = 3$;

comparing $x^2$ gives A = 1;

comparing x gives B = 1).

Question: Exercise 21.3, Question 1: Resolve the following into partial fractions. $(x^2 - x - 13)/((x^2 + 7)(x - 2))$

Answer: $(2x+3)/(x^2+7) - 1/(x-2)$ (because at x = 2, C = -1;

comparing $x^2$ gives A = 2;

comparing x gives B = 3).

Question: Exercise 21.3, Question 2: $(6x - 5)/((x^2 + 10)(x + 1))$

Answer: $(x+5)/(x^2+10) - 1/(x+1)$ (because at x = -1, C = -1;

comparing $x^2$ gives A = 1;

comparing x gives B = 5).

Question: Exercise 21.3, Question 3: $(15 + 5x + 5x^2 - 4x^3)/(x^2(x^2 + 5))$

Answer: $1/x + 3/x^2 + (-5x+2)/(x^2+5)$ (because at x = 0, B = 3;

comparing x gives A = 1;

comparing $x^2$ gives D = 2;

comparing $x^3$ gives C = -5).

Question: Exercise 21.3, Question 4: $(3x^2 - x + 1)/((x + 1)(x^2 - x + 3))$

Answer: $1/(x+1) + (2x-2)/(x^2-x+3)$ (because at x = -1, A = 1;

comparing $x^2$ gives B = 2;

comparing x gives C = -2).

Question: Exercise 21.3, Question 5: $(x^2 - x + 2)/((x + 1)(x^2 + 3))$

Answer: $1/(x+1) - 1/(x^2+3)$ (because at x = -1, A = 1;

comparing $x^2$ gives B = 0;

comparing x gives C = -1).

Question: Exercise 21.4, Question 1: Resolve the following into partial fractions. $x^2/((x^2 + 1)^2(1 - x))$

Answer: $(x+1)/(4(x^2+1)) - (x+1)/(4(x^2+1)^2) + 1/(4(1-x))$ (because at x = 1, E = 1/4, and comparing gives A = 1/4, B = 1/4, C = -1/4, D = -1/4).

Question: Exercise 21.4, Question 2: $(x^2 + x + 2)/(x^2(x^2 + 3)^2)$

Answer: $2/(9x^2) - 2/(9(x^2+3)) + (x+1)/(3(x^2+3)^2)$ (because at x = 0, B = 2/9, and comparing gives A = 0, C = 0, D = -2/9, E = 1/3, F = 1/3).

Question: Exercise 21.4, Question 3: $x^2/((1 + x)(1 + x^2)^2)$

Answer: $1/(4(1+x)) + (1-x)/(4(1+x^2)) + x/(4(1+x^2)^2)$ (because at x = -1, A = 1/4, and comparing gives B = -1/4, C = 1/4, D = 1/4, E = 0).

Question: Exercise 21.4, Question 4: $7/((x + 1)(2 + x^2)^2)$

Answer: $7/(9(x+1)) + 7(1-x)/(9(2+x^2)) + 7(1-5x)/(9(2+x^2)^2)$ (because at x = -1, A = 7/9, and comparing gives B = -7/9, C = 7/9, D = -35/9, E = 7/9).

Question: Exercise 21.4, Question 5: $49/((x - 2)(3 + x^2)^2)$

Answer: $1/(x-2) + (1-x)/(3+x^2) + (-7x-3)/(3+x^2)^2$ (because at x = 2, A = 1, and comparing gives B = -1, C = 1, D = -7, E = -3).

Exercise 21 Review, Question 1: Circle the correct answer. (i) An improper fraction can be resolved into a proper fraction by ___.

(a) addition

(b) multiplication

(c) subtraction

(d) division

Exercise 21 Review, Question 1: (ii) The partial fraction form of $x/((x-a)(x-b)(x-c))$ is ___.

(d) none

Exercise 21 Review, Question 1: (iii) The partial fractions of $(x-3)/(x^3+3x)$ are ___.

Exercise 21 Review, Question 1: (iv) $(x^3+1)/((x-1)(x+2))$ is a ___.

(a) proper fraction

✓ (b) improper fraction

(c) an identity element

(d) constant term

Exercise 21 Review, Question 1: (v) The fraction $(2x+5)/(x^2+5x+6)$ is called a ___.

✓ (a) proper fraction

(b) improper fraction

(c) both proper and improper

(d) none

Question: Exercise 21 Review, Question 2: Define proper, improper and rational fractions.

Answer: Proper fraction: degree of numerator less than degree of denominator. Improper fraction: degree of numerator equal to or greater than degree of denominator. Rational fraction: the quotient $P(x)/Q(x)$ of two polynomials, where $Q(x) \neq 0$.

Question: Exercise 21 Review, Question 3: Resolve the following into partial fractions. (i) $(5x + 8)/((x - 1)(x + 2))$

Answer: $(13/3)/(x-1) + (2/3)/(x+2)$ (because at $x = 1$, $13 = 3A$ and at $x = -2$, $-2 = -3B$).

Question: Exercise 21 Review, Question 3: (ii) $(9x^2 + 5x + 7)/(x(x + 2)(x -$

✓

5))

Answer: (-7/10)/x + (33/14)/(x+2) + (257/35)/(x-5) (because at x = 0, A = -7/10;

at x = -2, B = 33/14;

at x = 5, C = 257/35).

Question: Exercise 21 Review, Question 3: (iii) $(x^2 + 2x + 3)/((x^2 + 1)(x - 2))$

Answer: $(-(6x+2))/(5(x^2+1)) + 11/(5(x-2))$ (because at x = 2, C = 11/5;

comparing $x^2$ gives A = -6/5;

comparing x gives B = -2/5).

Question: Exercise 21 Review, Question 3: (iv) $(x^3 + 8x^2 + 9)/((x^2 + x + 1)(x + 1))$

Answer: Dividing: $1 + (6x^2-2x+8)/((x^2+x+1)(x+1)) = 1 + (-10x-8)/(x^2+x+1) + 16/(x+1)$.

Question: Exercise 21 Review, Question 3: (v) $(x + 5)/((x^2 + 1)^2(x - 3))$

Answer: $(-2x-6)/(25(x^2+1)) + (-4x-7)/(5(x^2+1)^2) + (2/25)/(x-3)$ (because at x = 3, E = 2/25, and comparing gives A = -2/25, B = -6/25, C = -4/5, D = -7/5).

Question: Exercise 21 Review, Question 3: (vi) $(7x + 3)/((x - 1)^2(x + 2))$

Answer: $(11/9)/(x-1) + (10/3)/(x-1)^2 - (11/9)/(x+2)$ (because at x = 1, B = 10/3;

at x = -2, C = -11/9;

comparing $x^2$ gives A = 11/9).

# 22. بنيادي شماريات

Question: Exercise 22.1, Question 1: Identify the type of the given data (discrete or continuous or count or categorical) and write the total number of observations. (a) The number of people entering a shopping mall for 30 days

Answer: It is count data, so it is discrete (count) data.

Total number of observations = 30.

Question: Exercise 22.1, Question 1: (b) The time spent by a person in a gym for one month (in minutes daily)

Answer: Time can take any value, so it is continuous data.

Total number of observations = 30 (daily measurements for a month).

Question: Exercise 22.1, Question 1: (c) The best age of 45 cars (in years)

Answer: Age can take any value, so it is continuous data.

Total number of observations = 45.

Question: Exercise 22.1, Question 1: (d) Electricity usage for 2 months (in Kwh)

Answer: Usage can take any value, so it is continuous data.

Total number of observations = 2 (measurements for two months).

Question: Exercise 22.1, Question 1: (e) The weight of 55 students of a class (in kilograms)

Answer: Weight can take any value, so it is continuous data.

Total number of observations = 55.

Question: Exercise 22.1, Question 1: (f) The eye colour of 20 people

Answer: Colour is a quality, so it is categorical (qualitative) data.

Total number of observations = 20.

Question: Exercise 22.1, Question 1: (g) The grades obtained by 70 students of a class

Answer: Grade is a quality, so it is categorical (qualitative) data.

Total number of observations = 70.

Question: Exercise 22.1, Question 1: (h) The results of the COVID diagnostic test of 10 people (positive or negative)

Answer: Results are a quality (positive/negative), so it is categorical (qualitative) data.

Total number of observations = 10.

Question: Exercise 22.1, Question 1: (i) The marital status of 30 teachers of a school

Answer: Status is a quality, so it is categorical (qualitative) data.

Total number of observations = 30.

Question: Exercise 22.1, Question 1: (j) The district of future residence of the students of your class

Answer: District is a quality, so it is categorical (qualitative) data. Total number of observations = the number of students in the class.

Question: Exercise 22.1, Question 2: Make a grouped frequency table of the education grades (current class of study) of 20 students of a school going on a given tour.

Answer: Table: IV: 1, V: 1, VI: 3, VII: 2, VIII: 4, IX: 4, X: 5.

Total = 20.

Question: Exercise 22.1, Question 3: The transport methods used by 20 students to attend a language-learning course are given. Make a grouped frequency table with percentage frequencies.

Answer: Table: Bus: 6 (30%), on foot: 7 (35%), motorcycle: 2 (10%), motor car: 5 (25%).

Total = 20.

Question: Exercise 22.1, Question 4: Make a grouped frequency table classifying the answers of 40 families asked how often they spend vacations outside the city. Also find the relative and percentage frequencies.

Answer: Table: Never: 8 (0.20, 20%), very rarely: 5 (0.125, 12.5%), sometimes: 10 (0.25, 25%), often: 8 (0.20, 20%), always: 9 (0.225, 22.5%).

Total = 40.

Question: Exercise 22.1, Question 5: Make a frequency table of the colours of the caps of prefects in different schools, arranged by colour.

Answer: Table: Green: 5, blue: 4, black: 4, brown: 3, khaki: 4.

Total = 20.

Question: Exercise 22.1, Question 6: The marks out of 5 obtained by 25 students in a class test are given. Make a discrete frequency table with frequencies and relative and percentage frequencies.

Answer: Table (marks: frequency, relative, percentage): 0.0: 1, 0.04, 4%;

0.5: 1, 0.04, 4%; 1.0: 3, 0.12, 12%; 1.5: 1, 0.04, 4%; 2.0: 3, 0.12, 12%; 2.5: 3, 0.12, 12%; 3.0: 2, 0.08, 8%; 3.5: 4, 0.16, 16%; 4.0: 3, 0.12, 12%; 4.5: 2, 0.08, 8%; 5.0: 2, 0.08, 8%.

Total = 25.

Question: Exercise 22.1, Question 7: The results of how many times a teacher's lectures were downloaded from a website in each class the day before a test are given. Classify the data in a discrete frequency table.

Answer: Table (downloads: frequency): 0: 2, 1: 7, 2: 7, 3: 5, 4: 3.

Total = 24.

Question: Exercise 22.1, Question 8: Make a table of the number of absent students in a class over 18 days.

Answer: Table (absent: frequency): 0: 1, 1: 1, 2: 1, 3: 2, 4: 2, 5: 1, 6: 1, 7: 1, 8: 1, 10: 1, 11: 1, 12: 2, 13: 1, 14: 1, 15: 1.

Total = 18.

Question: Exercise 22.1, Question 9: The weights of 40 college students in pounds are given approximately. Make a continuous frequency table of this data and also find the true class boundaries, class marks and cumulative frequencies.

Answer: Range = 57, classes = 7, interval = 9.

Table (class: frequency, true boundaries, mark, cumulative): 119-127: 3, 118.5-127.5, 123, 3; 128-136: 6, 127.5-136.5, 132, 9; 137-145: 10, 136.5-145.5, 141, 19; 146-154: 11, 145.5-154.5, 150, 30; 155-163: 5, 154.5-163.5, 159, 35; 164-172: 3, 163.5-172.5, 168, 38; 173-181: 2, 172.5-181.5, 177, 40.

Total = 40.

Question: Exercise 22.1, Question 10: The resistance values in ohms of a batch of 48 resistors of the same value are given. Make a continuous frequency table and also find the cumulative and percentage frequencies.

Answer: Range = 2.9, classes = 7, interval = 0.5.

Table (class: frequency, cumulative, percentage): 20.5-20.9: 3, 3, 6.25%; 21.0-21.4: 10, 13, 20.83%; 21.5-21.9: 11, 24, 22.92%; 22.0-22.4: 13, 37, 27.08%; 22.5-22.9: 9, 46, 18.75%; 23.0-23.4: 2, 48, 4.17%; 23.5-23.9: 0, 48, 0%.

Total = 48.

Question: Exercise 22.1, Question 11: The mass (in kg) of 50 pieces of metal was measured correctly to 0.1 kg and is given. Make a continuous frequency table. Also find the relative and percentage frequencies.

Answer: Range = 1.9, classes = 7, interval = 0.3.

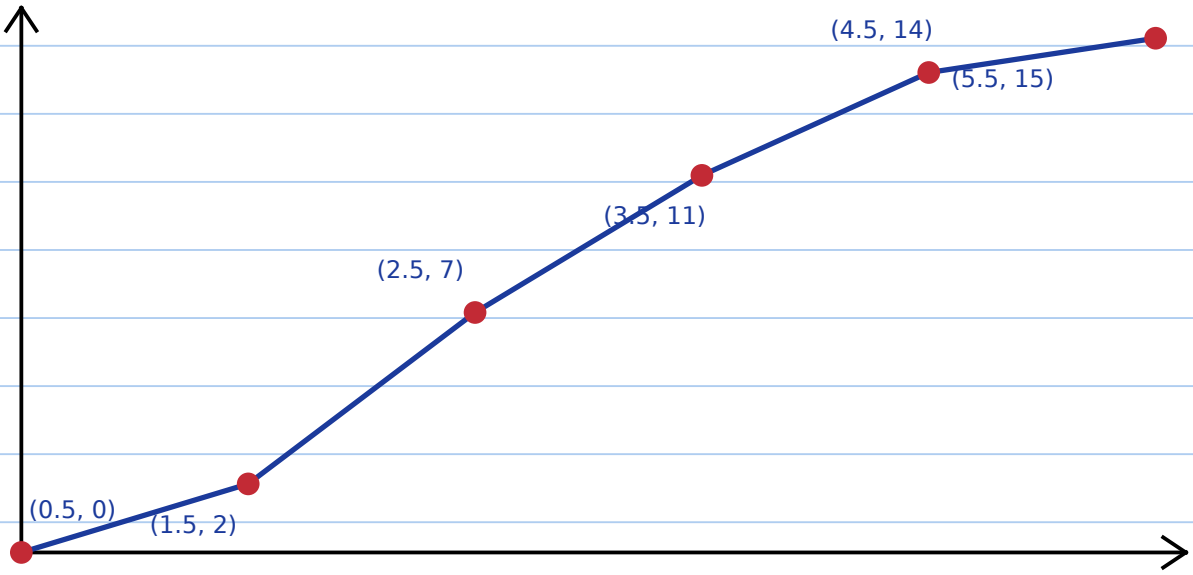
Table (class: frequency, relative, percentage): 7.1-7.3: 3, 0.06, 6%; 7.4-7.6: 5, 0.10, 10%; 7.7-7.9: 9, 0.18, 18%; 8.0-8.2: 13, 0.26, 26%; 8.3-8.5: 12, 0.24, 24%; 8.6-8.8: 6, 0.12, 12%; 8.9-9.1: 2, 0.04, 4%.

Total = 50.

Question: Exercise 22.2, Question 1: For the following data, find the percentage frequency, histogram, frequency polygon, cumulative frequency table and ogive. Marks 1, 2, 3, 4, 5; number of students 2, 5, 4, 3, 1.

Answer: Percentage frequencies: 13.33%, 33.33%, 26.67%, 20%, 6.67%. Cumulative frequencies: 2, 7, 11, 14, 15. Histogram: draw rectangles with

marks on the x-axis and frequencies on the y-axis. Frequency polygon: join the mid-points of the tops of the rectangles and bring both ends to the x-axis. Ogive: join the points (0.5, 0), (1.5, 2), (2.5, 7), (3.5, 11), (4.5, 14), (5.5, 15).




Ogive plotted from the points given in the answer.

Question: Exercise 22.2, Question 2: Make a histogram for the following data with unequal class intervals. Classes: 0-39, 40-49, 50-79, 80-99; frequencies: 6, 8, 12, 4.

Answer: True boundaries: -0.5-39.5, 39.5-49.5, 49.5-79.5, 79.5-99.5.

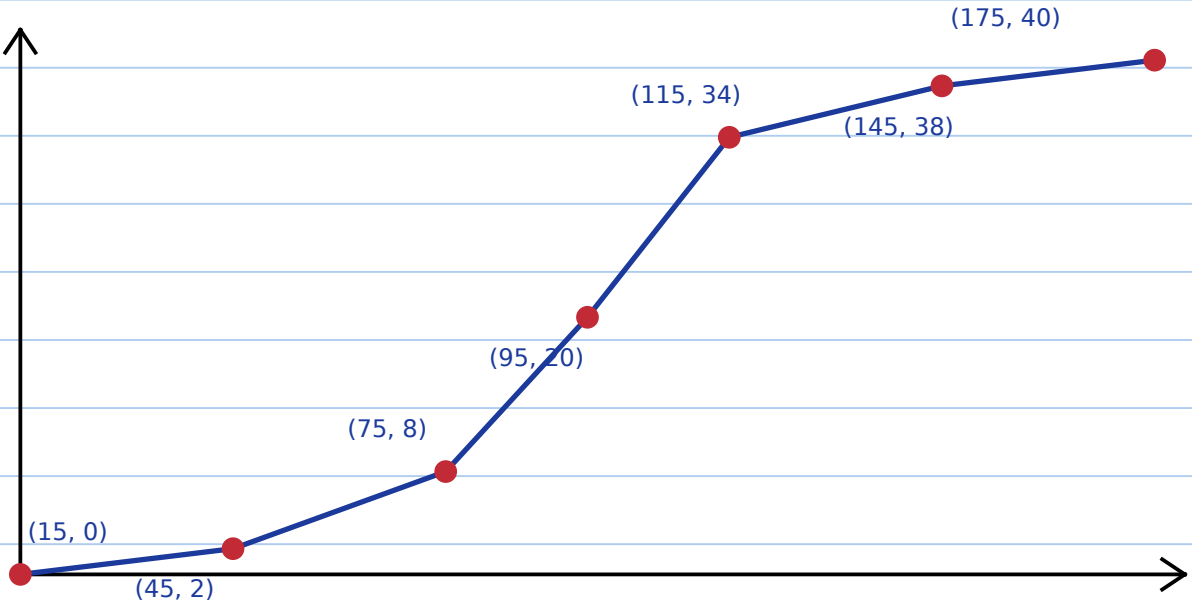
Adjusted heights (d = 10): 1.5, 8, 4, 2.

Draw rectangles with these heights.

Question: Exercise 22.2, Question 3: For the given data, make a histogram, frequency polygon, cumulative frequency table and ogive. Classes: 20-40, 50-70, 80-90, 100-110, 120-140, 150-170; frequencies: 2, 6, 12, 14, 4, 2.

Answer: True boundaries: 15-45, 45-75, 75-95, 95-115, 115-145, 145-175. Cumulative frequencies: 2, 8, 20, 34, 38, 40. Draw the histogram with

these boundaries and frequencies. Polygon: join the mid-points of the tops of the rectangles. Ogive: join the points (15, 0), (45, 2), (75, 8), (95, 20), (115, 34), (145, 38), (175, 40).




Ogive plotted from the points given in the answer.

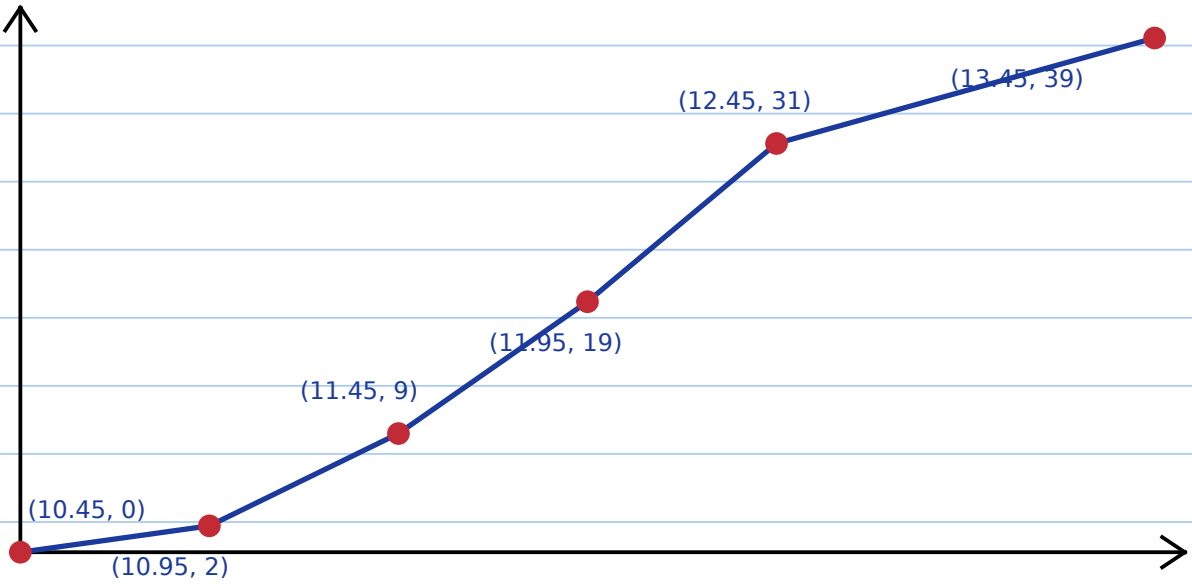
Question: Exercise 22.2, Question 4: For the following data, make a histogram, relative frequency polygon and ogive. Classes: 10.5-10.9, 11.0-11.4, 11.5-11.9, 12.0-12.4, 13.0-13.4; frequencies: 2, 7, 10, 12, 8.

Answer: Relative frequencies (n = 39): 0.051, 0.179, 0.256, 0.308, 0.205.

Draw the histogram with true boundaries.

Relative polygon: join the points of relative frequency at the class marks.

Ogive: join the points (10.45, 0), (10.95, 2), (11.45, 9), (11.95, 19), (12.45, 31), (13.45, 39).




Ogive plotted from the points given in the answer.

Question: Exercise 22.3, Question 1: Find the A.M., G.M., H.M. and median or mode of the following (wherever possible). (a) 3.2, 6, 10, 12, 12, -20, 25, 28, 30.8

Answer: A.M.

= 107/9 ≈ 11.89.

G.M.

and H.M.

are not possible because of the negative value.

Median = 12, mode = 12.

Question: Exercise 22.3, Question 1: (b) 14, 12, 18, 19, 0, -19, -18, -12, -14

Answer: A.M.

= 0/9 = 0.

G.M.

and H.M.

are not possible because of the zero and negative values.

Median = 0, there is no mode (all values are distinct).

Question: Exercise 22.3, Question 1: (c) 6.5, 11, 12.3, 9, 8.1, 16, 18, 20.5, 25

Answer: A.M.

= 126.4/9 ≈ 14.04, G.M.

≈ 12.84, H.M.

≈ 11.73, median = 12.3, there is no mode.

Question: Exercise 22.3, Question 1: (d) 51, 55, 52, 54, 58, 60, 61, 62, 52, 57, 52, 64

Answer: A.M.

= 678/12 = 56.5, G.M.

≈ 56.33, H.M.

≈ 56.18, median = 56, mode = 52.

Question: Exercise 22.3, Question 1: (e) Blood groups: A+, O-, AB+, O+, AB+, AB-, B+, AB+, O+, A-

Answer: It is categorical data, so A.M., G.M., H.M.

are not possible.

The median lies between AB+ and AB-, mode = AB+ (3 times).

Question: Exercise 22.3, Question 1: (f) Directions: east, south, east, east, south, north

Answer: It is categorical data, so A.M., G.M., H.M.

are not possible.

Median = east, mode = east (3 times).

Question: Exercise 22.3, Question 1: (g) Grades: B+, Fail, B+, A+, A-, C+, B+, A-, B-, Fail, A-, B+, C-

Answer: It is categorical data, so A.M., G.M., H.M.

are not possible.

Median = B+, mode = B+ (4 times).

Question: Exercise 22.3, Question 2: The daily earnings in rupees of ten workers are given. Find the A.M. (by direct, deviation and coding methods, A = 50, h = 10), G.M. (by direct and log methods), H.M., median and mode.

Answer: The printed data has 9 numbers (188, 170, 172, 125, 115, 195, 181, 190, 195) although the question mentions ten workers.

Direct: $\bar{x} = 1531/9 \approx 170.11$.

Deviation (A = 50): $\bar{x} = 50 + 1081/9 \approx 170.11$.

Coding (A = 50, h = 10): $\bar{x} = 50 + 10(108.1/9) \approx 170.11$.

G.M.

≈ 167.43, H.M.

≈ 164.37, median = 181, mode = 195.

[The number of values printed does not match the number of workers mentioned — noted.]

Question: Exercise 22.3, Question 3: Find the A.M., median and mode of the opinions of 60 people about dogs.

Answer: It is categorical data, so A.M.

is not possible.

Median = like dogs, mode = love dogs (20 people).

Question: Exercise 22.3, Question 4: Find the mode of the cause of death from a month's mortality data of a city.

Answer: The highest frequency is COVID (15), so the mode = COVID.

Question: Exercise 22.3, Question 5: Find the Q1, Q3, median and mode of the shoe sizes sold at 50% discount in a shop.

Answer: n = 191.

Q1 = 6.5, median = 7, Q3 = 7.5, mode = 7 (highest frequency 60).

Question: Exercise 22.3, Question 6: The daily wages (in rupees) of a thousand workers of a factory are given. Find the A.M., G.M., H.M., median, quartiles and modal wage.

Answer: n = 1000.

A.M.

= 32.4, G.M.

≈ 32.30, H.M.

≈ 32.10, Q1 = 30, median = 32, Q3 = 34, mode = 32.

Question: Exercise 22.3, Question 7: A summary of the profit earned by a company in 50 days is given. Find the A.M. of the profits by short-cut and coding methods: (a) A = 9000, h = 2000 (b) A = 11000, h = 2000.

Answer: By both methods $\bar{x}$ = 9640.

(a) Short-cut: $\bar{x}$ = 9000 + 32000/50 = 9640;

coding: $\bar{x}$ = 9000 + 2000(16/50) = 9640.

(b) Short-cut: $\bar{x}$ = 11000 - 68000/50 = 9640;

coding: $\bar{x}$ = 11000 + 2000(-34/50) = 9640.

Question: Exercise 22.3, Question 8: The marks (out of 50) of students are given in a grouped data table. Find the A.M., G.M., H.M., median and mode.

Answer: n = 84.

A.M.

= 3063/84 ≈ 36.46, G.M.

≈ 36.08, H.M.

≈ 35.70, median ≈ 36.64, mode ≈ 36.93.

Question: Exercise 22.3, Question 9: Find the A.M., G.M., H.M., median, quartiles and mode of the given data.

Answer: n = 39.

A.M.

≈ 11.92, G.M.

≈ 11.91, H.M.

≈ 11.89, Q1 ≈ 11.49, median ≈ 11.97, Q3 ≈ 12.38, mode ≈ 12.12.

Question: Exercise 22.3 (ii), Property 1: In ungrouped data, the sum of all deviations of the observations from the A.M. is zero.

Answer: True: $\Sigma(x - \bar{x}) = \Sigma x - n\,\bar{x} = n\,\bar{x} - n\,\bar{x} = 0$.

Question: Exercise 22.3 (ii), Property 2: The A.M. of a constant data set is the constant itself.

Answer: True: if all observations are a, then $\bar{x} = na/n = a$.

Question: Exercise 22.3 (ii), Property 3: If $\bar{x}$ is the A.M. of a set, then for $Y = aX + b$, $\bar{y} = a\,\bar{x} + b$, where a and b are real numbers.

Answer: True: $\bar{y} = \Sigma(aX + b)/n = a\,\bar{x} + b$.

Question: Exercise 22.3 (ii), Property 4: If all observations of a data set are the same or constant, then AM = GM = HM.

Answer: True: if all observations are a, then A.M. = a, G.M. = a, H.M. = a.

Question: Exercise 22.3 (ii), Property 5: For non-constant data, AM > GM > HM.

Answer: True: for example, for {4, 5, 6, 6, 6, 7, 7, 8}, A.M.

= 6.125 > G.M.

= 6.006 > H.M.

= 5.879.

Question: Exercise 22.3 (ii), Property 6: For non-constant data, (AM)(HM) ≈ $(G.M.)^2$.

Answer: True: in the same example, $(G.M.)^2$ = 36.072 and (A.M.)(H.M.) = 36.008, which are approximately equal.

Question: Exercise 22.3 (ii), Property 7: If the data is symmetric, AM = median; otherwise for non-symmetric data AM ≠ median.

Answer: True: for example, for {4, 5, 6, 6, 6, 7, 7, 7, 7, 8, 8, 8, 9, 10}, A.M.

= 7 = median (symmetric);

for {4, 8, 13, 14, 19, 20, 23}, A.M.

= 14.42 ≠ median = 14 (non-symmetric).

Question: Exercise 22.3 (ii), Property 8: For symmetric data, AM = median = mode.

Answer: True: in symmetric data, all three central measures are equal.

Question: Exercise 22.3 (ii), Property 9: For constant data, AM = GM = HM = median = mode.

Answer: True: for example, for {19, 19, 19, 19, 19}, all five measures are

19.

Question: Exercise 22.4, Question 1 (a): If W = {148, 145, 160, 157, 156, 160, 160, 165} and X = {-2, -2, -2, -2}, prove that the sum of all deviations of W from its A.M. is zero.

Answer: The A.M.

of W = 1251/8 = 156.375.

$\Sigma(W - \bar{x}) = \Sigma W - 8 \times 156.375 = 1251 - 1251 = 0.$

Hence proved.

Question: Exercise 22.4, Question 1 (b): Prove that the A.M. = G.M. = H.M. = median = mode of X.

Answer: X is constant data (-2, -2, -2, -2), so A.M.

= -2, median = -2, mode = -2, and by the book's constant-data property G.M.

= H.M.

= -2.

So all are equal.

Question: Exercise 22.4, Question 1 (c): Find the A.M. of Y = 3X.

Answer: $\bar{y} = 3 \times \bar{x} = 3 \times (-2) = -6.$

(Y = {-6, -6, -6, -6})

Question: Exercise 22.4, Question 1 (d): Find the A.M. of Z = 3W - 11.

Answer: $\bar{z} = 3 \times \bar{w} - 11 = 3 \times 156.375 - 11 = 469.125 - 11 = 458.125$.

Question: Exercise 22.4, Question 1 (e): Prove that for W, median < mode < H.M. < G.M. < A.M.

Answer: The actual values for W: H.M.

≈ 156.12, G.M.

≈ 156.24, A.M.

= 156.375, median = 158.5, mode = 160.

So the correct order is H.M.

< G.M.

< A.M.

< median < mode.

[The printed order puts the median and mode in the wrong place — noted.]

Question: Exercise 22.4, Question 2: Check whether the given data are symmetric. X = {7, 9, 3, 3, 3, 4, 1, 3, 2, 2} and Y = {2, 1, 4, 4, 4, 6, 6, 5, 7, 1}

Answer: For X, A.M.

= 3.7, median = 3, mode = 3.

A.M.

≠ median, so X is not symmetric.

For Y, A.M.

= 4, median = 4, mode = 4.

All three are equal, so Y is symmetric.

Question: Exercise 22.4, Question 3: The weights (in kg) and prices (in rupees per kg) of items for a household for a month are given. Using the A.M., G.M. and H.M., find the weighted average measures of price.

Answer: $\Sigma w = 45$, $\Sigma wx = 14850$, $\Sigma w \log x = 112.1672$, $\Sigma(w/x) = 0.151615$.

Weighted A.M.

= 14850/45 = 330, weighted G.M.

≈ 310.8, weighted H.M.

= 45/0.151615 ≈ 296.80.

Question: Exercise 22.4, Question 4: Of 50 purchased bags, the average fruit of 21 bags is 24.2 kg and the average fruit of 29 bags is 23.6 kg. Find the weighted average fruit of the 50 bags.

Answer: Weighted average = $(21 \times 24.2 + 29 \times 23.6)/50 = 1192.6/50 = 23.852$ kg.

Question: Exercise 22.4, Question 5: From the data of the number of COVID-19 deaths in the fourth week in a province, find the 2-, 3- and 4-day

moving averages.

Answer: 2-day: 116, 144, 173, 192, 227.5, 284.5. 3-day: 130, 158.67, 180.67, 214.33, 255. 4-day: 144.5, 168, 200.25, 238.25; centred: 156.25, 184.125, 219.25.

Question: Exercise 22.4, Question 6: The sales (in million rupees) of a company over 11 years are given. Find the 4- and 5-year moving averages.

Answer: 5-year: 154.8, 186.2, 222.2, 228.2, 229.8, 242.4, 252.6. 4-year: 144.5, 168, 200.25, 238.25, 238.25, 238.25, 238.25, 238.25; centred: 156.25, 184.125, 219.25, 238.25, 238.25, 238.25, 238.25.

Question: Exercise 22.4, Question 7: Estimate the median, quartiles and mode graphically for the given data.

Answer: n = 48.

Graphically (ogive and histogram): median ≈ 43.14, Q1 ≈ 36.64, Q3 = 48.25, mode = 45.5.

Question: Exercise 22.4, Question 8: Find the measures from the given graphs: (a) mode (b) median and quartiles.

Answer: (a) From the histogram: mode ≈ 113.21 kWh (class 108-127).

(b) From the ogive: median ≈ 2.022 mm, Q1 ≈ 2.018 mm, Q3 ≈ 2.024 mm.

Question: Exercise 22.5, Question 1: Find the range, variance, mean deviation and standard deviation of the number of absent students in a class. Numbers: 3, 5, 3, 2, 4, 1, 8.

Answer: Range = 7.

$\bar{x} = 26/7 \approx 3.714$.

Variance = 1540/343 ≈ 4.49.

Mean deviation = 82/49 ≈ 1.67.

Standard deviation ≈ 2.12.

Question: Exercise 22.5, Question 2: Find the range, variance, mean deviation and standard deviation of the scores of two players over 6 innings and decide which player performs more consistently.

Answer: A: range = 185, variance ≈ 4427.2, mean deviation ≈ 48.44, standard deviation ≈ 66.54.

B: range = 72, variance ≈ 601.5, mean deviation ≈ 21.11, standard deviation ≈ 24.53.

B has the smaller standard deviation, so B performs more consistently.

Question: Exercise 22.5, Question 3: Using the standard deviation of the income and expenditure of six families, draw a conclusion about their spread.

Answer: Income: mean = 35, standard deviation ≈ 17.08, relative = 0.488.

Expenditure: mean = 29, standard deviation ≈ 14.34, relative = 0.494.

The absolute spread of income is greater, but the relative spread of expenditure is slightly greater.

Question: Exercise 22.5, Question 4: The scores of teams A and B in a season are given. Use the relative mean deviation and relative standard deviation and decide which team performs more consistently.

Answer: Team A: relative mean deviation ≈ 1.037, relative standard

deviation ≈ 1.239.

Team B: relative mean deviation = 0.925, relative standard deviation ≈ 1.090.

Both relative measures of B are smaller, so team B performs more consistently.

Question: Exercise 22.5, Question 5: For the distribution of the masses of 50 pieces of metal, find the range, variance, mean deviation and standard deviation.

Answer: Range = 2.10.

$\bar{x}$ = 8.106.

Variance ≈ 0.196, standard deviation ≈ 0.443, mean deviation ≈ 0.343.

Review Exercise 22, Question 1 (i): The arithmetic mean depends on ___ observations of the data.

(a) middle

(b) extreme

✓ (c) all

(d) none

Review Exercise 22, Question 1 (ii): To find ___, ungrouped data must first be arranged in order.

(a) AM

(b) mode

✓ (c) median

(d) range

Review Exercise 22, Question 1 (iii): In a continuous frequency table, the

number of classes is between ___.

(a) 10

✓ (b) 15

(c) 20

(d) 25

Review Exercise 22, Question 1 (iv): Using the graph of ___, the mode of grouped data is found.

✓ (a) histogram

(b) polygon

(c) ogive

(d) bar chart

Review Exercise 22, Question 1 (v): If the data has no observations different from the others, then ___ is misleading.

(a) AM

✓ (b) range

(c) median

(d) mode

Review Exercise 22, Question 1 (vi): If the A.M. of {0, 90, k, 10, 100} is 40, then k = ___.

✓ (a) 0

(b) 90

(c) 10

(d) 100

Review Exercise 22, Question 1 (vii): The median and quartiles are ___ in nature.

(a) arithmetic

✓ (b) positional

(c) logical

(d) none

Review Exercise 22, Question 1 (viii): The median/quartiles divide the data in ___ ratio.

✓ (a) 50%-50%

(b) 25%-25%

(c) 75%-25%

(d) 40%-60%

Review Exercise 22, Question 1 (ix): If a number in the data is equal to zero, then ___ cannot be found.

(a) A.M.

(b) G.M.

✓ (c) H.M.

(d) median

Review Exercise 22, Question 1 (x): If all numbers in the data are equal, then ___.

(a) A.M. = G.M. = H.M.

(b) range = 0

(c) S.D. = 0

✓ (d) all of these

# 23. فیثا غورث ستایان

Question: Exercise 23, Question 1: The side measures of a triangle are given below; verify that the triangle is right-angled. (i) a = 16, b = 30, c = 34

Answer: $a^2 + b^2 = 256 + 900 = 1156 = 34^2 = c^2$.

So the triangle is right-angled.

Question: Exercise 23, Question 1: (ii) a = 6, b = 8, c = 10

Answer: $a^2 + b^2 = 36 + 64 = 100 = 10^2 = c^2$.

So the triangle is right-angled.

Question: Exercise 23, Question 1: (iii) a = 15, b = 20, c = 25

Answer: $a^2 + b^2 = 225 + 400 = 625 = 25^2 = c^2$.

So the triangle is right-angled.

Question: Exercise 23, Question 1: (iv) a = 13, b = $\sqrt{56}$, c = 15

Answer: $a^2 + b^2 = 169 + 56 = 225 = 15^2 = c^2$.

So the triangle is right-angled.

Question: Exercise 23, Question 2: Find the unknown value in the given shapes. (i) A right triangle with sides 7, 24 and x

Answer: $x^2 = 7^2 + 24^2 = 49 + 576 = 625$, so x = 25.

Question: Exercise 23, Question 2: (ii) A right triangle with sides x, 5 and 13

Answer: $x^2 = 13^2 - 5^2 = 169 - 25 = 144$, so x = 12.

Question: Exercise 23, Question 2: (iii) A right triangle with sides x, 2 and

$2\sqrt{2}$

Answer: $x^2 = (2\sqrt{2})^2 - 2^2 = 8 - 4 = 4$, so $x = 2$.

Question: Exercise 23, Question 2: (iv) A right triangle with sides 8, x and 17

Answer: $x^2 = 17^2 - 8^2 = 289 - 64 = 225$, so $x = 15$.

Question: Exercise 23, Question 3: A triangle has sides of measures 9.5 cm, 7.5 cm and x cm. For which value of x will these sides form a right triangle?

Answer: If x is the hypotenuse: $x = \sqrt{(9.5^2 + 7.5^2)} = \sqrt{146.5} \approx 12.10$ cm.

If 9.5 is the hypotenuse: $x = \sqrt{(9.5^2 - 7.5^2)} = \sqrt{34} \approx 5.83$ cm.

So $x \approx 12.10$ cm or $x \approx 5.83$ cm.

Question: Exercise 23, Question 4: ABC is an isosceles triangle with mAB = mAC = 12 cm and mBC = 10 cm. Find the perpendicular distance from A to BC.

Answer: The altitude divides BC into 5 and 5.

Perpendicular distance $= \sqrt{(12^2 - 5^2)} = \sqrt{119} \approx 10.91$ cm.

Question: Exercise 23, Question 5: The bottom of a ladder is 6 ft from a wall while its top rests 8 ft up the wall. Find the length of the ladder.

Answer: Length of the ladder $= \sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$ ft.

Question: Exercise 23, Question 6: Find the value of x in the given figure. (AB = 10, AC = 17, BD = x, DC = 15, AD is perpendicular)

Answer: From $\Delta ADC$: $AD^2 = 17^2 - 15^2 = 64$.

From $\Delta ABD$: $AD^2 = 10^2 - x^2$.

So 100 - $x^2$ = 64, $x^2$ = 36, x = 6.

Question: Exercise 23, Question 7: In ΔABC, ∠ADB is a right angle as shown in the figure. Find the lengths a, b and h if mCD = 5 units and mAD = 7 units.

Answer: From $h^2 = (mAC)^2 - (7^2 + 5^2)$: $h^2$ = 35, h = √35 ≈ 5.92.

b = √(49 + 35) = √84 = 2√21 ≈ 9.17.

a = √(25 + 35) = √60 = 2√15 ≈ 7.75.

Question: Exercise 23, Question 8: A rectangular swimming pool has sides 50 m and 30 m. Find the distance between two opposite corners.

Answer: Diagonal = $\sqrt{(50^2 + 30^2)} = \sqrt{3400} = 10\sqrt{34} \approx 58.31$ m.

Question: Exercise 23, Question 9: Each side of an equilateral triangle is 8 units. Find the length of any altitude.

Answer: Altitude = $\sqrt{(8^2 - 4^2)} = \sqrt{48} = 4\sqrt{3} \approx 6.93$ units.

Question: Exercise 23, Question 10: A triangle has sides of lengths x + 4, x and 20. If the largest side is 20, for which values of x will the triangle be right-angled?

Answer: From $(x + 4)^2 + x^2 = 20^2$: $2x^2 + 8x - 384 = 0$, $x^2 + 4x - 192 = 0$, x = 12 (the negative value -16 is not acceptable).

So the sides are 12, 16, 20.

Question: Exercise 23, Question 11: A man walks 18 m in one direction and then 24 m in the perpendicular direction. Find the distance from his starting point to his end point.

Answer: Distance = $\sqrt{(18^2 + 24^2)} = \sqrt{900} = 30$ m.

Question: Exercise 23, Question 12: In a rectangle ABCD, mBC + mCD = 17 cm and mBD + mAC = 26 cm. Find the length and width of the

rectangle.

Answer: The diagonals are equal: mBD = mAC = 13 cm.

If width = w and length = l, then w + l = 17 and $w^2 + l^2 = 169$.

So wl = 60, and w, l = 5, 12.

Length = 12 cm, width = 5 cm.

Review Exercise 23, Question 1 (i): The diagonal of a rectangle measures 6.5 cm. If its width is 2.5 cm, then its length is ___.

(a) 9 cm

(b) 4 cm

✓ (c) 6 cm

(d) 3 cm

Review Exercise 23, Question 1 (ii): Which of the following are the sides of a right triangle?

✓ (a) 3, 4, 5

(b) 2, 3, 4

(c) 5, 6, 7

(d) 4, 5, 6

Review Exercise 23, Question 1 (iii): The largest angle of a right triangle is ___.

(a) 100°

✓ (b) 90°

(c) 80°

(d) 110°

Review Exercise 23, Question 1 (iv): In a right triangle, the hypotenuse is the side opposite the ___ angle.

(a) acute

✓ (b) right

(c) obtuse

(d) none

Review Exercise 23, Question 1 (v): If a right triangle has sides a, b, c and c is the largest side, then ___.

✓ (a) $c^2 = a^2 + b^2$

(b) $b^2 = c^2 + a^2$

(c) $a^2 = b^2 + c^2$

(d) $c^2 = a^2 - b^2$

Review Exercise 23, Question 1 (vi): If two sides of a right triangle are 5 cm and 12 cm, then the hypotenuse is ___.

(a) 16

(b) 15

(c) 14

✓ (d) 13

Review Exercise 23, Question 1 (vii): If the hypotenuse of a right triangle is $3\sqrt{2}$ cm, then each leg is ___ long.

(a) 2 cm

(b) 5 cm

✓ (c) 3 cm

(d) 1 cm

Question: Review Exercise 23, Question 2: State Pythagoras theorem.

Answer: In a right triangle, the square of the hypotenuse equals the sum of the squares of the other two sides: $c^2 = a^2 + b^2$.

Question: Review Exercise 23, Question 3: A 25 m ladder is placed against a vertical wall. The foot of the ladder is 7 m from the wall. Find how high up the wall the ladder reaches.

Answer: Height = $\sqrt{(25^2 - 7^2)} = \sqrt{576} = 24$ m.

Question: Review Exercise 23, Question 4: Find the unknown value from each of the following shapes. (i) A right triangle with sides 2, 3 and x

Answer: $x^2 = 2^2 + 3^2 = 13$, so $x = \sqrt{13}$.

Question: Review Exercise 23, Question 4: (ii) A right triangle with sides 7, x and 10

Answer: $x^2 = 10^2 - 7^2 = 51$, so $x = \sqrt{51}$.

Question: Review Exercise 23, Question 4: (iii) A right triangle with sides x, 9 and 12

Answer: $x^2 = 12^2 - 9^2 = 63$, so $x = \sqrt{63} = 3\sqrt{7}$.

# 24. نسبت تناسب

Question: Exercise 24.1, Question 1: In ΔABC, DE ∥ AB. Find mCE if mDC = 12 cm, mBD = 6 cm, mAE = 5 cm.

Answer: By the basic proportionality theorem, mCE/mAE = mDC/mBD, so mCE = 5 × 12/6 = 10 cm.

Question: Exercise 24.1, Question 2: In ΔABC, PQ ∥ BC. Find the value of x if mAP = 5x - 3, mPB = 2, mAQ = 2x + 1, mQC = 3.

Answer: By the basic proportionality theorem, (5x - 3)/2 = (2x + 1)/3, so 15x - 9 = 4x + 2, 11x = 11, x = 1.

Question: Exercise 24.1, Question 3: Prove that a line parallel to one side of a triangle divides the other two sides in the same ratio.

Answer: Basic proportionality theorem: if DE ∥ BC then mAD/mDB = mAE/mEC. Proof: divide AD and AE into equal units; through the parallel lines the corresponding parts are equal, so the ratio of parts on both sides is the same.

Question: Exercise 24.1, Question 4: Prove that a line drawn through the midpoint of one side of a triangle parallel to another side bisects the third side.

Answer: Let D be the midpoint of AB and DE ∥ BC. By the basic proportionality theorem, mAD/mDB = mAE/mEC. But mAD = mDB, so mAE = mEC, i.e. E is the midpoint of AC.

Question: Exercise 24.1, Question 5: Prove that a line dividing two sides of a triangle in the same ratio is parallel to the third side.

Answer: Converse of the basic proportionality theorem: if mAP/mPB = mAQ/mQC then PQ ∥ BC. Proof: suppose PQ is not parallel to BC; then the line through Q parallel to BC meets AC at K, and by the basic

proportionality theorem mAQ/mQK = mAP/mPB. Comparing with the given ratio gives mQK = mQC, so K = C, and PQ ∥ BC.

Question: Exercise 24.2, Question 1: In the given figure, AD is the angle bisector of ∠A of ΔABC. Find the value of x if mAB = x + 9, mAC = 11 - x, mCD = 2x + 3 and mBD = 3x + 2.

Answer: By the angle bisector theorem, (x + 9)/(11 - x) = (3x + 2)/(2x + 3).

Solving: $5x^2 - 10x + 5 = 0$, $(x - 1)^2 = 0$, x = 1.

Question: Exercise 24.2, Question 2: In the given figure, ΔABC and ΔPQR are similar. Find the values of x and y if the lengths of the given sides are as shown in the figure.

Answer: From ΔABC ~ ΔPQR: 3/9 = x/6 = y/12, so x = 2 cm, y = 4 cm.

Question: Exercise 24.2, Question 3: Suppose A1 and A2 are the areas of two similar triangles ΔABC and ΔPQR respectively, as shown in the figure. Find A2 if A1 = 28 cm², mAB = 4x cm and mPQ = 2x cm.

Answer: $A1/A2 = (mAB/mPQ)^2 = (4x/2x)^2 = 4$, so A2 = 28/4 = 7 cm².

Question: Exercise 24.2, Question 4: The ratio of corresponding sides of two similar triangles is x - 5 : 2 and the ratio of their areas is 1 : 9. Find the value of x.

Answer: From $((x - 5)/2)^2 = 1/9$, (x - 5)/2 = 1/3 (length is positive), so x - 5 = 2/3, x = 17/3.

Question: Exercise 24.2, Question 5: Prove that the sides of two triangles are proportional if an acute angle of one triangle equals an acute angle of the other.

Answer: For right triangles: both triangles have a right angle and an equal acute angle, so by AA the triangles are similar, and corresponding sides are proportional.

Question: Exercise 24.2, Question 6: In a right triangle, the altitude drawn from the right angle to the hypotenuse divides the triangle into two triangles. Prove that each of them is similar to the original triangle.

Answer: Let $\Delta ABC$ have $\angle B = 90°$ and $BD \perp AC$.

In $\Delta ABD$ and $\Delta ABC$: $\angle A$ is common and $\angle ADB = \angle ABC = 90°$, so $\Delta ABD \sim \Delta ABC$ (AA).

Similarly $\Delta BCD \sim \Delta ABC$ (AA).

Review Exercise 24, Question 1 (i): In a proportion, the product of the means equals ___ of the product of the extremes.

(a) sum

(b) difference

(c) division

✓ (d) product

Review Exercise 24, Question 1 (ii): ___ are always similar.

(a) right triangles

(b) isosceles triangles

(c) acute triangles

✓ (d) equilateral triangles

Review Exercise 24, Question 1 (iii): The symbol for similarity of triangles is ___.

(a) =

(b) ≅

✓ (c) ~

(d) ↔

Review Exercise 24, Question 1 (iv): If a line parallel to one side of a triangle divides one side in the ratio 3 : 2, then it divides the other side in

___.

✓ (a) 3:2

(b) 2:3

(c) 2:6

(d) 5:3

Review Exercise 24, Question 1 (v): If a line cuts two sides of a triangle in the same ratio, then it will be ___ to the third side.

✓ (a) parallel line

(b) non-parallel line

(c) slanting line

(d) none

Review Exercise 24, Question 1 (vi): In ΔABC, the bisector of ∠A divides BC in ___ ratio if mAB = 6 cm and mAC = 8 cm.

(a) 5:8

✓ (b) 3:4

(c) 1:1

(d) 5:7

Review Exercise 24, Question 1 (vii): The bisector of an angle of an equilateral triangle divides the opposite side in ___.

(a) 2:3

(b) 3:2

✓ (c) 1:1

(d) 5:2

Review Exercise 24, Question 1 (viii): Corresponding sides of two similar triangles are ___.

(a) equal

(b) unequal

✓ (c) proportional

(d) none

Review Exercise 24, Question 1 (ix): If the ratio of corresponding sides of two similar triangles is 5:7, then the ratio of their areas equals ___.

(a) 5:7

(b) 7:5

(c) 25:7

✓ (d) 25:49

Review Exercise 24, Question 1 (x): If the ratio of the areas of two similar triangles is 36 : 121, then the ratio of their corresponding sides will be ___.

(a) 6:10

✓ (b) 6:11

(c) 11:6

(d) 10:6

Question: Review Exercise 24, Question 1 (xi): If corresponding sides of two similar triangles are in the ratio x : 2 and the ratio of areas is 4:9, then x = ___. (a) 3 (b) -3 (c) both a and b (d) none

Answer: (d) none — from the printed ratio, $(x/2)^2 = 4/9$, so $x = \pm 4/3$, which is none of the given options. [If the area ratio were 9:4, the answer would be (c) both 3 and -3 — noted.]

Review Exercise 24, Question 1 (xii): Two equilateral triangles are ___.

(a) identical triangles

✓ (b) similar triangles

(c) proportional

(d) equal-equal

# 25. گول جا زهه

Question: Exercise 25.1, Question 1: Can a circle pass through any three collinear points? Explain with reason.

Answer: No. The perpendicular bisectors of collinear points are parallel to each other and never meet, so no centre exists.

Question: Exercise 25.1, Question 2: Can you draw a circle through any four non-collinear points? Explain with reason.

Answer: Not necessarily. A circle passes through four points only when they are concyclic; in general the circle through three points does not pass through the fourth.

Question: Exercise 25.1, Question 3: Prove that one and only one circle passes through the vertices of a triangle.

Answer: The perpendicular bisectors of the sides of the triangle meet at a single point O. O is equidistant from every vertex, so the circle with centre O and radius OA passes through all three vertices, and it is the only such circle.

Question: Exercise 25.1, Question 4: Prove that one and only one circle passes through the vertices of a quadrilateral.

Answer: For a cyclic quadrilateral: the perpendicular bisectors of the sides meet at a single point O, equidistant from all four vertices, so the circle with centre O passes through all four vertices, and it is the only such circle.

Question: Exercise 25.1, Question 5: Three villages are located such that B is 6 km east of A and C is 8 km north of B. Find the location of the mosque so that it is equidistant from each village. Also find how much distance each villager must travel.

Answer: The mosque is the circumcentre of ΔABC.

AC = √(6² + 8²) = 10 km.

Since ∠B = 90°, the centre is the midpoint of AC, so the distance from each village = 10/2 = 5 km.

Each villager must travel 5 km.

Question: Exercise 25.2, Question 1: Prove that the diameters of a circle bisect each other.

Answer: Every diameter passes through the centre of the circle and the centre is the midpoint of every diameter, so the diameters bisect each other at the centre.

Question: Exercise 25.2, Question 2: Prove that the line from the centre of a circle to the midpoint of a chord is perpendicular to the chord.

Answer: If OC meets the midpoint C of chord AB, then ΔOCA ≅ ΔOCB (SSS), so m∠OCA = m∠OCB.

These angles are supplementary, so both are 90°, i.e. OC ⊥ AB.

Question: Exercise 25.2, Question 3: If a chord is 8 cm long and the perpendicular distance from the centre of the circle to the chord is 3 cm, find the circumference and the area of the circle.

Answer: Radius r = √(3² + 4²) = 5 cm.

Circumference = 2πr = 10π ≈ 31.42 cm.

Area = πr² = 25π ≈ 78.54 cm².

Question: Exercise 25.2, Question 4: Find the length of the chord at

distance k from the centre of a circle of radius r. (a) k = 4 cm, r = 9 cm

Answer: Chord = $2\sqrt{(r^2 - k^2)} = 2\sqrt{(81 - 16)} = 2\sqrt{65} \approx 16.12$ cm.

Question: Exercise 25.2, Question 4: (b) k = 3 cm, r = 6 cm

Answer: Chord = $2\sqrt{(r^2 - k^2)} = 2\sqrt{(36 - 9)} = 2\sqrt{27} = 6\sqrt{3} \approx 10.39$ cm.

Question: Exercise 25.2, Question 5: What is the radius of a circle if the chord is 10 cm long and the distance of the chord from the centre is 3 cm?

Answer: $r = \sqrt{(5^2 + 3^2)} = \sqrt{34} \approx 5.83$ cm.

Question: Exercise 25.3, Question 1: Prove that equal chords of two congruent circles are equidistant from their centres.

Answer: Theorem 25.4: if AB ≅ CD then from ΔAEO ≅ ΔCFO (RHS), OE ≅ OF, i.e. the chords are equidistant from the centres.

Question: Exercise 25.3, Question 2: Prove that two chords of two congruent circles equidistant from the centres are equal.

Answer: Theorem 25.5: if OE ≅ OF then from ΔAEO ≅ ΔCFO (RHS), AE = CF, so AB = 2AE = 2CF = CD.

Question: Exercise 25.3, Question 3: Find the distance from the centre of a chord of length α in a circle of radius r. (a) α = 7 cm and r = 6 cm

Answer: Distance $= \sqrt{(r^2 - (\alpha/2)^2)} = \sqrt{(36 - 12.25)} = \sqrt{23.75} \approx 4.87$ cm.

Question: Exercise 25.3, Question 3: (b) α = 5 cm and r = 4 cm

Answer: Distance $= \sqrt{(16 - 6.25)} = \sqrt{9.75} \approx 3.12$ cm.

Question: Exercise 25.3, Question 3: (c) α = 2 cm and r = 4 cm

Answer: Distance $= \sqrt{(16 - 1)} = \sqrt{15} \approx 3.87$ cm.

Question: Exercise 25.3, Question 3: (d) α = 4 cm and r = 9 cm

Answer: Distance $= \sqrt{(81 - 4)} = \sqrt{77} \approx 8.77$ cm.

Question: Exercise 25.3, Question 4: Find the distance between two parallel chords of lengths α and β in a circle of radius r. (a) α = 6 cm, β = 8 cm, r = 5 cm

Answer: Distances from the centre: $\sqrt{(25 - 9)} = 4$ cm and $\sqrt{(25 - 16)} = 3$

cm.

If the chords are on opposite sides of the centre, the distance = 4 + 3 = 7 cm;

on the same side, the distance = 4 - 3 = 1 cm.

Question: Exercise 25.3, Question 4: (b) α = 3 cm, β = 6 cm, r = 14 cm

Answer: Distances from the centre: √(196 - 2.25) ≈ 13.92 cm and √(196 - 9) ≈ 13.67 cm. On opposite sides: ≈ 27.59 cm; on the same side: ≈ 0.25 cm.

Question: Exercise 25.3, Question 5: If two chords AB and CD of a circle with centre O intersect at P, find the value of x. (a) Segments x, x and 2, 10

Answer: By the intersecting-chords property, $x^2 = 2 \times 10 = 20$, so $x = 2\sqrt{5} \approx 4.47$.

Question: Exercise 25.3, Question 5: (b) Segments 4, 8 and 3, x

Answer: $4 \times 8 = 3 \times x$, so $x = 32/3 \approx 10.67$.

Question: Exercise 25.3, Question 5: (c) Segments 3, 12 and x, 20

Answer: $3 \times 12 = x \times 20$, so $x = 36/20 = 1.8$.

Question: Exercise 25.3, Question 5: (d) Segments 7.5, 7.2 and 5, x

Answer: $7.5 \times 7.2 = 5 \times x$, so $x = 54/5 = 10.8$.

Question: Exercise 25.3, Question 5: (e) Segments 17, 12 and x, x

Answer: $x^2 = 17 \times 12 = 204$, so $x = 2\sqrt{51} \approx 14.28$.

Question: Exercise 25.3, Question 5: (f) Segments 7.2, x and 4.5, 9.6

Answer: $7.2 \times x = 4.5 \times 9.6 = 43.2$, so $x = 43.2/7.2 = 6$.

Review Exercise 25, Question 1 (i): All circles are always ___.

(a) congruent

✓ (b) similar

(c) small

(d) none

Question: Review Exercise 25, Question 1 (ii): In the given figure, the circles with centres D and E are ___. (a) congruent (b) similar (c) both (a) and (b) (d) none

Answer: (c) both (a) and (b) — in the figure the circles with centres D and E are of equal size, so they are both congruent and similar.

Review Exercise 25, Question 1 (iii): In the given figure, the length of chord AB is ___.

(a) 4 cm

(b) 6 cm

✓ (c) 8 cm

(d) 15 cm

Review Exercise 25, Question 1 (iv): One and only one circle passes through ___ points.

(a) collinear

✓ (b) non-collinear

(c) non-common

(d) none

Review Exercise 25, Question 1 (v): "If two chords of a circle are equal, then they are equidistant from the centre" — the hypothesis of this statement is:

(a) two chords of the circle are equidistant from the centre

✓ (b) two chords of the circle are equal

(c) a circle has two chords

(d) the centre of the circle is equidistant from the chord

Review Exercise 25, Question 1 (vi): "If two chords of a circle are equal, then they are equidistant from the centre" — the conclusion of this

statement is:

✓ (a) two chords of the circle are equidistant from the centre

(b) two chords of the circle are equal

(c) a circle has two chords

(d) the centre of the circle is equidistant from the chord

Review Exercise 25, Question 1 (vii): "One and only one circle passes through three non-collinear points" — the hypothesis of this statement is:

✓ (a) three points are non-collinear

(b) one and only one circle passes through three points

(c) two circles pass through three points

(d) three points are collinear

Review Exercise 25, Question 1 (viii): "One and only one circle passes through three non-collinear points" — the conclusion of this statement is:

(a) three points are non-collinear

✓ (b) one and only one circle passes through three points

(c) two circles pass through three points

(d) three points are collinear

Review Exercise 25, Question 1 (ix): A circle is an example of a ___ curve.

✓ (a) closed and simple

(b) open and simple

(c) closed and non-simple

(d) open and non-simple

# 26. گول جون چهٽي ليڪون

Question: Exercise 26.1, Question 1: Prove that a triangle circumscribing a circle is an equilateral triangle.

Answer: [As printed, this claim is not true in general — not every triangle circumscribing a circle is equilateral. The book's own related result is Example 1: a parallelogram circumscribing a circle is a rhombus.]

Question: Exercise 26.1, Question 2: Prove that a rectangle circumscribing a circle is a square.

Answer: All four sides of a rectangle circumscribing a circle touch the circle.

The distance between opposite tangent lines equals the diameter, so length = 2r and width = 2r, i.e. the rectangle is a square.

Question: Exercise 26.1, Question 3: Two concentric circles have diameters 10 cm and 5 cm respectively. A tangent is drawn from a point of the outer circle to the smaller circle. Find the length of the chord of the outer circle which touches the smaller circle.

Answer: Radii: R = 5 cm, r = 2.5 cm.

Half-chord = √(25 - 6.25) = √18.75, so chord = 2√18.75 = 5√3 ≈ 8.66 cm.

Question: Exercise 26.1, Question 4: A point is 5 cm from the centre of a circle. The tangent segment from the point to the circle is 4 cm. Find the diameter, circumference and area of the circle.

Answer: r = √(25 - 16) = 3 cm.

Diameter = 6 cm, circumference = 6π ≈ 18.85 cm, area = 9π ≈ 28.27

cm².

Question: Exercise 26.1, Question 5: A point is 25 cm from the centre of a circle of radius 7 cm. Find the length of the tangent from the point to the circle.

Answer: Tangent = $\sqrt{(25^2 - 7^2)} = \sqrt{576} = 24$ cm.

Question: Exercise 26.1, Question 6: How far from the centre of a circle of radius 3 cm can a tangent segment of length 10 cm be drawn to the circle?

Answer: Distance = $\sqrt{(3^2 + 10^2)} = \sqrt{109} \approx 10.44$ cm.

Question: Exercise 26.2, Question 1: Prove that the tangents drawn at the ends of a diameter of a circle are parallel to each other.

Answer: At both ends of the diameter, the radius is perpendicular to the tangent. Both tangents are perpendicular to the same diameter, so they are parallel to each other.

Question: Exercise 26.2, Question 2: Prove that if two tangents of a circle are parallel, then they are the ends of a diameter of the circle.

Answer: The tangents are parallel and each tangent is perpendicular to the radius at its point of contact. The radii of two parallel tangents lie on the same line, so the points of contact are the ends of a diameter.

Question: Exercise 26.2, Question 3: Prove that if two tangents of a circle are not parallel, then they meet at an external point.

Answer: Non-parallel lines meet at a point. If the meeting point were on the circle, both tangents would touch the circle at the same point, which is impossible; so the meeting point lies outside the circle.

Question: Exercise 26.2, Question 4: Find the value of the unknown x in the following figures. (a) Tangent x and secant with segments 4 and 5

Answer: $x^2 = 4 \times (4 + 5) = 36$, so $x = 6$.

Question: Exercise 26.2, Question 4: (b) Tangent x and secant with

segments 11 and 13

Answer: $x^2 = 11 \times (11 + 13) = 264$, so $x = 2\sqrt{66} \approx 16.25$.

Question: Exercise 26.2, Question 4: (c) Two secants with segments 6, 4 and 5, x

Answer: $6 \times (6 + 4) = 5 \times (5 + x)$, so $60 = 25 + 5x$, $x = 7$.

Question: Exercise 26.2, Question 4: (d) Two secants with segments 9, x and 12, 3

Answer: $9 \times (9 + x) = 12 \times (12 + 3) = 180$, so $9 + x = 20$, $x = 11$.

Question: Exercise 26.2, Question 4: (e) Two secants with segments 8, 5 and 7, x

Answer: $8 \times (8 + 5) = 7 \times (7 + x)$, so $104 = 49 + 7x$, $x = 55/7 \approx 7.86$.

Question: Exercise 26.2, Question 4: (f) Two secants with segments 18, x and 20, 9

Answer: $18 \times (18 + x) = 20 \times (20 + 9) = 580$, so $18 + x = 290/9$, $x = 128/9 \approx 14.22$.

Question: Exercise 26.2, Question 4: (g) Two secants with segments 20, x and 18, 9

Answer: $20 \times (20 + x) = 18 \times (18 + 9) = 486$, so $20 + x = 24.3$, $x = 4.3$.

Question: Exercise 26.2, Question 4: (h) Two secants with segments x, 5 and 15, 10

Answer: $x \times (x + 5) = 15 \times (15 + 10) = 375$, so $x^2 + 5x - 375 = 0$, $x = (-5 + 5\sqrt{61})/2 \approx 17.02$ (negative root rejected).

Question: Exercise 26.2, Question 5: Find the value of the unknown x in the following figures. (a) Two secants, one with external segment 4 and total 12, the other with external segment x and total 8

Answer: $4 \times 12 = x \times 8$, so $x = 6$.

Question: Exercise 26.2, Question 5: (b) Two secants with segments 8, x and 6, 10

Answer: $8 \times (8 + x) = 6 \times (6 + 10) = 96$, so $8 + x = 12$, $x = 4$.

Question: Exercise 26.2, Question 5: (c) Tangent x and secant with segments 3 and 12

Answer: $x^2 = 3 \times (3 + 12) = 45$, so $x = 3\sqrt{5} \approx 6.71$.

Question: Exercise 26.2, Question 5: (d) A secant with external segment 12, chord x, total 9x

Answer: $12 + x = 9x$, so $8x = 12$, $x = 3/2 = 1.5$.

Question: Exercise 26.2, Question 5: (e) Tangent 10 and secant with segments 5 and 3x

Answer: $10^2 = 5 \times (5 + 3x)$, so $100 = 25 + 15x$, $x = 5$.

Question: Exercise 26.2, Question 5: (f) Two tangents 13 and x

Answer: Tangents from the same external point are equal, so $x = 13$.

Question: Exercise 26.3, Question 1: If the distance between the centres of two circles equals the sum of their radii, prove that the two circles touch each other externally.

Answer: Theorem 26.4 (A): distance between centres = r + r' means the circles meet at a single point lying on the line joining the two centres, i.e. they touch each other externally.

Question: Exercise 26.3, Question 2: A triangle has the centres of three circles at its vertices. If the perimeter of the triangle equals the sum of the diameters of the circles, prove that the three circles touch each other externally.

Answer: Each side of the triangle joins the centres of two circles and the perimeter = 2(r1 + r2 + r3).

Each side equals the sum of two radii, so each pair of circles touches externally.

Question: Exercise 26.3, Question 3: Two congruent circles touch each

other externally. If the distance between their centres is 12 cm, find their radii and the circumference of the circles.

Answer: r + r = 12, so r = 6 cm.

Circumference = 2πr = 12π ≈ 37.70 cm.

Question: Exercise 26.3, Question 4: If PA and PB are tangents from an external point P to the given circle as shown in the figures, find the unknown angle. (a) ∠AOB = 110°

Answer: In quadrilateral OAPB, ∠OAP = ∠OBP = 90°, so ∠APB = 360° - 90° - 90° - 110° = 70°.

Question: Exercise 26.3, Question 4: (b) ∠APB = 80°

Answer: ∠AOB = 360° - 90° - 90° - 80° = 100°.

Question: Exercise 26.3, Question 5: Prove that if two circles touch externally, then the point of contact lies on the line joining the two centres.

Answer: By the proof of Theorem 26.4 (A): at the point of contact B, both radii are perpendicular to the common tangent, so O, B and O' are collinear.

Question: Exercise 26.3, Question 6: If two chords of a circle of radius 2.5 cm are 3.9 cm apart from each other and the length of one chord is 1.4 cm, find the length of the other chord.

Answer: Distance of the first chord from the centre = $\sqrt{(2.5^2 - 0.7^2)} = 2.4$ cm.

Distance of the second chord = 3.9 - 2.4 = 1.5 cm (chords are on opposite sides of the centre).

Second chord = $2\sqrt{(2.5^2 - 1.5^2)} = 2 \times 2 = 4$ cm.

Question: Exercise 26.3, Question 7: If two circles have radii 4 cm and 5 cm, do the circles touch each other when the distance between their centres is 35 cm? [The printed wording is slightly garbled — answered on this reading.]

Answer: No.

External touching needs distance = 4 + 5 = 9 cm and internal touching needs distance = 5 - 4 = 1 cm.

35 cm differs from both, so the circles are separate.

Review Exercise 26, Question 1 (i): If a point is outside a circle, then ___ tangent line(s) can be drawn from the point.

(a) one

✓ (b) two

(c) three

(d) none

Review Exercise 26, Question 1 (ii): The angle between a tangent line and the radius at the point of contact is ___.

(a) 45°

(b) 60°

✓ (c) 90°

(d) 120°

Review Exercise 26, Question 1 (iii): In the given figure, the circles with centres E and C ___.

(a) touch internally

✓ (b) touch externally

(c) intersect

(d) are congruent

Review Exercise 26, Question 1 (iv): In the given figure, the circles with centres B and C have ___ point(s) of contact.

✓ (a) none

(b) one

(c) two

(d) none of these

Review Exercise 26, Question 1 (v): In the given figure, the circles with centres E and C have ___ point(s) of contact.

(a) none

✓ (b) one

(c) two

(d) none of these

Review Exercise 26, Question 1 (vi): In the given figure, AB is a ___.

(a) tangent line

✓ (b) secant line

(c) chord

(d) none

Review Exercise 26, Question 1 (vii): In the given figure, CDE is a ___.

✓ (a) tangent line

(b) secant line

(c) chord

(d) none

Review Exercise 26, Question 1 (viii): In the given figure, ___ is the point of contact.

(a) A

(b) B

(c) C

✓ (d) D

Review Exercise 26, Question 1 (ix): Tangents drawn at the ends of a diameter of a circle are ___ to each other.

✓ (a) parallel

(b) perpendicular

(c) secant

(d) both

(b) and

Review Exercise 26, Question 1 (x): The maximum number of common tangents of two circles is ___ if they touch internally.

(a) 0

✓ (b) 1

(c) 2

(d) 3

Review Exercise 26, Question 1 (xi): The maximum number of common tangents of two circles is ___ if they touch externally.

(a) 0

(b) 1

(c) 2

✓ (d) 3

# 27. زهہ قوس

Question: Exercise 27.1, Question 1: Prove that in a circle, the arcs between two parallel and equal chords are equal.

Answer: If AB ∥ CD and AB = CD, then the perpendiculars from the centre to both chords are equal and the corresponding central angles are equal, so the arcs between the chords are equal.

Question: Exercise 27.1, Question 2: Prove that in two congruent circles, the arcs of two equal central angles are equal.

Answer: In two congruent circles, equal central angles cut off equal arcs, because arc length depends on the central angle and the radius, and both are equal.

Question: Exercise 27.1, Question 3: In the given figure, O is the centre of the circle. Find the value of x. [The figure shows central angles 40°, 25°, 100° and x.]

Answer: Angles around the centre sum to 360°: x = 360° - 40° - 25° - 100° = 195°.

[The figure labels are small — answered on this reading.]

Question: Exercise 27.1, Question 4: In the given figure, m∠PQR = 35° and m∠PRQ = 31°, find m∠QPR and m∠OQR.

Answer: m∠QPR = 180° - 35° - 31° = 114°.

Minor arc QR = 360° - 2 × 114° = 132°, so central angle m∠QOR = 132°.

In ΔOQR, OQ = OR, so m∠OQR = (180° - 132°)/2 = 24°.

Question: Exercise 27.1, Question 5: In the given figure, O is the centre of

the circle. m∠AOB = 120°, m∠BCD = 85°, find m∠OAD.

Answer: In ΔOAB, OA = OB and m∠AOB = 120°, so m∠OAB = 30°.

ABCD is cyclic, so m∠DAB = 180° - 85° = 95°.

So m∠OAD = 95° - 30° = 65°.

Question: Exercise 27.1, Question 6: In the given figure, AB is a diameter of the circle. m∠DAB = 30°, find m∠ACD.

Answer: AB is a diameter, so m∠ADB = 90°.

In ΔABD, m∠ABD = 180° - 90° - 30° = 60°.

Angles subtended by the same arc AD are equal, so m∠ACD = m∠ABD = 60°.

Review Exercise 27, Question 1 (i): If a chord subtends a central angle of 60°, then the chord and the radius segments are ___.

(a) parallel

(b) perpendicular

✓ (c) equal

(d) unequal

Review Exercise 27, Question 1 (ii): If an arc subtends a central angle of 45°, then the corresponding chord subtends an angle of ___ at the centre.

(a) 15°

(b) 30°

✓ (c) 45°

(d) 60°

Review Exercise 27, Question 1 (iii): If two chords of a circle subtend equal central angles, then they are ___.

(a) unequal

(b) perpendicular

✓ (c) equal

(d) none

Review Exercise 27, Question 1 (iv): The arcs opposite equal central angles of a circle are ___.

(a) parallel

✓ (b) equal

(c) perpendicular

(d) none

Review Exercise 27, Question 1 (v): A 6 cm chord subtends a central angle of 60°. The radius segment of the circle is ___ long.

(a) 4 cm

✓ (b) 6 cm

(c) 5 cm

(d) 8 cm

Review Exercise 27, Question 1 (vi): A chord of a circle subtending a central angle of 180° is always ___ in length.

(a) equal to the radius

(b) less than the radius

✓ (c) twice the radius

(d) half the radius

Review Exercise 27, Question 1 (viii): If the radius segments of a chord of a circle are equal to the chord, then the central angle opposite the chord will be ___.

✓ (a) 60°

(b) 75°

(c) 90°

(d) 45°

Review Exercise 27, Question 1 (ix): Of two equal arcs of a circle, one arc subtends a central angle of 30°, then the other arc subtends a central angle of ___.

(a) 60°

(b) 90°

(c) 75°

✓ (d) 30°

Review Exercise 27, Question 1 (x): A diameter divides a circle into ___ parts.

✓ (a) two

(b) three

(c) four

(d) all of these

# 28. گول جي سيگمينٽ ڪندون

Question: Exercise 28.1, Question 1: In Figure 1, O is the centre of the circle. Find the value of x given m∠C = 30°.

Answer: m∠AOB = 2 × m∠C = 60° (central angle).

In ΔOAB, OA = OB, so x = m∠OAB = (180° - 60°)/2 = 60°.

Question: Exercise 28.1, Question 2: In Figure 2, O is the centre of the circle. Find the value of x given m∠Q = 50°.

Answer: Reflex central angle m∠QOR = 240°.

Inscribed angle m∠QPR = 240°/2 = 120°.

In ΔPQR, x = m∠PRQ = 180° - 50° - 120° = 10°.

Question: Exercise 28.1, Question 3: In Figure 3, O is the centre of the circle. Find the value of x + y given m∠D = 30°.

Answer: m∠D = m∠ADB = 30° subtends arc AB.

Central angle x = m∠AOB = 2 × 30° = 60°.

In ΔOAB, OA = OB, so y = m∠OAB = (180° - 60°)/2 = 60°.

So x + y = 120°.

Question: Exercise 28.1, Question 4: Prove that angles subtended by equal arcs of two congruent circles are equal.

Answer: In congruent circles, equal arcs subtend equal central angles, and an inscribed angle is half the central angle, so the inscribed angles of

equal arcs are equal.

Question: Exercise 28.1, Question 5: Prove that in a circle, angles subtended at the circumference by the same arc — whether the minor arc or the major arc — are equal. [The printed wording is slightly garbled — answered on this reading.]

Answer: Angles in the same segment subtended by the same arc are equal (Theorem 28.2), because each is half the central angle.

Question: Exercise 28.2, Question 1: In Figure 1, O is the centre of the circle, AC is its diameter. Find the value of x given m∠ACB = 3x + 1 and m∠CAB = 2x - 1.

Answer: AC is a diameter, so m∠ABC = 90°.

In ΔABC, (3x + 1) + (2x - 1) + 90° = 180°, so 5x = 90°, x = 18.

Question: Exercise 28.2, Question 2: In Figure 2, ∠ACB = 2x - 2. Find the value of x. [As printed, this question cannot be solved — only one angle expression is given, with no second condition, so x cannot be determined. This item was skipped and logged in skipped-items.md.]

Answer: [As printed, this question cannot be solved — only one angle expression is given, with no second condition, so x cannot be determined.]

Question: Exercise 28.2, Question 3: In a circle, the measure of the inscribed angle of the minor arc is 7x - 1. Find the value of x. [As printed, this question cannot be solved — only one angle expression is given, with no second condition, so x cannot be determined. This item was skipped and logged in skipped-items.md.]

Answer: [As printed, this question cannot be solved — only one angle expression is given, with no second condition, so x cannot be determined.]

Question: Exercise 28.2, Question 4: Prove that a rhombus inscribed in a circle is a square.

Answer: Opposite angles of a cyclic quadrilateral are supplementary and opposite angles of a rhombus are equal, so each angle is 90°. A rhombus with equal sides and 90° angles is a square.

Question: Exercise 28.2, Question 5: Prove that the exterior angle of a cyclic quadrilateral equals the interior opposite angle.

Answer: Exterior angle + adjacent interior angle = 180° (straight line).

Adjacent interior angle + opposite interior angle = 180° (cyclic quadrilateral).

So exterior angle = interior opposite angle.

Review Exercise 28, Question 1 (i): The inscribed angle of ___ is an obtuse angle.

(a) minor arc

✓ (b) major arc

(c) semicircle

(d) all of these

Review Exercise 28, Question 1 (ii): The circular area enclosed by ___ arc and chord is ___.

(a) sector

✓ (b) segment

(c) major arc

(d) circumference of the circle

Review Exercise 28, Question 1 (iii): If 2x is the measure of the inscribed angle of an arc, then the central angle of the corresponding major arc is ___.

(a) x

(b) 2x

(c) 3x

✓ (d) 4x

Review Exercise 28, Question 1 (iv): If 2x and 60° are inscribed angles of the same segment, then x = ___.

(a) 120°

(b) 60°

(c) 20°

✓ (d) 30°

Review Exercise 28, Question 1 (v): The inscribed angle of the minor arc of a circle is ___.

✓ (a) acute angle

(b) obtuse angle

(c) right angle

(d) reflex angle

Review Exercise 28, Question 1 (vi): The inscribed angle of the major arc of a circle is ___.

(a) acute angle

✓ (b) obtuse angle

(c) right angle

(d) reflex angle

Review Exercise 28, Question 1 (vii): The sum of the opposite angles of a cyclic quadrilateral is ___.

(a) 90°

✓ (b) 180°

(c) 270°

(d) 360°

Review Exercise 28, Question 1 (viii): A parallelogram inscribed in a circle is ___.

(a) kite

(b) trapezium

✓ (c) rectangle

(d) rhombus

Review Exercise 28, Question 1 (ix): The central angle of an arc is ___ the corresponding inscribed angle.

(a) smaller than

✓ (b) bigger than

(c) equal to or smaller than

(d) equal to or bigger than

Review Exercise 28, Question 1 (x): The sum of the central angles of all arcs of a circle is ___.

(a) 90°

(b) 180°

✓ (c) 360°

(d) 1000°

# 29. عملي جاميٹري - گول

Question: Exercise 29.1, Question 1: With the help of a circular object, draw a circle. Locate its centre.

Answer: Draw two non-parallel chords AB and CD in the circle. Draw the perpendicular bisector of each chord. The point O where the two perpendicular bisectors meet is the centre of the circle.

Question: Exercise 29.1, Question 2: Draw three non-collinear points X, Y, Z. Draw a circle passing through them.

Answer: Join the points to draw XY and YZ. Draw the perpendicular bisector of XY and the perpendicular bisector of YZ. Their meeting point O is the centre. With O as centre and radius equal to OX, draw the circle passing through X, Y and Z.

Question: Exercise 29.1, Question 3: With a circular object, draw a minor arc AB. Complete the circle by finding its centre.

Answer: Take a third point C on the arc. Draw chords AC and BC. Draw the perpendicular bisectors of both chords, meeting at point O — O is the centre. With O as centre and radius equal to OA, draw the rest of the circle to complete the circle of arc AB.

Question: Exercise 29.1, Question 4: With a circular object, draw a major arc ABC. Complete the circle by finding its centre.

Answer: Take points A, B, C on the arc. Draw chords AB and BC. Draw the perpendicular bisectors of both chords, meeting at point O — O is the centre. With O as centre and radius equal to OA, draw the rest of the circle to complete the circle of arc ABC.

Question: Exercise 29.1, Question 5: With a circular object, draw a minor arc AB. Complete the circle WITHOUT finding its centre.

Answer: Mark equal-length chords PQ, QR, RS, ST, TU, UV on the arc. Get each next point by making the exterior angle equal (each new chord makes the same exterior angle with the previous chord). The arc keeps extending this way until the circle is complete — without needing to find the centre.

Question: Exercise 29.2, Question 1 (i): Draw ΔABC and draw its circumcircle given mAB = 5.5 cm, mAC = 6 cm, m∠A = 50°.

Answer: First draw ΔABC. Draw the perpendicular bisectors of AB and AC, meeting at point O — O is the circumcentre. With O as centre and radius equal to OA, draw the circle passing through A, B and C.

Question: Exercise 29.2, Question 1 (ii): Draw ΔABC and draw its circumcircle given mAB = 6 cm, mBC = 4.5 cm, mAC = 6 cm.

Answer: Draw ΔABC from the three sides. Draw the perpendicular bisectors of any two sides, meeting at point O — O is the circumcentre. With O as centre and radius equal to OA, draw the circle passing through A, B and C.

Question: Exercise 29.2, Question 2 (i): Draw ΔPQR and draw its incircle given mPQ = 5 cm, mQR = 6.5 cm, mRP = 5.5 cm.

Answer: Draw ΔPQR from the three sides. Draw the interior bisectors of two angles, meeting at point O — O is the incentre. From O draw a perpendicular to any one side, meeting it at L. With O as centre and radius equal to OL, draw the circle touching all three sides.

Question: Exercise 29.2, Question 2 (ii): Draw ΔPQR and draw its incircle given mPQ = 6 cm, m∠P = 60°, m∠Q = 50°.

Answer: Draw ΔPQR from the given side and angles. Draw the interior bisectors of two angles, meeting at point O — O is the incentre. From O draw a perpendicular to any one side, meeting it at L. With O as centre and radius equal to OL, draw the circle touching all three sides.

Question: Exercise 29.2, Question 3 (i): Draw ΔXYZ and draw the excircle opposite ∠Y given mXY = 4.5 cm, mYZ = 5 cm, m∠Y = 30°.

Answer: Draw ΔXYZ from the given sides and angle. Extend sides XY and ZY. Draw the exterior bisectors of ∠X and ∠Z, meeting at point I — I is the excentre. From I draw a perpendicular to XZ, meeting it at L. With I as centre and radius equal to IL, draw the circle touching XZ and the extensions of XY and ZY.

Question: Exercise 29.2, Question 3 (ii): Draw ΔXYZ and draw the excircle opposite ∠Y given mXY = 5.5 cm, mYZ = 6 cm, mXZ = 7.5 cm.

Answer: Draw ΔXYZ from the three sides. Extend sides XY and ZY. Draw the exterior bisectors of ∠X and ∠Z, meeting at point I — I is the excentre. From I draw a perpendicular to XZ, meeting it at L. With I as centre and radius equal to IL, draw the circle touching XZ and the extensions of XY and ZY.

Question: Exercise 29.2, Question 4: Draw a circle with centre O and radius 3.5 cm and construct its circumscribed equilateral triangle.

Answer: Draw a circle with centre O and radius 3.5 cm. Take any point A on the circle. Mark three equal arcs from A, dividing the circle into three equal parts (each central angle 120°). Draw tangents at the three points, which meet each other to form the equilateral triangle.

Question: Exercise 29.2, Question 5: Draw ΔPQR with mPQ = 4.5 cm, mQR = 5.5 cm, mPR = 6 cm, and construct the equilateral triangle circumscribed about its circumcircle.

Answer: Draw ΔPQR from the three sides and draw its circumcircle (join the perpendicular bisectors of the sides to get centre O). Then divide the circle into three equal arcs (each central angle 120°) and draw tangents at the three points, which meet each other to form the equilateral triangle.

Question: Exercise 29.2, Question 6: Construct the circumscribed

equilateral triangle of a circle with centre O and radius 3.8 cm.

Answer: Draw a circle with centre O and radius 3.8 cm. Divide the circle into three equal arcs (each central angle 120°) and draw tangents at the three points, which meet each other to form the equilateral triangle.

Question: Exercise 29.3, Question 1: Construct the circumscribed square of a circle with centre C and radius 4.2 cm.

Answer: Draw a circle with centre C and radius 4.2 cm. Draw two mutually perpendicular diameters meeting the circle at four points. Draw tangents at those four points, which meet each other to form the circumscribed square.

Question: Exercise 29.3, Question 2: Construct an inscribed square in a circle with centre O and radius 3.6 cm.

Answer: Draw a circle with centre O and radius 3.6 cm. Draw two mutually perpendicular diameters meeting the circle at four points. Join those four points with straight lines to form the inscribed square.

Question: Exercise 29.3, Question 3: Construct the circumscribed regular hexagon of a circle with centre O and radius 4.8 cm. [The printed question says "six-sided circumscribed triangle" — it means the circumscribed hexagon.]

Answer: Draw a circle with centre O and radius 4.8 cm. Divide the circle into six equal arcs (each central angle 60°). Draw tangents at the six points, which meet each other to form the circumscribed hexagon.

Question: Exercise 29.3, Question 4: Construct an inscribed regular hexagon in a circle with centre C and radius 4.4 cm. [It means the inscribed hexagon.]

Answer: Draw a circle with centre C and radius 4.4 cm. Divide the circle into six equal arcs (each central angle 60°, and each chord equals the radius). Join the six points with straight lines to form the inscribed

hexagon.

Question: Exercise 29.3, Question 5: Construct an inscribed regular pentagon in a circle with centre O and radius 4.3 cm. (Hint: make five equal angles at the centre.)

Answer: Draw a circle with centre O and radius 4.3 cm.

Make five equal angles at the centre (each 360°/5 = 72°) meeting the circle at five points.

Join those five points with straight lines to form the inscribed pentagon.

Question: Exercise 29.4, Question 1: Draw a minor arc PQ; without using the centre, draw the tangent to PQ passing through its midpoint.

Answer: Find the midpoint M of the arc (divide the arc into two equal parts). Draw the perpendicular to the chord of the arc at M — that perpendicular is the tangent to the arc at M, because a tangent is perpendicular to the radius and the perpendicular to a chord passes through the centre.

Question: Exercise 29.4, Question 2: Draw a major arc PQR. Without using the centre, draw the tangent to PQR passing through its midpoint Q.

Answer: Q is the midpoint of the arc. Draw the perpendicular to the chord PR of the arc at Q — that perpendicular is the tangent to the arc at Q.

Question: Exercise 29.4, Question 3: Draw a minor arc XY; without using the centre, draw the tangent to XY passing through point Z outside the arc.

Answer: Find the midpoint M of arc XY and draw the tangent at M (the method of Question 1). Then draw through Z the line parallel to that tangent which touches the arc — that is the tangent passing through Z.

Question: Exercise 29.4, Question 4: With centre C, the circle's radius is

2.9 cm. Draw the tangent to the circle from point P on it.

Answer: Draw a circle with centre C and radius 2.9 cm, and take point P on the circle. Join CP. Draw the perpendicular to CP at P — that perpendicular is the tangent to the circle at P.

Question: Exercise 29.4, Question 5: With centre P, a circle has radius 3.2 cm. Draw the tangent to the circle from point Q which is 8 cm from P.

Answer: Draw a circle with centre P and radius 3.2 cm. Take point Q 8 cm from P. Find the midpoint O of PQ. With O as centre and radius equal to OP, draw another circle cutting the first circle at two points A and B. QA and QB are the two tangents.

Question: Exercise 29.4, Question 6: Draw two tangents to a circle with centre C and radius 3.3 cm meeting each other at the following angles: (i) 50° (ii) 63°

Answer: Draw a circle with centre C and radius 3.3 cm.

(i) If the angle between the two tangents is 50°, the meeting angle and the central angle are supplementary, so the central angle = 180° - 50° = 130°.

Draw two radii with a 130° central angle and draw perpendiculars (tangents) at their ends, which meet at 50°.

(ii) In the same way take central angle = 180° - 63° = 117° and draw the tangents, which meet at 63°.

Question: Exercise 29.5, Question 1 (a): Two equal circles each of radius 3.3 cm with centres at points A and B such that mAB = 7.8 cm. Draw the direct common tangents of these circles.

Answer: Join A and B with a 7.8 cm segment. Draw circles of radius 3.3 cm

at A and B. At A draw the perpendicular to AB meeting the circle at P and S. At B draw the perpendicular to AB meeting the circle at Q and T. Join PQ and ST — these are the two direct common tangents.

Question: Exercise 29.5, Question 1 (b): The same two equal circles. Draw the transverse common tangents of these circles.

Answer: Find the midpoint O of AB. With O as centre and radius equal to OA, draw a circle cutting both circles at points. The lines through those points are the transverse common tangents, which cut AB at O.

Question: Exercise 29.5, Question 2 (a): Two unequal circles with radii 2.1 cm and 3.3 cm respectively and centres at A and B such that mAB = 8 cm. Draw the direct common tangents of these circles.

Answer: Join A and B with an 8 cm segment.

Draw circles of radius 3.3 cm at A and 2.1 cm at B.

Draw a small circle at A with the difference radius (3.3 - 2.1 = 1.2 cm).

Draw tangents from B to that small circle and extend them.

With set squares draw parallels to them touching both circles — these are the direct common tangents.

Question: Exercise 29.5, Question 2 (b): The same two unequal circles. Draw the transverse common tangents of these circles.

Answer: Join A and B with an 8 cm segment.

Draw circles of radius 3.3 cm at A and 2.1 cm at B.

Draw a large circle at A with the sum radius (3.3 + 2.1 = 5.4 cm).

Draw tangents from B to that large circle and draw parallels to them touching both circles — these are the transverse common tangents.

Question: Exercise 29.5, Question 3 (i): Draw a common tangent of two unequal circles. Their centres are A and B, radii 2.2 cm and 3.8 cm respectively, when the circles touch internally.

Answer: If the circles touch internally, the distance between centres = 3.8 - 2.2 = 1.6 cm.

Place A and B 1.6 cm apart and draw both circles, which touch internally at one point.

Draw the common perpendicular (tangent) to both circles at that point.

Question: Exercise 29.5, Question 3 (ii): The same two circles, when the circles touch externally.

Answer: If the circles touch externally, the distance between centres = 3.8 + 2.2 = 6 cm.

Place A and B 6 cm apart and draw both circles, which touch externally at one point.

Draw the common perpendicular (tangent) to both circles at that point.

Question: Exercise 29.5, Question 3 (iii): The same two circles, when the circles intersect each other and mAB = 5.6 cm.

Answer: Place A and B 5.6 cm apart and draw circles of radii 3.8 cm and 2.2 cm, which cut each other at two points.

Use the difference-radius (3.8 - 2.2 = 1.6 cm) method for the direct common tangent and the sum-radius (3.8 + 2.2 = 6 cm) method for the

transverse common tangent.

Question: Exercise 29.5, Question 4 (i): Draw a circle touching both sides of a given angle of 35°.

Answer: Draw a 35° angle. Draw the bisector of the angle. Take any point O on the bisector. From O draw a perpendicular to one side of the angle, meeting it at L. With O as centre and radius equal to OL, draw the circle touching both sides of the angle.

Question: Exercise 29.5, Question 4 (ii): The same task when the angle is 40°.

Answer: Draw a 40° angle. Draw the bisector of the angle. Take any point O on the bisector. From O draw a perpendicular to one side of the angle, meeting it at L. With O as centre and radius equal to OL, draw the circle touching both sides of the angle.

Question: Exercise 29.5, Question 5: Draw a circle touching two given converging lines and passing through a given point M.

Answer: Draw the bisector of the angle between the two converging lines — the centre of the circle lies on it. Draw the line through M perpendicular to one line, meeting the bisector at point O — O is the centre. With O as centre and radius equal to OM, draw the circle touching both lines and passing through M.

Review Exercise 29, Question 1 (i): The perpendicular bisectors of ___ non-parallel meet at the centre of the circle.

(a) radii

✓ (b) chords

(c) tangents

(d) none

Review Exercise 29, Question 1 (ii): Through one point ___ circles can pass.

(a) one

(b) two

(c) three

✓ (d) unlimited

Review Exercise 29, Question 1 (iii): Through three non-collinear points ___ circle(s) can pass.

✓ (a) one

(b) two

(c) three

(d) unlimited

Review Exercise 29, Question 1 (iv): The ___ angles of a regular polygon are equal in measure.

(a) interior

(b) exterior

✓ (c) both a and b

(d) none

Review Exercise 29, Question 1 (v): Each interior angle of a regular hexagon is equal to ___.

(a) 90°

(b) 108°

✓ (c) 120°

(d) 135°

Review Exercise 29, Question 1 (vi): A circle touching all three sides of a triangle is called ___.

(a) circumcircle

✓ (b) incircle

(c) excircle

(d) tricircle

Review Exercise 29, Question 1 (vii): A circle touching one side of a

triangle and the extensions of the other two sides from inside is called ___.

✓ (a) excircle

(b) circumcircle

(c) incircle

(d) tricircle

Review Exercise 29, Question 1 (viii): The centre of the incircle is called ___.

(a) excentre

✓ (b) incentre

(c) midpoint centre

(d) perpendicular centre

Review Exercise 29, Question 1 (ix): The perpendicular bisectors of the sides of a triangle meet each other at ___.

✓ (a) circumcentre

(b) midpoint centre

(c) incentre

(d) excentre

Review Exercise 29, Question 1 (x): The meeting point of the interior angle bisectors of a triangle is called ___.

(a) perpendicular centre

(b) midpoint centre

✓ (c) incentre

(d) excentre

Review Exercise 29, Question 1 (xi): If a regular hexagon is inscribed in a circle, then each side of the hexagon is ___ the radius of the circle:

(a) <

(b) >

✓ (c) =

(d) ≤

Review Exercise 29, Question 1 (xii): The angle between the tangent and the radius at the point of contact is ___.

(a) reflex

(b) acute

(c) obtuse

✓ (d) right

Review Exercise 29, Question 1 (xiii): The line joining the point of contact to the centre of the circle is the ___ of the angle of the tangent line.

(a) cube

(b) square

(c) supplement

✓ (d) complement [The printed wording is slightly garbled — the radius is perpendicular to the tangent, so the angle between radius and tangent is 90°.]

Review Exercise 29, Question 1 (xiv): ___ tangent lines do not cut the line joining the centres of the circles.

(a) perpendicular

✓ (b) external

(c) transverse

(d) internal

Review Exercise 29, Question 1 (xv): The direct common tangents of two equal circles are ___.

(a) cutting

(b) non-cutting

(c) equal

✓ (d) parallel

Review Exercise 29, Question 1 (xvi): The common tangents of two equal

circles cut the line joining the centres of the circles at ___.

(a) directly

✓ (b) transversely

(c) externally

(d) parallel [The printed options do not match — the correct fact is that the transverse tangents meet the line of centres at its midpoint.]

Review Exercise 29, Question 1 (xvii): If two circles of radii 5 cm and 2 cm touch each other externally, the distance between their centres is ___.

(a) 5 cm

(b) 10 cm

(c) 3 cm

✓ (d) 7 cm

Review Exercise 29, Question 1 (xviii): If two circles of radii 5 cm and 2 cm touch each other internally, the distance between their centres is ___.

(a) 5 cm

(b) 10 cm

✓ (c) 3 cm

(d) 7 cm

Review Exercise 29, Question 1 (xix): Two or more converging lines always meet the inclined lines at ___.

(a) one more point

✓ (b) one point

(c) two points

(d) more points

Review Exercise 29, Question 1 (xx): A closed shape with three or more sides is called ___.

(a) pentagon

(b) hexagon

(c) heptagon

✓ (d) polygon

# 30. ٹرگنومیٹري جو تعارف

Question: Exercise 30.1, Question 1 (i): Write 32° 15′ in decimal form in degrees.

Answer: 32° 15′ = (32 + 15/60)° = (32 + 0.25)° = 32.25°.

Question: Exercise 30.1, Question 1 (ii): Write 10° 30′ in decimal form in degrees.

Answer: 10° 30′ = (10 + 30/60)° = (10 + 0.5)° = 10.5°.

Question: Exercise 30.1, Question 1 (iii): Write 8° 15′ 30″ in decimal form in degrees.

Answer: 8° 15′ 30″ = (8 + 15/60 + 30/3600)° = (8 + 0.25 + 0.0083)° = 8.2583°.

Question: Exercise 30.1, Question 1 (iv): Write 45° 21′ 36″ in decimal form in degrees.

Answer: 45° 21′ 36″ = (45 + 21/60 + 36/3600)° = (45 + 0.35 + 0.01)° = 45.36°.

Question: Exercise 30.1, Question 1 (v): Write 25° 30′ in decimal form in degrees.

Answer: 25° 30′ = (25 + 30/60)° = (25 + 0.5)° = 25.5°.

Question: Exercise 30.1, Question 1 (vi): Write 18° 6′ 21″ in decimal form in degrees.

Answer: 18° 6′ 21″ = (18 + 6/60 + 21/3600)° = (18 + 0.1 + 0.0058)° = 18.1058°.

Question: Exercise 30.1, Question 2 (i): Write 32.25° in D° M′ S″ form.

Answer: 32.25° = 32° + (0.25 × 60)′ = 32° + 15′ = 32° 15′ 0″.

Question: Exercise 30.1, Question 2 (ii): Write 47.36° in D° M′ S″ form.

Answer: 47.36° = 47° + (0.36 × 60)′ = 47° + 21.6′ = 47° + 21′ + (0.6 ×

60)″ = 47° 21′ 36″.

Question: Exercise 30.1, Question 2 (iii): Write 57.325° in D° M′ S″ form.

Answer: 57.325° = 57° + (0.325 × 60)′ = 57° + 19.5′ = 57° + 19′ + (0.5 × 60)″ = 57° 19′ 30″.

Question: Exercise 30.1, Question 2 (iv): Write 67.58° in D° M′ S″ form.

Answer: 67.58° = 67° + (0.58 × 60)′ = 67° + 34.8′ = 67° + 34′ + (0.8 × 60)″ = 67° 34′ 48″.

Question: Exercise 30.1, Question 2 (v): Write 22.5° in D° M′ S″ form.

Answer: 22.5° = 22° + (0.5 × 60)′ = 22° + 30′ = 22° 30′ 0″.

Question: Exercise 30.1, Question 2 (vi): Write 225.60° in D° M′ S″ form.

Answer: 225.60° = 225° + (0.60 × 60)′ = 225° + 36′ = 225° 36′ 0″.

Question: Exercise 30.1, Question 3 (i): Express π/4 radians in degree measure.

Answer: π radians = 180°, so π/4 radians = 180°/4 = 45°.

Question: Exercise 30.1, Question 3 (ii): Express π/3 radians in degree measure.

Answer: π radians = 180°, so π/3 radians = 180°/3 = 60°.

Question: Exercise 30.1, Question 3 (iii): Express −3π/4 radians in degree measure.

Answer: π radians = 180°, so −3π/4 radians = −3 × 180°/4 = −135°.

Question: Exercise 30.1, Question 3 (iv): Express 3 radians in degree measure.

Answer: 1 radian = (180/π)° ≈ 57.2958°, so 3 radians = 3 × 57.2958° ≈ 171.89°.

Question: Exercise 30.1, Question 3 (v): Express 3/π radians in degree measure.

Answer: 3/π radians = (3/π) × (180/π)° = 540/π² ≈ 540/9.8696 ≈ 54.72°.

Question: Exercise 30.1, Question 3 (vi): Express 4.5 radians in degree

measure.

Answer: 1 radian = (180/π)° ≈ 57.2958°, so 4.5 radians = 4.5 × 57.2958° ≈ 257.83°.

Question: Exercise 30.1, Question 4 (i): Convert 30° into radian measure.

Answer: 30° = 30 × (π/180) radians = π/6 radians.

Question: Exercise 30.1, Question 4 (ii): Convert 45° into radian measure.

Answer: 45° = 45 × (π/180) radians = π/4 radians.

Question: Exercise 30.1, Question 4 (iii): Convert 60° into radian measure.

Answer: 60° = 60 × (π/180) radians = π/3 radians.

Question: Exercise 30.1, Question 4 (iv): Convert (22 1/2)° into radian measure.

Answer: (22 1/2)° = 22.5° = 22.5 × (π/180) radians = π/8 radians.

Question: Exercise 30.1, Question 4 (v): Convert −225° into radian measure.

Answer: −225° = −225 × (π/180) radians = −5π/4 radians.

Question: Exercise 30.1, Question 4 (vi): Convert 60° 35′ 48″ into radian measure.

Answer: First in degrees: 60° 35′ 48″ = (60 + 35/60 + 48/3600)° = (18179/300)° = 60.5967°.

Then in radians: 60.5967 × (π/180) = 18179π/54000 radians ≈ 1.0576 radians.

Question: Exercise 30.2, Question 1 (i): Find the value of θ given r = 5 cm and l = 20 cm.

Answer: From l = rθ, θ = l/r = 20/5 = 4 radians.

Question: Exercise 30.2, Question 1 (ii): Find the value of θ given r = 2 cm and l = 30.2 cm.

Answer: θ = l/r = 30.2/2 = 15.1 radians.

Question: Exercise 30.2, Question 1 (iii): Find the value of θ given r = 2.87 cm and l = 6 cm.

Answer: θ = l/r = 6/2.87 ≈ 2.09 radians.

Question: Exercise 30.2, Question 1 (iv): Find the value of θ given r = 2.5 cm and l = 4.5 cm.

Answer: θ = l/r = 4.5/2.5 = 1.8 radians.

Question: Exercise 30.2, Question 2 (i): Find the value of l given r = 1.01 cm and θ = 2.1 radians.

Answer: l = rθ = 1.01 × 2.1 = 2.121 cm.

Question: Exercise 30.2, Question 2 (ii): Find the value of l given r = 5.1 cm and θ = 2 radians.

Answer: l = rθ = 5.1 × 2 = 10.2 cm.

Question: Exercise 30.2, Question 2 (iii): Find the value of l given r = 6 cm and θ = 30°.

Answer: θ = 30° = π/6 radians, so l = rθ = 6 × π/6 = π ≈ 3.14 cm.

Question: Exercise 30.2, Question 2 (iv): Find the value of l given r = 15 cm and θ = 60° 30′.

Answer: θ = 60° 30′ = 60.5° = 60.5 × π/180 = 121π/360 radians ≈ 1.056 radians, so l = rθ = 15 × 121π/360 = 121π/24 ≈ 15.84 cm.

Question: Exercise 30.2, Question 3 (i): Find the value of r given θ = 60° and l = 2 m.

Answer: θ = 60° = π/3 radians, so r = l/θ = 2/(π/3) = 6/π ≈ 1.91 m.

Question: Exercise 30.2, Question 3 (ii): Find the value of r given θ = 3.5 radians and l = 7/4 m.

Answer: r = l/θ = (7/4)/3.5 = (7/4)/(7/2) = 1/2 m = 0.5 m.

Question: Exercise 30.2, Question 3 (iii): Find the value of r given θ = 1/4 radian and l = 4 cm.

Answer: r = l/θ = 4/(1/4) = 16 cm.

Question: Exercise 30.2, Question 3 (iv): Find the value of r given θ = 180° and l = 15.4 cm.

Answer: θ = 180° = π radians, so r = l/θ = 15.4/π ≈ 4.90 cm.

Question: Exercise 30.2, Question 4 (i): In a unit circle, find the arc length whose central angle is 30°.

Answer: In a unit circle r = 1.

θ = 30° = π/6 radians, so l = rθ = π/6 ≈ 0.5236.

Question: Exercise 30.2, Question 4 (ii): In a unit circle, find the arc length whose central angle is 45°.

Answer: θ = 45° = π/4 radians, so l = rθ = π/4 ≈ 0.7854.

Question: Exercise 30.2, Question 4 (iii): In a unit circle, find the arc length whose central angle is 60°.

Answer: θ = 60° = π/3 radians, so l = rθ = π/3 ≈ 1.0472.

Question: Exercise 30.2, Question 4 (iv): In a unit circle, find the arc length whose central angle is 90°.

Answer: θ = 90° = π/2 radians, so l = rθ = π/2 ≈ 1.5708.

Question: Exercise 30.2, Question 5: An arc of a circle subtends π/6 at the centre. The circle's radius is 5 cm. Find: (i) the arc length (ii) the area of the sector of the circle.

Answer: (i) l = rθ = 5 × π/6 = 5π/6 ≈ 2.62 cm.

(ii) Area of sector = $(1/2)r^2\theta = (1/2) \times 25 \times \pi/6 = 25\pi/12 \approx 6.54$ cm².

Question: Exercise 30.2, Question 6: A point moves on a circle of radius 10 cm at 3.5 revolutions per second. Find the distance covered by the point in one second.

Answer: Distance in one revolution = circumference = 2πr = 2π × 10 = 20π cm.

In one second there are 3.5 revolutions, so distance = 3.5 × 20π = 70π ≈ 219.9 cm.

Question: Exercise 30.2, Question 7: Find the area of a sector whose central angle is π/4 radians and the circle's radius is 4 cm.

Answer: Area of sector = $(1/2)r^2\theta = (1/2) \times 16 \times \pi/4 = 2\pi \approx 6.28$ cm².

Question: Exercise 30.2, Question 8: A point on the rim of a wheel of diameter 21 cm covers a distance of 5040 metres per minute. What is its angular speed in radians per second? [The printed wording is slightly garbled.]

Answer: Linear speed v = 5040/60 = 84 m/s.

Radius r = 21/2 cm = 0.105 m.

Angular speed ω = v/r = 84/0.105 = 800 radians per second.

Question: Exercise 30.2, Question 9: A car is moving on a circular path. The radius of the circle is twice the arc which the car covers on the circular path. Find the central angle of the arc. [The printed wording is slightly garbled — read as "twice".]

Answer: r = 2l.

θ = l/r = l/(2l) = 1/2 radian ≈ 28.65°.

Question: Exercise 30.2, Question 10: In a circle of radius 12 cm, each arc subtends 84° at the centre. Find the arc length and the area of the sector of that arc.

Answer: θ = 84° = 84 × π/180 = 7π/15 radians.

Arc length l = rθ = 12 × 7π/15 = 28π/5 ≈ 17.59 cm.

Area of sector = $(1/2)r^2\theta = (1/2) \times 144 \times 7\pi/15 = 168\pi/5 \approx 105.56$ cm².

Question: Exercise 30.3, Question 1 (i): Find the co-terminal angles of 55°.

Answer: Co-terminal angle = 55° + 360°n.

For example: 55° + 360° = 415° and 55° − 360° = −305°.

Question: Exercise 30.3, Question 1 (ii): Find the co-terminal angles of π/6.

Answer: Co-terminal angle = π/6 + 2πn.

For example: π/6 + 2π = 13π/6 and π/6 − 2π = −11π/6.

Question: Exercise 30.3, Question 1 (iii): Find the co-terminal angles of −45°.

Answer: Co-terminal angle = −45° + 360°n.

For example: −45° + 360° = 315° and −45° − 360° = −405°.

Question: Exercise 30.3, Question 1 (iv): Find the co-terminal angles of −3π/4.

Answer: Co-terminal angle = −3π/4 + 2πn.

For example: −3π/4 + 2π = 5π/4 and −3π/4 − 2π = −11π/4.

Question: Exercise 30.3, Question 2 (i): Identify the quadrant of the angle 8π/5.

Answer: 8π/5 = 288°, which lies between 270° and 360°, so it is in the fourth quadrant.

Question: Exercise 30.3, Question 2 (ii): Identify the quadrant of the angle 75°.

Answer: 75° lies between 0° and 90°, so it is in the first quadrant.

Question: Exercise 30.3, Question 2 (iii): Identify the quadrant of the angle −818°.

Answer: $-818° + 2 \times 360° = -818° + 720° = -98°$, which is in the third quadrant (on the negative side, $-90°$ to $-180°$).

Question: Exercise 30.3, Question 2 (iv): Identify the quadrant of the angle $1090°$.

Answer: $1090° - 2 \times 360° = 370°$, and $370° - 360° = 10°$, which is in the first quadrant.

Question: Exercise 30.3, Question 2 (v): Identify the quadrant of the angle $7\pi/6$.

Answer: $7\pi/6 = 210°$, which lies between $180°$ and $270°$, so it is in the third quadrant.

Question: Exercise 30.3, Question 2 (vi): Identify the quadrant of the angle $-5\pi/4$.

Answer: $-5\pi/4 = -225°$, and $-225° + 360° = 135°$, which is in the second quadrant.

Question: Exercise 30.3, Question 3 (i): Find the sign of $\cos 120°$.

Answer: $120°$ is in the second quadrant, where cos is negative.

Question: Exercise 30.3, Question 3 (ii): Find the sign of $\sin 340°$.

Answer: $340°$ is in the fourth quadrant, where sin is negative.

Question: Exercise 30.3, Question 3 (iii): Find the sign of $\sec 200°$.

Answer: $200°$ is in the third quadrant, where sec is negative.

Question: Exercise 30.3, Question 3 (iv): Find the sign of $\operatorname{cosec} 198°$.

Answer: $\operatorname{cosec} 198°$ is in the third quadrant, where cosec is negative.

Question: Exercise 30.3, Question 3 (v): Find the sign of $\tan(-\pi/3)$.

Answer: $-\pi/3 = -60°$, which is in the fourth quadrant, where tan is negative.

Question: Exercise 30.3, Question 3 (vi): Find the sign of $\cot(-2\pi/3)$.

Answer: $-2\pi/3 = -120°$, which is in the third quadrant, where cot is positive.

Question: Exercise 30.3, Question 4 (i): In which quadrant is θ given sin θ > 0 and cos θ < 0?

Answer: sin > 0 in the first or second quadrant, cos < 0 in the second or third quadrant, so θ is in the second quadrant.

Question: Exercise 30.3, Question 4 (ii): In which quadrant is θ given cos θ < 0 and tan θ > 0?

Answer: cos < 0 in the second or third quadrant, tan > 0 in the first or third quadrant, so θ is in the third quadrant.

Question: Exercise 30.3, Question 4 (iii): In which quadrant is θ given sec θ < 0 and sin θ < 0?

Answer: sec < 0 in the second or third quadrant, sin < 0 in the third or fourth quadrant, so θ is in the third quadrant.

Question: Exercise 30.3, Question 4 (iv): In which quadrant is θ given sec θ < 0 and cot θ > 0?

Answer: sec < 0 in the second or third quadrant, cot > 0 in the first or third quadrant, so θ is in the third quadrant.

Question: Exercise 30.3, Question 4 (v): In which quadrant is θ given tan θ < 0 and cos θ < 0?

Answer: tan < 0 in the second or fourth quadrant, cos < 0 in the second or third quadrant, so θ is in the second quadrant.

Question: Exercise 30.3, Question 4 (vi): In which quadrant is θ given 0 < cot θ < 1?

Answer: cot > 0 in the first or third quadrant, and cot < 1 means tan θ > 1, which happens in the first quadrant from 45° to 90° and in the third quadrant from 225° to 270°. So the given condition does not fix the quadrant uniquely — θ may be in the first or third quadrant.

Question: Exercise 30.3, Question 5: If cos θ = 3/5 and π/2 < θ < π, find the values of the remaining trigonometric ratios. [The printed question is

self-contradictory: in the second quadrant ($\pi/2 < \theta < \pi$) cos is negative, but the printed value 3/5 is positive. The solution below keeps the stated interval and uses $\cos\theta = -3/5$.]

Answer: In the second quadrant sin θ is positive.

$\sin^2\theta = 1 - \cos^2\theta = 1 - 9/25 = 16/25$, so $\sin\theta = 4/5$.

$\tan\theta = \sin\theta/\cos\theta = (4/5)/(-3/5) = -4/3$.

$\cot\theta = -3/4$.

$\sec\theta = -5/3$.

$\operatorname{cosec}\theta = 5/4$.

Question: Exercise 30.3, Question 6 (i): Find the remaining trigonometric ratios given $\sin\theta = \sqrt{3}/2$ and θ is in the second quadrant.

Answer: In the second quadrant cos is negative.

$\cos^2\theta = 1 - 3/4 = 1/4$, so $\cos\theta = -1/2$.

$\tan\theta = (\sqrt{3}/2)/(-1/2) = -\sqrt{3}$.

$\cot\theta = -1/\sqrt{3}$.

$\sec\theta = -2$.

$\operatorname{cosec}\theta = 2/\sqrt{3}$.

Question: Exercise 30.3, Question 6 (ii): Find the remaining trigonometric

ratios given cos θ = 2/3 and θ is in the fourth quadrant.

Answer: In the fourth quadrant sin is negative.

$\sin^2\theta = 1 - 4/9 = 5/9$, so $\sin\theta = -\sqrt{5}/3$.

$\tan\theta = (-\sqrt{5}/3)/(2/3) = -\sqrt{5}/2$.

$\cot\theta = -2/\sqrt{5}$.

$\sec\theta = 3/2$.

$\text{cosec}\,\theta = -3/\sqrt{5}$.

Question: Exercise 30.3, Question 6 (iii): Find the remaining trigonometric ratios given tan θ = −1/2 and θ is in the second quadrant.

Answer: In the second quadrant sin is positive and cos is negative.

From $\tan\theta = -1/2$, $\sin\theta = 1/\sqrt{5}$ and $\cos\theta = -2/\sqrt{5}$ (since $\sin^2 + \cos^2 = 1$ and $\sin/\cos = -1/2$).

$\cot\theta = -2$.

$\sec\theta = -\sqrt{5}/2$.

$\text{cosec}\,\theta = \sqrt{5}$.

Question: Exercise 30.3, Question 6 (iv): Find the remaining trigonometric ratios given sec θ = cosec θ = √2 and θ is in the first quadrant.

Answer: $\cos\theta = 1/\sec\theta = 1/\sqrt{2}$, $\sin\theta = 1/\text{cosec}\,\theta = 1/\sqrt{2}$.

tan θ = sin θ/cos θ = 1.

cot θ = 1.

Question: Exercise 30.3, Question 6 (v): Find the remaining trigonometric ratios given cos θ = 1/2 and tan θ is positive.

Answer: tan > 0 in the first or third quadrant, cos > 0 in the first or fourth quadrant, so θ is in the first quadrant.

$\sin^2\theta = 1 - 1/4 = 3/4$, so sin θ = √3/2.

tan θ = √3.

cot θ = 1/√3.

sec θ = 2.

cosec θ = 2/√3.

Question: Exercise 30.3, Question 7 (i): Find the value of tan 30° tan 60° + tan 45° cot 45°.

Answer: = (1/√3)(√3) + (1)(1) = 1 + 1 = 2.

Question: Exercise 30.3, Question 7 (ii): Find the value of (tan 60° − tan 30°)/(1 + tan 60° tan 30°).

Answer: = (√3 − 1/√3)/(1 + √3 × 1/√3) = ((3 − 1)/√3)/(1 + 1) = (2/√3)/2 = 1/√3.

Question: Exercise 30.3, Question 7 (iii): Find the value of sin 45°/(sin 45° + cot 45°).

Answer: = (1/√2)/(1/√2 + 1) = (1/√2)/((1 + √2)/√2) = 1/(1 + √2) = √2 − 1.

Question: Exercise 30.3, Question 7 (iv): Find the value of cos(π/6)·cos(π/3)

+ sin(π/6)·cos(π/3).

Answer: = (√3/2)(1/2) + (1/2)(1/2) = √3/4 + 1/4 = (√3 + 1)/4.

Question: Exercise 30.3, Question 7 (v): Find the value of (tan(π/4) − tan(π/6))/(1 + tan(π/4)·tan(π/6)).

Answer: = (1 − 1/√3)/(1 + 1 × 1/√3) = ((√3 − 1)/√3)/((√3 + 1)/√3) = (√3 − 1)/(√3 + 1) = 2 − √3.

Question: Exercise 30.3, Question 7 (vi): Find the value of (tan 45° + cot 45°)/(sin 60° cos 30° − cos 60° sin 30°).

Answer: Numerator = 1 + 1 = 2.

Denominator = sin(60° − 30°) = sin 30° = 1/2.

So the value = 2/(1/2) = 4.

Question: Exercise 30.4, Question 1 (i): Prove $\sin^2\theta = (\sec^2\theta - 1)\cos^2\theta$, where $\cos\theta \neq 0$.

Answer: Right side = $(\sec^2\theta - 1)\cos^2\theta = \tan^2\theta \cdot \cos^2\theta = (\sin^2\theta/\cos^2\theta) \cdot \cos^2\theta = \sin^2\theta$ = left side.

Question: Exercise 30.4, Question 1 (ii): Prove $(1 + \sin\theta)/\cos\theta = \tan\theta + \sec\theta$, where $\cos\theta \neq 0$.

Answer: Right side = $\tan\theta + \sec\theta = \sin\theta/\cos\theta + 1/\cos\theta = (1 + \sin\theta)/\cos\theta$ = left side.

Question: Exercise 30.4, Question 1 (iii): Prove $\tan\theta = \sin\theta\sqrt{(1 + \tan^2\theta)}$, where $\cos\theta \neq 0$. [The printed identity is only correct when $\cos\theta > 0$, because $\sqrt{(1 + \tan^2\theta)} = |\sec\theta|$.]

Answer: Right side = $\sin\theta \cdot \sqrt{(\sec^2\theta)} = \sin\theta \cdot |\sec\theta|$.

When $\cos\theta > 0$, $|\sec\theta| = \sec\theta$, so right side = $\sin\theta \cdot \sec\theta = \sin\theta/\cos\theta = \tan\theta$ = left side.

Question: Exercise 30.4, Question 1 (iv): Prove $\cos^4\theta - \sin^4\theta = \cos^2\theta - \sin^2\theta$.

Answer: Left side = $(\cos^2\theta - \sin^2\theta)(\cos^2\theta + \sin^2\theta) = (\cos^2\theta - \sin^2\theta) \times 1 = \cos^2\theta - \sin^2\theta$ = right side.

Question: Exercise 30.4, Question 1 (v): Prove $\sin^3\theta = \sin\theta - \sin\theta\cos^2\theta$.

Answer: Right side = $\sin\theta(1 - \cos^2\theta) = \sin\theta \cdot \sin^2\theta = \sin^3\theta$ = left side.

Question: Exercise 30.4, Question 1 (vi): Prove $\sqrt{\frac{1 + \cos\theta}{1 - \cos\theta}} = \frac{\sin\theta}{1 - \cos\theta}$, where $\cos\theta \neq 1$. [The printed identity is only correct when $\sin\theta > 0$.]

Answer: Square of left side = $\frac{1 + \cos\theta}{1 - \cos\theta} = \frac{(1 + \cos\theta)^2}{(1 - \cos\theta)(1 + \cos\theta)} = \frac{(1 + \cos\theta)^2}{\sin^2\theta}$.

When $\sin\theta > 0$, left side = $\frac{1 + \cos\theta}{\sin\theta}$, and right side = $\frac{\sin\theta}{1 - \cos\theta} = \frac{\sin\theta(1 + \cos\theta)}{\sin^2\theta} = \frac{1 + \cos\theta}{\sin\theta}$.

Both sides are equal.

Question: Exercise 30.4, Question 1 (vii): Prove $\sqrt{\frac{\sec\theta + 1}{\sec\theta - 1}} = \frac{\sec\theta + 1}{\tan\theta}$, where $\tan\theta \neq 0$. [The printed identity is only correct when $\frac{\sec\theta + 1}{\tan\theta} \geq 0$.]

Answer: Square of left side = $\frac{\sec\theta + 1}{\sec\theta - 1} = \frac{(\sec\theta + 1)^2}{(\sec\theta - 1)(\sec\theta + 1)} = \frac{(\sec\theta + 1)^2}{\tan^2\theta}$.

So left side = $\frac{|\sec\theta + 1|}{|\tan\theta|}$, which equals the right side $\frac{\sec\theta + 1}{\tan\theta}$ when $\frac{\sec\theta + 1}{\tan\theta} \geq 0$.

Question: Exercise 30.4, Question 1 (viii): Prove $\frac{\sin^2\theta}{\cos\theta} + \cos\theta = \sec\theta$, where $\tan\theta \neq 0$.

Answer: Left side = $\frac{\sin^2\theta + \cos^2\theta}{\cos\theta} = \frac{1}{\cos\theta} = \sec\theta$ = right side.

Question: Exercise 30.4, Question 1 (ix): Prove $\sec\theta - \cos\theta = (\cot\theta +$

cos θ)(tan θ − sin θ), where cos θ ≠ 0 or sin θ ≠ 0. [The printed identity is wrong as printed — see the counterexample below.]

Answer: This identity is not correct as printed.

At θ = 45°: left side = √2 − 1/√2 = 1/√2 ≈ 0.707, but right side = (1 + 1/√2)(1 − 1/√2) = 1 − 1/2 = 0.5.

The two sides are not equal.

The related correct identity is: sec θ − cos θ = (1 − cos²θ)/cos θ = sin²θ/cos θ = sin θ tan θ.

Question: Exercise 30.4, Question 1 (x): Prove (cot θ + cosec θ)/(sin θ + tan θ) = cosec θ cot θ, where sin θ ≠ 0.

Answer: Numerator of left side = cos θ/sin θ + 1/sin θ = (1 + cos θ)/sin θ.

Denominator = sin θ + sin θ/cos θ = sin θ(1 + cos θ)/cos θ.

So left side = [(1 + cos θ)/sin θ] × [cos θ/(sin θ(1 + cos θ))] = cos θ/sin²θ = cosec θ cot θ = right side.

Question: Exercise 30.4, Question 1 (xi): Prove sin²θ = tan²θ/(1 + tan²θ), where cos θ ≠ 0.

Answer: Right side = tan²θ/sec²θ = (sin²θ/cos²θ)/(1/cos²θ) = sin²θ = left side.

Question: Exercise 30.4, Question 1 (xii): Prove cos²θ = cot²θ/(1 + cot²θ), where sin θ ≠ 0.

Answer: Right side = cot²θ/cosec²θ = (cos²θ/sin²θ)/(1/sin²θ) = cos²θ = left side.

Question: Exercise 30.4, Question 1 (xiii): Prove sin θ cos θ tan θ + sin θ

cos θ cot θ = 1, where sin θ ≠ 0 or cos θ ≠ 0.

Answer: Left side = sin θ cos θ(sin θ/cos θ) + sin θ cos θ(cos θ/sin θ) = sin²θ + cos²θ = 1 = right side.

Question: Exercise 30.4, Question 2 (i): Convert (sec θ + 1)(sec θ − 1) into tan²θ.

Answer: (sec θ + 1)(sec θ − 1) = sec²θ − 1 = tan²θ.

Question: Exercise 30.4, Question 2 (ii): Convert tan θ into cos θ/sin θ − cosec θ/cos θ. [The printed expression has a sign error — see the correct form below.]

Answer: The printed expression cos θ/sin θ − cosec θ/cos θ does not equal tan θ but −tan θ: cos θ/sin θ − (1/sin θ)/cos θ = (cos²θ − 1)/(sin θ cos θ) = −sin²θ/(sin θ cos θ) = −tan θ.

The correct form is: tan θ = cosec θ/cos θ − cos θ/sin θ, because (1/sin θ)/cos θ − cos θ/sin θ = (1 − cos²θ)/(sin θ cos θ) = sin²θ/(sin θ cos θ) = tan θ.

Question: Exercise 30.4, Question 2 (iii): Convert cosec θ + cot θ into (1 + cos θ)/sin θ.

Answer: cosec θ + cot θ = 1/sin θ + cos θ/sin θ = (1 + cos θ)/sin θ.

Question: Exercise 30.5, Question 1: From the top of a 102-metre-high lighthouse, the angle of depression of a ship is 30°18′. How far is the ship from the lighthouse?

Answer: Angle of depression = angle of elevation from the ship = 30°18′ = 30.3°.

In the triangle tan(30.3°) = 102/x, where x is the ship's distance from the lighthouse.

tan(30.3°) ≈ 0.5843, so x = 102/0.5843 ≈ 174.6 m.

Question: Exercise 30.5, Question 2: A ladder makes a 60° angle with the ground and reaches a height of 6 m. Find the length of the ladder.

Answer: sin 60° = 6/L, where L is the length of the ladder.

So L = 6/sin 60° = 6/(√3/2) = 12/√3 = 4√3 ≈ 6.93 m.

Question: Exercise 30.5, Question 3: Find the angle of elevation when a 6 m tall pole casts a shadow of length 2√3 m.

Answer: tan θ = 6/(2√3) = 3/√3 = √3, so θ = 60°.

Question: Exercise 30.5, Question 4: The angle of elevation of the top of a hill is 30°. After walking 210 metres towards the hill from a point, the angle of elevation is 45°. Find the height of the hill.

Answer: Let the height of the hill be h and the distance from the second point be x.

tan 45° = h/x, so x = h.

tan 30° = h/(x + 210), so 1/√3 = h/(h + 210).

This gives h + 210 = h√3, i.e. 210 = h(√3 − 1), so h = 210/(√3 − 1) = 105(√3 + 1) ≈ 286.9 m.

Question: Exercise 30.5, Question 5: An observer is 4280 m above the ground and 9613 m away from a farmhouse. Find the angle of depression of the farmhouse from the observer. [9613 m is read as the horizontal distance.]

Answer: tan θ = 4280/9613 ≈ 0.4452, so θ = arctan(0.4452) ≈ 24.03° ≈ 24°.

Review Exercise 30, Question 1 (i): The system of measurement in which

an angle is measured in radians is called ___.

(a) CGS system

(b) base-60 system

(c) MKS system

✓ (d) circular system

Review Exercise 30, Question 1 (ii): The union of two rays with a common endpoint is called ___.

✓ (a) an angle

(b) a degree

(c) a minute

(d) a radian

Review Exercise 30, Question 1 (iii): 10° = ___.

(a) 600″

(b) 3600″

✓ (c) 600′

(d) 600

Review Exercise 30, Question 1 (iv): 5π/4 radians = ___.

(a) 115°

✓ (b) 225°

(c) 135°

(d) 45°

Review Exercise 30, Question 1 (v): ___ = (1/2) sec 45°.

(a) √2

✓ (b) √2/2

(c) 2/√2

(d) 1/√2

Review Exercise 30, Question 1 (vi): cosec θ · sin θ = ___.

✓ (a) 1

(b) 0

(c) −1

(d) 0.5

Review Exercise 30, Question 1 (vii): $\sec^2\theta - \tan^2\theta$ = ___.

(a) 0

✓ (b) 1

(c) −1

(d) $\cos^2\theta$

Review Exercise 30, Question 1 (viii): In a unit circle, for the point (x,y), sin θ = ___.

(a) x

(b) −x

✓ (c) y

(d) −y

Review Exercise 30, Question 1 (ix): If $\sin\theta = -1/\sqrt{2}$ and θ is in the third quadrant, then tan θ = ___.

(a) −1

✓ (b) 1

(c) +1

(d) $1/\sqrt{2}$ [Printed options

✓ (b) and

(c) are the same value — both are correct.]

Review Exercise 30, Question 1 (x): If an object is above the observer's level, the angle between the horizontal ray and the line of sight is called ___.

(a) angle of depression

✓ (b) angle of elevation

(c) an obtuse angle

(d) none

Question: Review Exercise 30, Question 2: Define angle and angle in standard position.

Answer: An angle is the union of two rays with a common endpoint. An angle in standard position is one whose vertex is at the origin and whose initial side is the positive part of the x-axis.

Question: Review Exercise 30, Question 3: Convert 90° 30′ 70″ into radians.

Answer: 70″ = 1′ 10″, so 90° 30′ 70″ = 90° 31′ 10″ = (90 + 31/60 + 10/3600)° = 90.5194°.

In radians: 90.5194 × π/180 ≈ 1.5799 radians.

Question: Review Exercise 30, Question 4: How many degrees are there in (7200)′?

Answer: 1° = 60′, so 7200′ = 7200/60 = 120°.

Question: Review Exercise 30, Question 5: Convert the measure 2π/3 radians into degrees.

Answer: 2π/3 radians = (2π/3) × (180/π)° = 120°.

Question: Review Exercise 30, Question 6: Write 3/π radians in D° M′ S″ form.

Answer: 3/π radians = (3/π) × (180/π)° = 540/π² ≈ 54.7179°.

0.7179° × 60 = 43.074′, so 43′ + 0.074 × 60″ = 43′ 4.4″.

Answer ≈ 54° 43′ 4″.

Question: Review Exercise 30, Question 7: Find the value of r given l = 7 cm and θ = π/4 radians. [The printed question says "find θ", but θ is already given — the question means r.]

Answer: r = l/θ = 7/(π/4) = 28/π ≈ 8.91 cm.

Question: Review Exercise 30, Question 8: Find the value of θ when the length of an arc equals the radius of the circle.

Answer: l = r, so θ = l/r = r/r = 1 radian.

Question: Review Exercise 30, Question 9: Prove $(1 - \cos^2\theta)(1 + \cot^2\theta) = 1$.

Answer: Left side = $\sin^2\theta \cdot \operatorname{cosec}^2\theta = \sin^2\theta \times (1/\sin^2\theta) = 1$ = right side.

Question: Review Exercise 30, Question 10: Find the area of the sector of a circle when r = 2 cm and θ = 3 radians.

Answer: Area of sector = $(1/2)r^2\theta = (1/2) \times 4 \times 3 = 6$ cm².